# Targeted Bottlenecking of Self-Supervised Speech Layers for Generalizable Audio Deepfake Detection

Jakob Kauffmann

2026

**Abstract**

Audio deepfake detectors trained on one corpus often generalize poorly to unfamiliar recording channels, codecs, and synthesizers. This project examines where in a self-supervised speech model the channel and codec nuisance is concentrated, and introduces GLADA, an architecture that aggregates SSL hidden states by layer group, applies a Variational Information Bottleneck and an adversarial domain discriminator to each group, and encourages group diversity through a CKA penalty. A multi-seed cross-dataset study on In-the-Wild, ASVspoof 2021 LA, ASVspoof 2021 DF, WaveFake, and CodecFake+ finds that restricting the bottleneck to the lowest layer group outperforms both a single post-pooling bottleneck and the full per-group placement on every secondary out-of-domain dataset. Per-group linear probes and per-group CKA against handcrafted acoustic descriptors both localize codec and channel signal to that lowest group, which supports the placement.

**Keywords:** audio deepfake detection; self-supervised speech models; variational information bottleneck; per-layer-group bottleneck; adversarial domain adaptation; cross-domain generalization.

# Targeted Bottlenecking of Self-Supervised Speech Layers for Generalizable Audio Deepfake Detection

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Master of Science

by

Jakob Kauffmann

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The Designated Project Committee Approves the Project Titled

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## ABSTRACT

Targeted Bottlenecking of Self-Supervised Speech Layers for Generalizable Audio Deepfake Detection

by Jakob Kauffmann

Audio deepfake detectors trained on one corpus often generalize poorly to unfamiliar recording channels, codecs, and synthesizers. This project examines where in a self-supervised speech model the channel and codec nuisance is concentrated, and introduces GLADA, an architecture that aggregates SSL hidden states by layer group, applies a Variational Information Bottleneck and an adversarial domain discriminator to each group, and encourages group diversity through a CKA penalty. A multi-seed cross-dataset study on In-the-Wild, ASVspoof 2021 LA, ASVspoof 2021 DF, WaveFake, and CodecFake+ finds that restricting the bottleneck to the lowest layer group outperforms both a single post-pooling bottleneck and the full per-group placement on every secondary out-of-domain dataset. Per-group linear probes and per-group CKA against handcrafted acoustic descriptors both localize codec and channel signal to that lowest group, which supports the placement.



**Keywords:** audio deepfake detection, self-supervised speech models, variational information bottleneck, per-layer-group bottleneck, adversarial domain adaptation, cross-domain generalization

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# CHAPTER 1

# Introduction

## 1.1 Motivation

Advances in deep generative models have made it possible to synthesize speech that is perceptually indistinguishable from genuine recordings. Text-to-speech systems such as Variational Inference with adversarial learning for end-to-end Text-to-Speech (VITS) [1] and VALL-E [2], along with voice conversion techniques, can produce convincing imitations of any target speaker from only a few seconds of reference audio. These capabilities pose serious threats to trust in audio communications. In January 2024, thousands of New Hampshire voters received robocalls with an AI-generated imitation of President Biden urging them not to vote in the primary [3]; the voice clone was created with a commercial tool in minutes. The same month, a Hong Kong finance worker transferred $25 million after a video call in which the CFO and multiple colleagues were all real-time deepfake audio-visual clones [4].

Audio deepfake detection (the task of determining whether a speech recording is genuine or synthetically generated) has grown into an active research area, driven by the ASVspoof challenge series and the proliferation of zero-shot TTS systems. Detection systems trained and evaluated on standard benchmarks have achieved strong performance: on the ASVspoof 2019 Logical Access (LA) dataset, state-of-the-art models achieve sub-1% Equal Error Rate (EER), with systems such as AASIST [5] reaching 0.83% EER. These results suggest that the detection problem might be largely solved.

These results do not hold on real-world audio. Müller et al. [6] demonstrated this gap using the In-the-Wild dataset, a collection of audio from diverse online sources including social media, podcasts, and video platforms. Models that achieve sub-1% EER on ASVspoof 2019 LA suffer catastrophic degradation on In-the-Wild:

RawNet2 rises to approximately 37.8% EER, AASIST to approximately 31% EER, and RawGAT-ST to approximately 37.2% EER.

The cause is *domain shift*: the conditions under which training data was collected and the conditions encountered in deployment do not match. ASVspoof 2019 LA recordings are produced in controlled studio environments at 16 kHz sampling rate in FLAC (Free Lossless Audio Codec) format, using a small set of known synthesis algorithms. Real-world audio, by contrast, may contain background noise, reverberation, variable microphone quality, lossy compression, and deepfakes generated by unknown methods.

### 1.2 Research Gap

Recent work has demonstrated two complementary strategies for improving cross-domain generalization in audio deepfake detection. First, *hierarchical layer aggregation* exploits the observation that different layers of self-supervised speech models encode different types of artifact information. Liang et al. [7] grouped SSL hidden states and applied hierarchical contrastive learning across groups, achieving 6.87% EER on the In-the-Wild dataset, a substantial improvement over flat aggregation baselines. El-Kheir et al. [8] provided empirical support for this design by showing that lower SSL layers capture acoustic and spectral cues while upper layers encode higher-level linguistic and synthesis-method information, and that different deepfake attacks are best detected by different layer subsets.

Second, *adversarial domain adaptation* has proven effective for aligning source- and target-domain feature distributions without requiring target-domain labels. Huang et al. [9] combined a Variational Information Bottleneck with adversarial domain alignment in IB-CAAN, achieving 4.93% ITW EER, the strongest published result on that benchmark. These adversarial approaches trace back to the Domain-Adversarial

Neural Network (DANN) framework of Ganin et al. [10], which uses gradient reversal to learn domain-invariant representations.

In the broader deep-learning literature, multi-layer domain adaptation (separate domain discriminators placed at multiple representation depths) has been explored in computer vision and multimodal sensing. Fang et al. [11] proposed MLADA, which attaches domain discriminators to several intermediate layers of a visual feature extractor; Huang et al. [12] developed MLAN with multi-layer adversarial networks; Meegahapola et al. [13] proposed M3BAT for multimodal mobile sensing with multi-branch adversarial training. Natural language processing has the closest direct analogue in multi-domain adversarial neural networks for text classification [14], which separate private and shared features per layer. Other NLP extensions of adversarial DA, such as the meta self-paced framework MSP-DA [15], refine the adversarial training procedure on top of DANN through meta-learning rather than the per-layer placement question. The architectural pattern of multi-layer adversarial domain adaptation therefore has precedents across vision, multimodal sensing, and NLP; what has not been characterised for SSL audio anti-spoofing is the placement of an information bottleneck co-located with each per-group discriminator inside a grouped SSL aggregation pipeline.

My starting point was to combine the grouped-layer structure that has proved beneficial for hierarchical feature extraction [7] with per-group adversarial domain adaptation drawn from the multi-layer DA literature in vision [11, 12, 13] and the stratified adversarial setting in NLP [14], and to add a per-group Variational Information Bottleneck inspired by IB-CAAN [9] and SIBNet [16]. The hypothesis was that attaching per-group domain discriminators and bottlenecks to grouped SSL representations would produce finer-grained domain alignment than a single global discriminator, since each group captures a distinct kind of information that may need its own alignment. Experiments confirmed the per-group placement claim under the

diagnostic ablation (Chapter 5), but the full GLADA configuration with all auxiliary losses produced the worst cross-domain result of all 11 configurations tested at 19.71% ITW EER, against 12.98% for the best ablation and 13.60% for a backbone-only baseline. The per-group adversarial objectives interacted destructively with the remaining auxiliary losses under monolithic post-pooling VIB, producing negative transfer rather than the anticipated improvement; the per-group VIB variant introduced in Chapter 3 recovers this, with a multi-seed mean of $11.97 \pm 1.31\%$ ITW EER.

## 1.3 Research Objectives

This project pursues the following four research objectives:

1. **Design and implement the GLADA framework.** An architecture that combines grouped-layer attention pooling over SSL hidden states with per-group adversarial domain discriminators, a Variational Information Bottleneck, a speaker adversary, and a CKA-based diversity penalty.

2. **Evaluate whether per-group domain adaptation outperforms global domain adaptation.** The full GLADA configuration is compared against a global-DA baseline and a backbone-only baseline on six evaluation datasets spanning controlled, in-the-wild, and codec-based conditions.

3. **Conduct an ablation study isolating each component's contribution.** Eleven configurations are evaluated across all six datasets with bootstrap confidence intervals, quantifying the marginal contribution of each GLADA component.

4. **Compare GLADA against published state-of-the-art methods.** The best configurations are benchmarked against IB-CAAN [9], HierCon [7], and other recent systems across five SSL backbone architectures.

## 1.4 Contributions

This project makes the following contributions, structured as one headline framing (A0), four substructure claims (A1--A4), and nine confirmations of precedent findings carried forward into the SSL anti-spoofing setting (B1--B9).

**Headline.**

**A0. Targeted per-layer-group information bottlenecking is more effective than a single global post-pooling bottleneck for cross-domain audio deepfake detection.** Lower self-supervised speech model layer groups carry most of the cross-domain channel and acoustic information; the central evidence is a multi-seed leave-one-out per-group VIB ablation and a per-group CKA-to-acoustic-features plus linear domain/codec probing analysis. Layer-wise SSL analyses for anti-spoofing [8, 17, 18, 19, 20] and SSL representation studies [21, 22] establish lower-layer importance and hierarchical specialization; the contribution here is the placement-comparison evidence inside a grouped SSL aggregation pipeline.

**Substructure.**

**A1. Architectural design.** A per-group Variational Information Bottleneck co-located with each per-layer-group adversarial discriminator, one bottleneck per group, with CKA-diversity regularisation across groups. The individual components are not novel: multi-layer adversarial domain adaptation appears in vision (MLADA [11], MLAN [12]) and multimodal sensing (M3BAT [13]); VIB combined with adversarial alignment exists for ASR/SV/anti-spoofing (IB-CAAN [9], SIBNet [16], VBDA [23], variational DA + MI maximisation [24, 25], MIM-DG [26]); distributed/multi-task/multi-view per-component VIB exists in

vision and communications (Distributed VIB [27], Multi-Task VIB [28], Multi-view IB [29], Cell-VIB [30], FrankenSplit [31], adversarial bottleneck injection [32], Structured IB [33]). The bundled per-group VIB-before-per-group-adversary co-location applied to grouped SSL speech features for anti-spoofing has not appeared in the consensus literature.

**A2. Diagnostic finding.** Monolithic post-pooling VIB underperforms per-group pre-discriminator VIB inside a grouped SSL aggregation pipeline on the In-the-Wild benchmark. On the across-seed mean ($n = 3$), the placement comparison gives 14.66% ITW EER (monolithic) versus 11.97 $\pm$ 1.31% (per-group). The directional finding is consistent on every paired seed. Across the cross-domain axis, the g0-only configuration (bottleneck active only on the lowest layer group) further generalises the placement claim: it is the lowest-mean configuration on each of the four secondary OOD benchmarks (ASVspoof 2021 LA, ASVspoof 2021 DF, WaveFake, CodecFake+), beating both monolithic and full per-group on every cross-dataset column. Conversely, bottlenecking only the upper groups (`drop_vib_g0`, with $g_0$ unbottlenecked) is the worst-ranked variant on ITW and below g0-only on every secondary OOD dataset; the contrast localises the cross-domain signal to the lowest SSL layer group at the level of EER rather than just representations. This specific bottleneck-placement comparison, including the cross-dataset ranking, has not been characterised for SSL anti-spoofing; the closest precedents (IB-CAAN [9], SIBNet [16], VBDA [23]) study a single bottleneck rather than co-located per-group placement.

**A3. Mechanism evidence.** A leave-one-out per-group VIB ablation isolates each group's contribution. Dropping the lowest-group bottleneck ($g_0$) raises ITW EER by 5.94 pp; dropping the middle group ($g_1$) by 2.13 pp; dropping the top group

($g_2$) by 0.32 pp, within seed noise. $g_0$ dominates the cross-domain benefit, but $g_0$-only is 1.45 pp behind the full per-group VIB on the across-seed mean: $g_1$ and $g_2$ contribute non-additively. The acoustic-feature analysis (per-group probes and CKA) localises codec discriminability monotonically with depth, in line with layer-wise analyses for ADD [8, 18, 19] and multi-branch interpretability studies [34, 35].

**A4. Methodological diagnostic.** A combined per-layer-group CKA-to-handcrafted-acoustic-features and per-layer-group linear domain/codec probing analysis, applied jointly as mechanism evidence for bottleneck-placement design choices. CKA against acoustic features exists for ASR and general SSL [21, 22, 36, 37, 38]; linear probing of nuisance signals exists for speaker/demographic identity [39, 40, 41, 42, 43, 44]; the joint per-group application of both to a bottleneck-placement design choice in audio anti-spoofing is the contribution.

**Confirmations of precedent findings.** Carried forward into the SSL anti-spoofing setting with multi-seed evidence and honest accounting:

**B1. VIB combined with adversarial domain alignment improves cross-domain anti-spoofing.** Established by IB-CAAN [9], SIBNet [16], anti-spoofing transfer with VIB, VBDA [23], and MI-maximising variational domain adversarial learning for speaker verification [24, 25, 26, 45, 46]. Confirmed here with multi-seed validation; the residual 6.74 pp gap to IB-CAAN is attributed to confidence-aware attention weighting and stronger augmentation rather than to the bottleneck placement itself.

**B2. Lower SSL transformer layers carry more anti-spoof signal.** Established by [8, 18, 17, 19, 20, 46, 47]. Confirmed here under a different probe (leave-one-out

per-group VIB) and quantified by cross-domain attribution.

**B3. Hierarchical SSL aggregation outperforms last-layer features.** Established by HierCon [7], HM-CONFORMER [48], Attentive Merging [47], Wav2DF-TSL [49], and hierarchical structure learning in Poincaré sphere [50]. Confirmed here.

**B4. Speaker GRL supports anti-spoof generalisation.** Established by adversarial speaker verification [51], channel-invariant speaker embedding via joint multi-task and adversarial training [52], joint optimisation of ASV and anti-spoofing in the embedding space [53], and speaker disentanglement of SSL via interpretability [54]. Confirmed here.

**B5. RawBoost and waveform augmentation help.** Established by RawBoost [55], WavLM-back-end recipes with combined augmentation [56], WavLM + advanced RawBoost [57], multilingual augmentation strategies [58], the Vicomtech wav2vec2 baseline [59], and RIR augmentation analyses [60, 61, 62]. Confirmed here; the absence of MUSAN/RIR is part of our engineering-gap discussion.

**B6. Cross-dataset evaluation reveals dataset-specific overfit.** Established across cross-corpus literature; confirmed here on six datasets (ASVspoof 2019 LA, ITW, 2021 LA, 2021 DF, WaveFake, CodecFake+).

**B7. Linear probes quantify residual nuisance after invariance training.** Established by causally disentangled contrastive speaker embeddings [39], multi-source DA for forensic speaker recognition [40], the SSL review of Mohamed et al. [41], the unified invariance framework of Plachouras et al. [42], patch-token probing [43], and larger probing-head benchmarks [44]. Confirmed here per layer

group: domain signal remains linearly recoverable at 0.96--0.98 accuracy on the post-VIB representations despite the adversarial alignment objective.

**B8. CKA reveals hierarchical specialisation in SSL audio.** Established by Pasad et al. [21], Riera et al. [22], the equivalence of CKA, RSA, and CCA [36], the multi-level SSL feature gating analysis [19], CKA-based distillation [37], and comparisons of hand-crafted versus deep features in audio scenes [38]. Confirmed here per layer group.

**B9. Staged SSL backbone unfreezing is an open stability problem.** Established by the BERT fine-tuning stability paper [63], scheduled unfreezing with Fisher information [64], Speech-FT [65], less-forgetting fine-tuning for SSL [66], WavLM's own delayed-unfreezing schedule [67], child-ASR transfer [68], and continual layer-specific fine-tuning [69]. Confirmed here; the Phase 2 ramp instability documented in Chapter 6 is one more anti-spoof data point, not a GLADA-specific bug.

**Companion deliverable.** A systematic survey of audio deepfake detection methods, conducted as a component of this master's writing project, accompanies this report [70]. The survey covers feature representations, model architectures, and domain adaptation strategies, with PRISMA-based selection and cross-dataset evaluation experiments that fed directly into the GLADA design. A condensed version of the earlier full-stack negative-transfer analysis (the result reported in Section 5.1) was accepted as a full paper at the 50th IEEE Computer Software and Applications Conference (COMPSAC 2026), held in Madrid, Spain; the conference paper has since been withdrawn from the formal venue but remains an archival snapshot of the negative-transfer analysis on which the present report builds.

## 1.5 Report Organization

- **Chapter 2** surveys the technical background: audio deepfake generation and detection, self-supervised speech models, domain adaptation theory, multi-layer DA, attention-based layer aggregation, and representation similarity.
- **Chapter 3** works through the GLADA architecture itself: hierarchical grouped-layer attention pooling, the Variational Information Bottleneck, per-group domain discriminators, the speaker adversary, and the two-phase training strategy.
- **Chapter 4** describes the datasets, implementation, ablation design, evaluation protocol, and baselines.
- **Chapter 5** reports the experiments: full GLADA performance, the ablation study, the backbone comparison, cross-dataset generalisation, and training dynamics.
- **Chapter 6** interprets those results, focusing on per-group vs. global DA, the role of VIB, and the limitations of the study.
- **Chapter 7** pulls the findings together and lays out the directions for future work that follow from them.

# CHAPTER 2

## Background and Related Work

Earlier in the writing project I compiled a 60-source survey of audio deepfake detection covering architectures, cross-dataset evaluation, and the factors that drive generalization; the survey is included with the submitted deliverables [70]. The sections below cover the narrower subset of that material that GLADA directly builds on.

### 2.1 Audio Deepfake Generation

Audio deepfakes come from two synthesis families: text-to-speech (TTS) and voice conversion (VC). TTS has progressed substantially over the past decade. Tacotron 2 [71] generates mel-spectrograms from text through a sequence-to-sequence model with near-human output quality. FastSpeech 2 [72] replaces autoregressive decoding with a parallel architecture, gaining a large speedup. VITS [1] pairs variational inference with adversarial training, closing more of the gap to human speech. The most striking recent step is VALL-E [2]: it frames synthesis as neural codec language modelling and can clone an arbitrary speaker from three seconds of reference audio.

Voice conversion addresses the complementary problem: change the speaker identity while preserving what was said. TTS starts from text; VC starts from an existing utterance and remaps it onto the acoustic profile of a target speaker. Both paradigms rely on neural vocoders (WaveNet [73], WaveRNN [74], HiFi-GAN [75]), and HiFi-GAN has become the default in nearly every recent TTS and VC pipeline.

The net effect is that contemporary synthesis artefacts now lie below what a human listener can reliably flag. HiFi-GAN spectrograms no longer exhibit the smearing and spectral seams of older unit-selection or parametric vocoders, and zero-shot systems such as VALL-E reproduce prosodic detail that used to require speaker-specific training data. With those surface cues effectively gone, detection has

to depend on learned representations rather than handcrafted acoustic features.

## 2.2 Audio Deepfake Detection

### 2.2.1 Traditional and End-to-End Approaches

Early detectors fed handcrafted acoustic features into generative or discriminative classifiers. Typical pairings combined MFCCs [76], LFCCs, or CQCCs [77] with a Gaussian mixture or support-vector classifier [78]. On constrained benchmarks these systems were competitive, but they assumed the spectral signature of each synthesis algorithm was already known and representable through the chosen features.

End-to-end architectures removed the feature-engineering step. Tak et al. [79] introduced RawNet2, processing raw waveforms through learned sinc-band filters, residual blocks, and a GRU sequence encoder. Operating on raw samples lets the model capture codec-related artefacts that a fixed filterbank would discard. Jung et al. [5] pushed the line further with AASIST, whose graph attention over spectro-temporal features captures long-range artefact dependencies and drives the EER below 1%. Both systems defined the state of the art on ASVspoof 2019 LA [80].

### 2.2.2 The Generalization Crisis

In-domain numbers are strong, yet the same detectors fail on real-world audio. The In-the-Wild (ITW) benchmark [6], collected from social media and news broadcasts, makes this concrete: RawNet2 rises to roughly 37.8% EER and AASIST to roughly 31%, both close to a random classifier. The controlled ASVspoof training distribution and real-world acoustic conditions simply do not match.

The shift responsible is not one phenomenon. Acoustically, ITW audio contains environmental noise, room reverberation, microphone variation, and lossy codec compression that the studio training data never contained. Generatively, part of ITW comes from synthesis methods released after the ASVspoof 2019 cutoff (neural vocoders and zero-shot TTS in particular), whose artefacts differ qualitatively from

the training attacks. The combined gap maps the current landscape: end-to-end systems sit in the 31--38% ITW EER range, SSL fine-tuning reduces that to 5--19%, and SSL plus specialised domain adaptation reaches 4--7% [9, 17].

### 2.2.3 Self-Supervised Learning Approaches

Self-supervised learning (SSL) narrows the generalisation gap because pre-trained acoustic features transfer across domains with much less degradation than supervised features trained from scratch. Huang et al. [9] combined an XLSR backbone with information-bottleneck conditioning and confidence-aware adversarial alignment, reaching 4.93% ITW EER, well ahead of end-to-end baselines. Serrano et al. [17] took an ensemble route, fusing scores from BEATs, WavLM-Large, MMS, and XLSR for 6.4% ITW EER; the gain comes from backbones that capture complementary artefact cues. Pan et al. [47] showed that attentively merging every SSL hidden state outperforms using only the final one; useful artefact signal sits at intermediate depth, not the top of the network. El-Kheir et al. [8] pushed this further by showing that the domain shift visible at one layer is qualitatively different from the shift at another. That observation is the empirical basis for layer-stratified adaptation (Section 2.6).

## 2.3 Self-Supervised Learning for Speech

### 2.3.1 Wav2Vec 2.0

Wav2Vec 2.0 [81] learns from raw audio by masking latent time steps and training a Transformer context network to identify the correct quantised token through a contrastive objective. A convolutional feature encoder feeds into the Transformer context network. Two variants exist (Base with 12 layers, 95M parameters; Large with 24 layers, 317M parameters), both pre-trained on 960 hours of LibriSpeech. The resulting representations capture rich phonetic structure without labelled supervision.

### 2.3.2 XLS-R (XLSR-300M)

XLS-R [82] extends the Wav2Vec 2.0 framework to a massively multilingual setting. Pre-training uses roughly 436,000 hours of unlabelled speech across 128 languages, which enables cross-lingual transfer without language-specific supervision. XLSR-300M reuses the Wav2Vec 2.0-Large CNN encoder and Transformer, but the diversity of its training data pushes the model toward language-agnostic acoustic and phonetic features. That is particularly useful for audio deepfake detection, where test audio routinely comes from speakers and recording conditions that the labelled training set never anticipated. A 1B-parameter variant extends coverage further, at substantially higher computational cost.

### 2.3.3 WavLM

WavLM [67] modifies the Wav2Vec 2.0 pre-training objective in one important respect: it adds a denoising masked speech prediction task. WavLM has to recover clean speech representations from inputs corrupted with overlapping speakers and additive noise, on top of predicting masked tokens. The denoising pressure pushes the model to separate genuine speech structure from acoustic clutter, a capability that carries over directly to deepfake detection, where synthesis artefacts have to be distinguished from environmental noise. Two variants exist: Base-Plus (12 Transformer layers, 94M parameters, 94,000 hours of pre-training data) and Large (24 Transformer layers, 317M parameters). WavLM-Large achieves state-of-the-art performance on the SUPERB benchmark across every speech understanding task, and it is the primary backbone used in GLADA.

## 2.4 Domain Adaptation

### 2.4.1 Domain-Adversarial Neural Networks (DANN)

Domain-Adversarial Neural Networks (DANN) [10] cast unsupervised domain adaptation as a minimax game. The feature extractor has to produce representations

that solve the main task (in our case, deepfake detection) while remaining invariant to the input domain. An adversarial domain discriminator sits on top of the representation and tries to identify whether a feature originated from the source or target domain; the extractor is penalised whenever the discriminator succeeds.

What makes this optimisable in a single backward pass is the Gradient Reversal Layer (GRL) [10]. The GRL acts as the identity in the forward pass; in backpropagation it negates the gradient from the domain discriminator and scales it by $\alpha$:

$$\mathrm{GRL}_\alpha(x) = x \quad \text{(forward)}, \qquad \left.\frac{\partial \mathcal{L}}{\partial x}\right|_{\mathrm{GRL}} = -\alpha \cdot \frac{\partial \mathcal{L}}{\partial \mathrm{GRL}_\alpha(x)} \quad \text{(backward)} \tag{1}$$

The negated gradient forces the feature extractor to maximise the discriminator's loss, which pushes the source and target feature distributions toward alignment. The reversal strength $\alpha$ is typically warmed up linearly from 0 to 1 over the first few epochs to keep early training stable. The same GRL mechanism applies to any adversarial alignment objective, including speaker disentanglement.

### 2.4.2 Distributional Matching Methods

Non-adversarial domain adaptation approaches such as Maximum Mean Discrepancy (MMD) [83] and Correlation Alignment (CORAL) [84] directly minimise statistical divergences between source and target feature distributions using fixed kernel functions. They sidestep the instability of adversarial training, but they impose fixed functional forms on the alignment objective and cannot adapt to complex, multimodal distribution mismatches. Adversarial alignment via the GRL can learn task-specific notions of invariance that fixed-kernel statistics cannot capture, which is why GLADA uses the adversarial approach.

## 2.5 The Information Bottleneck Principle

### 2.5.1 Information Bottleneck Objective

The Information Bottleneck (IB) principle, introduced by Tishby et al. [85], gives an information-theoretic framework for representation learning. Given an input variable $X$ and a target variable $Y$, the IB seeks a compressed stochastic representation $Z$ that retains as much information about $Y$ as possible while discarding irrelevant information from $X$. The corresponding objective is:

$$\mathcal{L}_{\text{IB}} = -I(Z;Y) + \beta \cdot I(Z;X) \tag{2}$$

$I(\cdot;\cdot)$ is mutual information; $\beta > 0$ trades prediction accuracy (large $I(Z;Y)$) for compression (small $I(Z;X)$).

### 2.5.2 Deep Variational Information Bottleneck (VIB)

Computing the mutual information terms exactly is intractable for high-dimensional data and deep neural networks. Alemi et al. [86] proposed the Deep Variational Information Bottleneck (VIB), which approximates the IB objective with variational inference. A parametric encoder $q_\phi(z|x)$ approximates the true posterior, and a fixed marginal prior $p(z)$ produces a variational upper bound on $I(Z;X)$:

$$I(Z;X) \leq \mathbb{E}_{x\sim p(x)}\left[\text{KL}\left(q_\phi(z|x) \parallel p(z)\right)\right] \tag{3}$$

With the encoder parameterized as a multivariate Gaussian with diagonal covariance, $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag}(\sigma^2_\phi(x)))$, and a standard normal prior $p(z) = \mathcal{N}(0, I)$, the Kullback-Leibler (KL) divergence has a closed-form solution [87]:

$$\text{KL}(q\|p) = -\frac{1}{2}\sum_{j=1}^{d_z}\left[1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right] \tag{4}$$

VIB uses the reparameterisation trick [87] to make the stochastic sampling step differentiable, writing the sampled representation as $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$ and $\odot$ element-wise multiplication.

### 2.5.3 Information Bottleneck for Domain Adaptation

For domain adaptation the information bottleneck has a dual purpose. Domain-specific attributes (recording environment, microphone characteristics, codec artefacts) tend to be high-dimensional nuisances. A compression penalty ($\beta > 0$) discards them disproportionately, leaving the lower-dimensional task-relevant features intact. Huang et al. [9] showed empirically that the information bottleneck stabilises adversarial domain adaptation for audio deepfake detection. Without it, the adversarial min-max game produces destructive gradients that interfere with classification.

## 2.6 Multi-Layer Domain Adaptation

### 2.6.1 MLADA (Fang et al., CVPR 2021)

Standard domain adaptation aligns source and target distributions exclusively at the final layer of a neural network. Fang et al. [11] demonstrated in computer vision that this global approach is suboptimal. Their Multi-Layer Adversarial Domain Adaptation (MLADA) attaches independent domain discriminators to multiple intermediate stages of a Convolutional Neural Network (CNN). Domain shift manifests differently at different levels of abstraction (low-level textures versus high-level semantic shapes), and applying adversarial alignment progressively across multiple layers outperformed single-layer global adaptation by a wide margin, establishing a precedent for level-specific alignment.

### 2.6.2 Stratified Adversarial DA in NLP

The closest analogue to per-layer adversarial alignment in NLP is the multi-domain adversarial network of Ding et al. [14] for text classification. It separates private and shared features at each layer of the encoder using orthogonality constraints, so that every layer carries its own domain-specific and domain-invariant subspaces. The discriminator structure is layer-stratified rather than fully replicated per layer, but the per-layer message of MLADA carries through. Other NLP work extends adversarial

DA along the procedural axis instead of the placement axis: MSP-DA [15] builds on DANN by using meta-learning to schedule a self-paced training procedure end-to-end and to tune the domain-sensitive coefficients of the adversarial loss automatically for cross-domain text classification. Stratified or scheduled extensions of DANN consistently improve over a single static final-layer discriminator in NLP, but the strongest empirical evidence for per-discriminator alignment at multiple depths still comes from the vision literature.

### 2.6.3 Layer-Wise SSL Analysis for Deepfake Detection

What makes multi-layer domain adaptation interesting for audio deepfake detection comes from recent layer-wise studies of SSL speech models. El-Kheir et al. [8] performed a layer-by-layer analysis on WavLM and XLSR and showed that lower layers predominantly encode pitch, formants, and spectral shape (all sensitive to recording conditions), while higher layers encode more abstract linguistic and synthesis-related patterns. The domain shift induced by background noise is qualitatively different from the shift induced by an unseen generator. A single global discriminator has to address both at once; a per-group discriminator does not, which is the architectural opening GLADA targets.

## 2.7 Attention-Based Layer Aggregation

### 2.7.1 Attentive Statistics Pooling (ASP)

Okabe et al. [88] introduced Attentive Statistics Pooling (ASP) for speaker verification, collapsing a frame-level sequence into a single utterance-level vector. Mean pooling throws away the information about spread; ASP replaces it with an attention-weighted mean *and* standard deviation over time:

$$e_t = v^\top \tanh(W x_t + b), \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{5}$$

$$\mu = \sum_{t=1}^{T} \alpha_t \cdot x_t, \qquad \sigma = \sqrt{\sum_{t=1}^{T} \alpha_t \cdot (x_t - \mu)^2 + \epsilon} \tag{6}$$

The output is the concatenation $[\mu; \sigma]$. ASP is a useful fit for deepfake detection because vocoder outputs tend to exhibit anomalously smooth temporal variance, and the $\sigma$ channel makes that pattern visible.

### 2.7.2 Attentive Layer Merging

SSL speech models expose hidden states at every Transformer layer. Pan et al. [47] showed that taking only the final layer throws away artefact cues that sit in the intermediate representations. Their attentive merging learns a softmax-weighted sum over all layers, letting the network shift focus toward whichever depth is most informative for the detection task.

### 2.7.3 HierCon: Grouped Layer Attention

Liang et al. [7] extended attentive merging with HierCon, which organises contiguous SSL layers into blocks. Each block has its own attention mechanism, targeted at a specific abstraction level (acoustic, phonetic, semantic), and the per-block outputs feed a hierarchical contrastive objective. Their central claim is that SSL layers behave more like semantically distinct groups than like a flat stack, and pooling that respects this grouping outperforms pooling that does not.

## 2.8 Representation Similarity

### 2.8.1 Centered Kernel Alignment (CKA)

Once SSL layers are processed in groups, nothing stops the groups from collapsing into redundant copies of one another without an explicit penalty. Kornblith et al. [89] introduced Centred Kernel Alignment (CKA) for the closely related problem of comparing neural representations with a metric stable under the scalings and rotations that ordinarily distort cosine similarity. Given two centred feature matrices $X$ and $Y$, linear CKA is computed through the Hilbert-Schmidt Independence Criterion (HSIC) [83]:

$$\mathrm{HSIC}(X, Y) = \frac{1}{B^2}\mathrm{tr}(XX^\top YY^\top) \tag{7}$$

$$\mathrm{CKA}(X, Y) = \frac{\mathrm{HSIC}(X, Y)}{\sqrt{\mathrm{HSIC}(X, X) \cdot \mathrm{HSIC}(Y, Y)}} \tag{8}$$

CKA returns a value between 0 (completely dissimilar) and 1 (identical up to linear transformation). It is invariant under orthogonal transformations and isotropic scaling, which makes it an appropriate similarity measure across parallel network branches.

### 2.8.2 CKA as Diversity Loss

Tran et al. [19] adapted CKA into a diversity loss for audio deepfake detection. Penalising high CKA between parallel multi-scale feature branches forced each branch to specialise on a different set of spoofing cues. Applied to grouped SSL features, the same penalty pushes each layer group toward a distinct role: acoustic versus phonetic features, for instance.

## 2.9 Data Augmentation for Robustness

A persistent challenge for audio deepfake detection is the acoustic mismatch between pristine, studio-recorded training audio and the noisier audio that shows up at evaluation. Augmenting source-domain audio with noise, codec effects, and reverberation is a direct, low-cost way to narrow that gap.

### 2.9.1 RawBoost

RawBoost [55] operates directly on the raw waveform and simulates the transmission and encoding artefacts that real-world audio typically accumulates. Three interference types are mixed in stochastically: linear and non-linear convolutive noise (room acoustics and microphone frequency response), impulsive signal-dependent additive noise (codec artefacts and transmission clicks), and stationary signal-independent additive noise (constant hum or white noise). Training against these perturbations forces the network to separate synthesis artefacts from channel effects.

### 2.9.2 MUSAN and RIR

MUSAN additive noise [90] and RIR convolution [91] cover the complementary axis that RawBoost does not: the environmental variability absent from studio recordings. Combining all three augmentation strategies yields substantial improvements on In-the-Wild evaluation [9].

## 2.10 Layer-Wise Learning Rate Decay

Fine-tuning large pre-trained models requires careful optimization to prevent catastrophic forgetting. Clark et al. [92] introduced Layer-Wise Learning Rate Decay (LLRD), in which the learning rate decays exponentially from the top layers down to the bottom:

$$\eta_i = \eta_{\text{base}} \times \rho^{(L-i)} \tag{9}$$

where $\eta_i$ is the learning rate for layer $i$, $L$ is the total number of layers, $\eta_{\text{base}}$ is the learning rate for the topmost layer, and $\rho \in (0, 1)$ is the decay factor. Lower layers receive smaller updates to preserve general acoustic features, while higher layers adapt more rapidly to the classification task.

## 2.11 Domain Adaptation for Audio Deepfake Detection

The strongest published results in cross-domain audio deepfake detection come from SSL representations combined with domain adaptation. Three representative systems mark out the current performance range.

Huang et al. [9] proposed IB-CAAN, which applies a Variational Information Bottleneck to the final-layer representation of an XLSR backbone and augments it with confidence-aware adversarial alignment between source and target. The VIB compresses task-irrelevant acoustic nuisances before the adversarial objective is applied, stabilising the min-max training dynamic. IB-CAAN further combines RawBoost, MUSAN, and RIR augmentation during training. The full system reaches 4.93% EER on the ITW benchmark, the strongest single-model result among published methods

at the time of writing.

Serrano et al. [17] took an ensemble route. Their system fuses score outputs from four independently trained models (BEATs, WavLM-Large, MMS, and XLSR) through calibrated score-level fusion. Each model captures different aspects of synthesis artefacts because the four pre-training objectives and data are different. The ensemble reaches 6.4% ITW EER, a useful reference point: no single backbone captures every relevant artefact type, and architectural diversity is an effective complementary strategy.

Liang et al. [7] proposed HierCon, which puts SSL hidden layers into contiguous groups and runs hierarchical contrastive learning over the group representations. By treating different layer ranges as capturing different levels of abstraction (acoustic, phonetic, semantic), HierCon forces the model to use artefact signals at multiple representational depths. HierCon reaches 6.87% ITW EER and, importantly, demonstrates that layer-stratified processing of SSL hidden states provides a consistent advantage over single-level aggregation.

GLADA takes a different route. Instead of a single global adversarial discriminator at the final layer, GLADA attaches an independent domain discriminator to each SSL layer group through a gradient reversal layer. The per-group alignment addresses the finding of El-Kheir et al. [8] that lower layers require acoustic-level domain alignment while higher layers require synthesis-level alignment, two objectives a global discriminator has to conflate. GLADA also incorporates a VIB bottleneck (following IB-CAAN [9]) and a CKA diversity penalty to keep group representations from collapsing to redundant features. To my knowledge, GLADA is the first framework to combine per-group adversarial domain adaptation with hierarchical SSL layer aggregation for audio deepfake detection, porting the multi-layer adversarial alignment principle from MLADA [11] in vision and the stratified adversarial setting of multi-domain text

classification [14] in NLP into the speech domain.

# CHAPTER 3

# GLADA Methodology

GLADA (Grouped Layer Adversarial Domain Adaptation) is built around three coordinated mechanisms: grouped attention over SSL hidden states, per-group adversarial alignment with the target domain, and a per-group Variational Information Bottleneck co-located with each discriminator. A speaker adversary and a CKA diversity penalty enter as auxiliary objectives. The development below begins with the underlying domain-adaptation problem and works outward to the full loss and the two-phase training schedule.

## 3.1 Problem Formulation

Cross-domain audio deepfake detection fits naturally into the unsupervised domain adaptation (UDA) framework [10]. Let $\mathcal{D}_s = \{(x_i^s, y_i^s, s_i^s)\}_{i=1}^{n_s}$ denote the *source domain*: labelled audio recorded under controlled studio conditions. Each source sample carries a task label $y_i^s \in \{0, 1\}$ (bonafide vs. spoofed), a speaker identity $s_i^s$, and an implicit domain label $d_i^s = 0$. The *target domain* $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$ is a collection of real-world audio samples with only the domain label $d_j^t = 1$; no task labels are available during training. Source and target share a label space but have different marginal distributions, $P_s(X) \neq P_t(X)$.

In my setup the source is the ASVspoof 2019 LA training set [80]: studio-recorded audio from 20 speakers with six known TTS and VC attack algorithms. The target is the In-the-Wild dataset [6], which collects audio from many real-world sources under unknown synthesis methods and variable acoustic conditions.

The goal is to learn a feature extractor $F : \mathcal{X} \to \mathcal{Z}$ and a task classifier $C : \mathcal{Z} \to \{0, 1\}$ such that $C(F(x))$ correctly classifies samples from either domain. The representation produced by $F$ has to satisfy several constraints at once. It has to be domain-invariant, $P_s(F(x)) \approx P_t(F(x))$, so the classifier cannot rely on domain-

specific shortcuts. It has to stay discriminative enough that the bonafide/spoofed distinction survives the invariance constraint. And because only 20 speakers appear in the source domain, the representation also has to be speaker-invariant: encoding those speaker identities would overfit the model to a sample that is not representative of real-world speaker diversity.

Standard UDA performs a single global alignment between source and target feature distributions. GLADA aligns each SSL layer group separately instead. The reason for the finer-grained adaptation is empirical: different SSL layers encode fundamentally different kinds of deepfake-relevant information [8].

### 3.2 Architecture Overview

Figure 1 illustrates the GLADA architecture. The pipeline processes raw audio through eight stages:

1. **Input:** Raw audio waveform $x \in \mathbb{R}^{64{,}000}$ (16 kHz sampling rate, 4 seconds).

2. **Augmentation:** RawBoost [55] is applied stochastically during training. The codebase also supports MUSAN noise [90] and room-impulse-response convolution [91]; only RawBoost was active for the experiments reported here.

3. **SSL backbone:** A self-supervised learning model (WavLM [67]) extracts all hidden states $\mathbf{H} = [h_0, h_1, \ldots, h_L]$, where each $h_i \in \mathbb{R}^{T \times D}$ and $L$ is the number of Transformer layers.

4. **Per-layer normalization:** Each hidden state is independently normalized: $h'_i = \text{LayerNorm}_i(h_i)$.

5. **Hierarchical attention pooling:** The $L$+1 normalized hidden states are divided into $K$ groups. Within each group, a learnable weighted sum followed

by Attentive Statistics Pooling (ASP) [88] produces a group representation $g_k \in \mathbb{R}^{2D}$.

6. **Projection and VIB:** The $K$ group representations are concatenated and projected to a unified representation, which is then passed through a Variational Information Bottleneck [86]: $z \sim \mathcal{N}(\mu(\text{pooled}), \sigma^2(\text{pooled}))$.

7. **Task classifier:** A multilayer perceptron (MLP) predicts bonafide versus spoofed from the bottleneck representation $z$.

8. **Adversarial heads:** Two adversarial components operate via gradient reversal [10]:

   - *$K$ per-group domain discriminators*, each operating on a group representation $g_k$ to encourage per-group domain invariance.
   - A *speaker adversary* operating on the bottleneck $z$ to encourage speaker-invariant features.

The per-group domain discriminators (step 8) are the primary contribution of GLADA. Each individual component draws on prior work (ASP from Okabe et al. [88], VIB from Alemi et al. [86], GRL from Ganin et al. [10], and layer grouping from HierCon [7]), but to my knowledge, GLADA is the first framework to apply per-group adversarial domain adaptation to self-supervised speech representations for deepfake detection.

## 3.3 SSL Backbone and Per-Layer Normalization

### 3.3.1 Backbone Selection

WavLM [67] is used as the primary SSL backbone. Its denoising pre-training objective (Section 2.3) separates synthesis artifacts from acoustic noise, a capability that maps directly onto deepfake detection.

Two WavLM variants are evaluated; their characteristics are in Table 1.

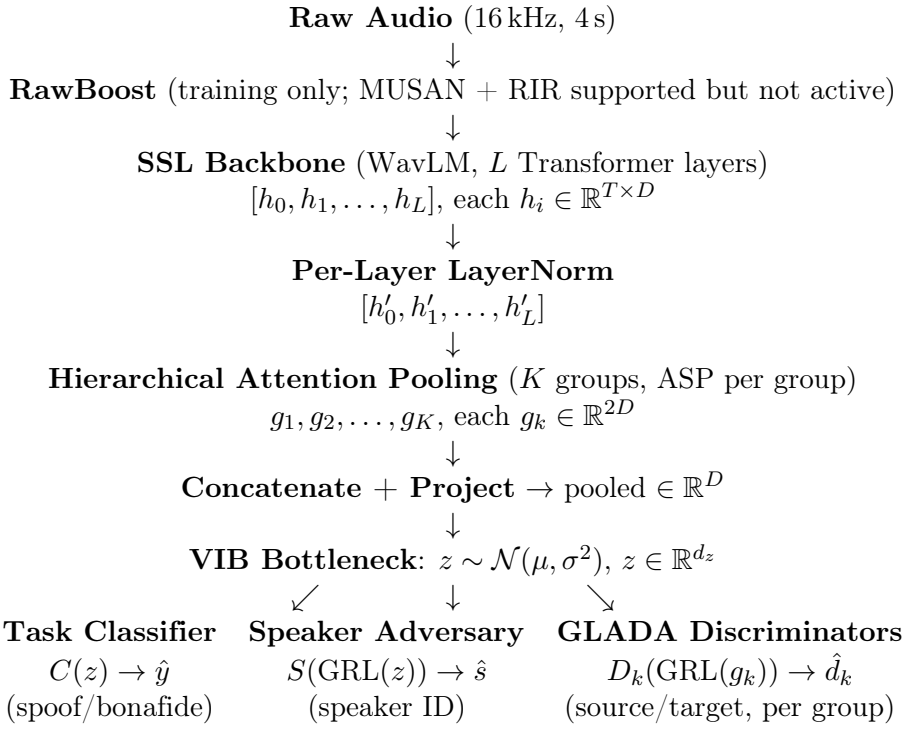



Figure 1: GLADA architecture overview. Raw audio is processed through an SSL backbone that extracts all hidden states. Hierarchical attention pooling groups the layers and produces per-group representations $g_k$, which feed into per-group domain discriminators via gradient reversal. The concatenated groups are projected and compressed through a VIB bottleneck before the task classifier and speaker adversary. Solid arrows indicate the forward pass; gradient reversal layers (GRL) negate gradients during backpropagation for the adversarial heads.

Table 1: SSL backbones evaluated in this work. WavLM-Base-Plus is used for ablation studies due to faster training; WavLM-Large is the primary backbone.

| **Backbone** | **Layers** | **Dim ($D$)** | **Params** | **Groups ($K$)** |
|---|---|---|---|---|
| WavLM-Base-Plus | 12 | 768 | 94M | 3 |
| WavLM-Large | 24 | 1024 | 317M | 4 |

### 3.3.2 Multi-Layer Hidden State Extraction

Unlike prior work that uses only the last hidden state for classification [93], GLADA extracts *all* hidden states from the SSL backbone. For a backbone with

$L$ Transformer layers this yields $L$+1 hidden states: the CNN feature extractor output $h_0$ plus the outputs of each Transformer layer $h_1, \ldots, h_L$. Each hidden state $h_i \in \mathbb{R}^{T \times D}$, where $T$ is the number of time frames and $D$ is the model dimension. For WavLM-Large that produces 25 hidden states (1 CNN + 24 Transformer), each with $D = 1024$.

Without all hidden states the grouped-layer architecture has nothing to group: each group needs a contiguous span of backbone depths so its attention head, bottleneck, and discriminator can act on features specific to that depth range.

### 3.3.3 Per-Layer Normalization

El-Kheir et al. [8] demonstrated that SSL hidden states differ dramatically in activation magnitude across layers, by as much as 10--100×. Without normalisation, higher-magnitude layers dominate the gradient signal during backpropagation, and the model never learns useful attention weights for the lower-magnitude layers.

The fix is to apply standard LayerNorm [94] independently to each hidden state before grouping: $h'_i = \text{LayerNorm}_i(h_i)$. Each of the $L$+1 layers has its own learnable affine parameters $\gamma_i, \beta_i \in \mathbb{R}^D$, so the model can learn an appropriate scaling for each layer's activation distribution.

The CNN feature extractor that produces $h_0$ stays frozen throughout training. It functions as a fixed acoustic frontend that converts raw waveforms into latent representations.

## 3.4 Hierarchical Grouped-Layer Attention Pooling

The hierarchical attention pooling module converts the $L$+1 frame-level hidden states into $K$ utterance-level group representations. This involves four steps: layer grouping, per-group attentive weighted sum, attentive statistics pooling, and group projection.

### 3.4.1 Layer Grouping

The $L+1$ hidden states are divided into $K$ contiguous groups of approximately equal size. For WavLM-Large ($L = 24$, yielding 25 hidden states with $K = 4$ groups), the assignments are:

- **Group 0** (layers 0--5): Acoustic features (pitch, formants, spectral shape)
- **Group 1** (layers 6--11): Phonetic and phonological features
- **Group 2** (layers 12--17): Word-level and semantic features
- **Group 3** (layers 18--23): Task-specific patterns, including synthesis artifact signatures

Two lines of work motivate this grouping. El-Kheir et al. [8] (Section 2.6) showed that different SSL layers encode different kinds of deepfake-relevant information, ranging from acoustic features in lower layers to abstract synthesis patterns in higher layers. Liang et al. [7] demonstrated the effectiveness of grouping SSL layers for hierarchical contrastive learning in audio deepfake detection. Since different layer groups encode different types of information, they may also exhibit different types of domain shift, which is what motivates per-group domain adaptation in the first place rather than a single global alignment.

### 3.4.2 Per-Group Attentive Weighted Sum

Within each group $k$, the model learns which layers are most informative through a set of learnable layer weights $\theta_k \in \mathbb{R}^{|G_k|}$, where $|G_k|$ is the number of layers in group $k$. After softmax normalisation, the group's frame-level representation is a weighted sum:

$$w_k = \text{softmax}(\theta_k), \qquad \tilde{h}_k = \sum_{i \in G_k} w_{k,i} \cdot h'_i \quad \in \mathbb{R}^{T \times D} \tag{10}$$

The model can therefore focus on the most relevant layers within each group automatically. Weights are initialised uniformly, placing no prior bias on any particular layer.

### 3.4.3 Attentive Statistics Pooling

Each group's frame-level representation $\tilde{h}_k \in \mathbb{R}^{T \times D}$ has to be collapsed to a fixed-dimensional utterance-level vector. I use Attentive Statistics Pooling (ASP) [88], which computes an attention-weighted mean and standard deviation along the time dimension.

The attention mechanism assigns importance weights to each frame:

$$e_t = v^\top \tanh(W\tilde{h}_{k,t} + b), \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{11}$$

where $W \in \mathbb{R}^{D \times D}$, $b \in \mathbb{R}^D$, and $v \in \mathbb{R}^D$ are learnable parameters. The attention-weighted mean and standard deviation are then:

$$\mu_k = \sum_{t=1}^{T} \alpha_t \cdot \tilde{h}_{k,t} \quad \in \mathbb{R}^D \tag{12}$$

$$\sigma_k = \sqrt{\sum_{t=1}^{T} \alpha_t \cdot (\tilde{h}_{k,t} - \mu_k)^2 + \epsilon} \quad \in \mathbb{R}^D \tag{13}$$

The group representation is the concatenation of the weighted mean and standard deviation:

$$g_k = [\mu_k; \sigma_k] \quad \in \mathbb{R}^{2D} \tag{14}$$

ASP is used instead of simple mean pooling because it captures both the centre and the spread of the frame-level distribution. Synthesis artefacts frequently appear as unusual variance patterns in SSL features; an overly smooth mel-spectrogram from a neural vocoder, for instance, produces lower variance than natural speech in certain layers, and the $\sigma$ channel exposes that. The attention mechanism also lets the model concentrate on the most informative frames within an utterance.

### 3.4.4 CKA Diversity Loss

Without explicit regularisation the $K$ groups can converge on similar representations, defeating the purpose of grouping. To keep the groups distinct, I use a diversity

loss built on Centred Kernel Alignment (CKA) [89], following Tran et al. [19], who applied CKA as a diversity penalty for audio deepfake detection.

Given centred group representations $G_i, G_j \in \mathbb{R}^{B\times 2D}$ (a batch of group outputs), linear CKA is computed through the Hilbert-Schmidt Independence Criterion (HSIC) [83]:

$$\mathrm{HSIC}(G_i, G_j) = \frac{1}{B^2}\mathrm{tr}(G_i G_i^\top G_j G_j^\top) \tag{15}$$

$$\mathrm{CKA}(G_i, G_j) = \frac{\mathrm{HSIC}(G_i, G_j)}{\sqrt{\mathrm{HSIC}(G_i, G_i)\cdot \mathrm{HSIC}(G_j, G_j)}} \tag{16}$$

CKA equals 1 when the representations are identical up to linear transformation, and approaches 0 when they are completely dissimilar. The diversity loss averages CKA across all pairs of groups:

$$\mathcal{L}_{\mathrm{CKA}} = \frac{1}{\binom{K}{2}} \sum_{i<j} \mathrm{CKA}(G_i, G_j) \tag{17}$$

Minimising this loss pushes each group toward a different aspect of the input, so the per-group domain discriminators operate on meaningfully distinct representations rather than near-duplicates.

### 3.4.5 Group Projection

The $K$ group representations are concatenated, then linearly projected to a unified representation:

$$\mathrm{pooled} = W_{\mathrm{proj}} \cdot [g_1; g_2; \ldots; g_K] + b_{\mathrm{proj}} \quad \in \mathbb{R}^D \tag{18}$$

$W_{\mathrm{proj}} \in \mathbb{R}^{D\times 2KD}$ and $b_{\mathrm{proj}} \in \mathbb{R}^D$. The projection reduces the concatenated $2KD$-dimensional vector back to $D$ dimensions, so the VIB bottleneck receives a single compact representation that integrates information across all $K$ layer groups.

## 3.5 Variational Information Bottleneck

The Variational Information Bottleneck (VIB) compresses the pooled representation, keeping the task-relevant information and discarding domain-specific noise.

The underlying Information Bottleneck (IB) framework [85] seeks a compressed representation $Z$ of an input $X$ that preserves as much information about the target $Y$ as possible:

$$\min_{Z} \quad -I(Z;Y) + \beta \cdot I(Z;X) \tag{19}$$

$I(\cdot;\cdot)$ is mutual information; $\beta > 0$ trades prediction (large $I(Z;Y)$) for compression (small $I(Z;X)$). The useful observation for domain adaptation is that domain-specific information (recording environment, microphone characteristics, compression artefacts) is high-dimensional and distributed, whereas the task-relevant signal (genuine vs. synthesised speech) lives in a much lower-dimensional subspace. Compression therefore discards the domain nuisances disproportionately.

Computing $I(Z;X)$ exactly is intractable for neural networks, so I follow Alemi et al. [86] and optimise a variational upper bound using a diagonal Gaussian encoder $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma^2_\phi(x)))$ with a standard normal prior $p(z) = \mathcal{N}(0, I)$, trained through the reparameterisation trick [87]. At inference time the deterministic encoding $z = \mu_\phi(x)$ is used for reproducible predictions. For this choice of encoder and prior, the KL divergence has a closed-form solution [87]:

$$\mathrm{KL}(q\|p) = -\frac{1}{2}\sum_{j=1}^{d_z}\left[1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right] \tag{20}$$

where $d_z$ is the bottleneck dimensionality.

### 3.5.1 Implementation Details

The VIB is implemented as a two-headed MLP. Both heads share the same architecture:

$$\mu_\phi(x) = W_{\mu_2} \cdot \mathrm{GELU}(W_{\mu_1} \cdot x + b_{\mu_1}) + b_{\mu_2} \tag{21}$$

$$\log\sigma^2_\phi(x) = W_{\sigma_2} \cdot \mathrm{GELU}(W_{\sigma_1} \cdot x + b_{\sigma_1}) + b_{\sigma_2} \tag{22}$$

$W_{\mu_1}, W_{\sigma_1} \in \mathbb{R}^{D\times D}$ and $W_{\mu_2}, W_{\sigma_2} \in \mathbb{R}^{d_z\times D}$, with bottleneck dimensionality $d_z = 256$. Two numerical safeguards prevent training instability:

1. **Log-variance clamping.** The output of the log-variance head is clamped to $[-20, 20]$, bounding the standard deviation to approximately $[4.5\times10^{-5},\ 2.2\times10^{4}]$ and preventing numerical overflow in the exponential.

2. **KL clamping.** The per-batch KL loss is clamped at a maximum of 10.0 before being multiplied by $\beta$. The clamp keeps the KL term from dominating the total loss during Phase 2 (Section 3.10), when unfreezing the backbone can cause sudden changes in the representation distribution.

### 3.5.2 Role in GLADA

The VIB plays a specific architectural role in GLADA that differs from prior work. In IB-CAAN [9] the information bottleneck operates on the same representation that feeds the domain discriminator. In GLADA the VIB compresses the *pooled* representation (after group projection) before the task classifier and speaker adversary, while the per-group domain discriminators operate on the *pre-VIB* group representations $g_k$. The separation has three consequences. The task classifier receives a compressed, domain-noise-reduced representation through $z$. The domain discriminators receive richer, uncompressed group-level features, which enables more effective per-group alignment. And the VIB shields the classification pathway from adversarial gradient damage, consistent with the finding by Huang et al. [9] that the information bottleneck is the critical component for stable adversarial domain adaptation.

## 3.6 Per-Group Variational Information Bottleneck

The standard GLADA architecture in Sections 3.5 and 3.7 places a single, monolithic VIB after the group projection step, while the per-group domain discriminators operate on the *pre-VIB* group representations $g_k$. The bottleneck and the discriminators therefore sit on separate gradient paths: the VIB compresses the *pooled* representation, but the domain discriminators receive raw, uncompressed group features. This

is the architectural design behind the primary GLADA results in Chapter 5.

A natural follow-up question is whether co-locating compression and adversarial alignment (applying one VIB *per group*, so each discriminator receives a compressed $z_k$ rather than the raw $g_k$) improves domain invariance. This architectural variant, **Per-Group VIB** (`PerGroupVIBModel`), is explored in the ablation study of Section 5.2.

### 3.6.1 Motivation

In the original GLADA forward pass the data flow for group $k$ is:

$$z = \text{VIB}(\text{proj}([g_1; \ldots; g_K])), \qquad \hat{d}_k = D_k(\text{GRL}(g_k)) \tag{23}$$

The bottleneck acts on the *merged* representation; the discriminator sees the raw group output. Because $g_k$ is never passed through a compression module before reaching the discriminator, domain-identifying information in $g_k$ goes unpenalised by the IB objective.

In the per-group variant the data flow becomes:

$$z_k = \text{VIB}_k(g_k), \qquad \hat{d}_k = D_k(\text{GRL}(z_k)), \qquad z = W_{\text{proj}}\,[z_1; \ldots; z_K] \tag{24}$$

Each group is independently compressed to a bottleneck representation $z_k \in \mathbb{R}^{d_z}$ before its domain discriminator sees it. The $K$ compressed representations are then concatenated and linearly projected to produce the final representation $z$ that feeds the task classifier and speaker adversary.

Co-location like this puts the IB compression pressure and the adversarial alignment pressure on the *same* representation at each level of the hierarchy, rather than on two different representations connected only through the global projection.

### 3.6.2 Architecture

The per-group VIB model is implemented as `PerGroupVIBModel`, a subclass of the base `GLADAModel`. It replaces the single `VIBBottleneck` and the global projection with $K$ independent VIB modules plus a projection layer:

- $K$ `VIBBottleneck` modules: `group_vibs[`$k$`]`: $\mathbb{R}^{2D} \to \mathbb{R}^{d_z}$, one per group.
- A linear projection `group_compressed_proj`: $\mathbb{R}^{K \cdot d_z} \to \mathbb{R}^{d_z}$, which merges the $K$ compressed representations into a single bottleneck $z$.
- $K$ per-group discriminators `glada_disc`: each now operates on $z_k \in \mathbb{R}^{d_z}$ (bottleneck dimension) rather than $g_k \in \mathbb{R}^{2D}$ (ASP dimension).

All other components (the SSL backbone, per-layer normalization, hierarchical attention pooling, task classifier, and speaker adversary) remain identical to the base model.

### 3.6.3 KL Aggregation

Each VIB module $k$ contributes its own KL divergence $\mathcal{L}_{\mathrm{KL},k} = \mathrm{KL}(q_{\phi_k}(z_k|g_k) \,\|\, \mathcal{N}(0, I))$. These are aggregated into a single scalar using a uniform mean:

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{KL}\Big(q_{\phi_k}(z_k|g_k) \,\Big\|\, \mathcal{N}(0, I)\Big) \tag{25}$$

The closed-form KL for each group uses the same diagonal Gaussian formula as Equation (20), applied independently to each $q_{\phi_k}$. The scalar $\mathcal{L}_{\mathrm{KL}}$ then enters the combined loss (Equation (31)) with the same weight $\beta = 10^{-3}$. Because the mean is taken over $K$ groups, the effective compression pressure per group is $\beta/K$, weaker than in the monolithic case. Empirically that prevents KL collapse in the groups whose information the model finds less useful.

### 3.6.4 Leave-One-Out Ablation and `vib_disabled_groups`

To identify which group's VIB contributes most to domain invariance, the `PerGroupVIBModel` supports a `vib_disabled_groups` flag. When group $k$ is listed in this set, the model disables stochastic sampling and KL pressure for that group: $z_k = \mu_{\phi_k}(g_k)$ (the mean of the posterior, without the reparameterisation noise) and $\mathcal{L}_{\mathrm{KL},k} = 0$. All other groups continue to operate normally.

The flag enables the leave-one-out experiment reported in Section 5.2: by disabling

one group's VIB at a time, the experiment isolates which layer group benefits most from the bottleneck's compression. A large EER increase when group $k$ is disabled means the bottleneck on that group is doing meaningful work; a small increase means the group's compression is largely redundant. The `g0-only` configuration sets `vib_disabled_groups=(1, 2)` (for a 3-group backbone) or `(1, 2, 3)` (for a 4-group backbone), keeping only the acoustic group's VIB active.

### 3.6.5 Relationship to Prior Work

The per-group VIB is an architectural variant of GLADA, not a standalone novel contribution. Each component has prior precedent: VIB applied to domain adaptation appears in IB-CAAN [9] for speech, and applying separate information compression at each stage of a hierarchical network is a natural extension of the IB framework of Tishby et al. [85]. The specific combination (one VIB per SSL layer group, feeding per-group domain discriminators, with leave-one-out disabling support) is an architectural investigation motivated by GLADA's own grouped structure, not a claim of a fundamentally new method.

## 3.7 Grouped-Layer Adversarial Domain Adaptation

### 3.7.1 Motivation

Standard domain-adversarial neural networks [10] apply a single domain discriminator to the final feature representation, performing global alignment of source and target distributions. That treats the feature space as homogeneous, with every feature benefiting equally from the same alignment signal. SSL backbones, however, produce a *hierarchy* of representations, and different layers encode fundamentally different types of information.

As El-Kheir et al. [8] showed (Section 2.6), different SSL layers encode different types of deepfake-relevant information. Crucially, these different types of information may also exhibit different types of domain shift. Acoustic features in lower layers

shift due to microphone quality, background noise, and compression, factors that differ between studio-recorded ASVspoof data and real-world In-the-Wild audio. Task-specific features in higher layers shift due to the use of unknown synthesis methods in deployment.

A single global discriminator operating on the pooled representation cannot resolve these distinct per-level domain shifts. It has to find a single alignment that addresses acoustic-level mismatch and task-level mismatch at once, potentially over-aligning one while under-aligning the other.

Domain adaptation at multiple levels of a neural network has been explored in other modalities. Fang et al. [11] introduced Multi-Layer Adversarial Domain Adaptation (MLADA) for convolutional neural networks in computer vision, attaching domain discriminators at multiple CNN stages and demonstrating that per-layer adaptation outperforms single-layer global adaptation. The closest NLP analogue is the multi-domain adversarial network of Ding et al. [14] for text classification, which separates private and shared features per encoder layer. Other NLP work has refined the adversarial training procedure on top of DANN: MSP-DA [15] introduces meta self-paced learning that schedules training and tunes domain-loss coefficients automatically. To my knowledge, however, the combination of per-group adversarial domain adaptation, per-group variational information bottlenecks, and grouped hierarchical SSL aggregation has not been investigated for audio anti-spoofing. Component-wise precedents exist in vision (MLADA) and NLP (Ding et al. multi-domain adversarial networks).

### 3.7.2 Gradient Reversal Layer

Adversarial alignment in GLADA is implemented through the Gradient Reversal Layer (GRL) [10]: an identity function during the forward pass that negates the

gradient (scaled by a factor $\alpha$) during backpropagation:

$$\mathrm{GRL}_\alpha(x) = x \quad \text{(forward)}, \qquad \left.\frac{\partial \mathcal{L}}{\partial x}\right|_{\mathrm{GRL}} = -\alpha \cdot \frac{\partial \mathcal{L}}{\partial \mathrm{GRL}_\alpha(x)} \quad \text{(backward)} \tag{26}$$

The result is a minimax game between the domain discriminator and the feature extractor. The discriminator minimises the domain classification loss, learning to distinguish source from target samples. The feature extractor, receiving the negated gradient, is pushed to produce representations that *maximise* the discriminator's error, i.e., representations that are domain-invariant. At equilibrium the feature extractor produces representations from which the domain of origin cannot be determined.

The reversal strength $\alpha$ is warmed up linearly from 0 to 1 over the first $E_w$ epochs:

$$\alpha(t) = \min\left(1,\ \frac{t}{E_w}\right) \tag{27}$$

$t$ is the current epoch and $E_w = 5$. The warmup keeps the adversarial gradient from destabilising the feature extractor before it has learned reasonable task representations.

### 3.7.3 Per-Group Discriminator Architecture

GLADA attaches an independent domain discriminator to each of the $K$ layer groups. Each discriminator $D_k$ is a lightweight two-layer MLP that receives the group's ASP output (passed through a GRL) and predicts the domain label:

$$\hat{d}_k = D_k(\mathrm{GRL}_\alpha(g_k)) = W_{k,2} \cdot \mathrm{ReLU}(W_{k,1} \cdot \mathrm{GRL}_\alpha(g_k) + b_{k,1}) + b_{k,2} \tag{28}$$

$W_{k,1} \in \mathbb{R}^{128 \times 2D}$, $W_{k,2} \in \mathbb{R}^{2 \times 128}$, and dropout with probability 0.1 is applied after the ReLU activation. The hidden dimension of 128 is deliberately small: each discriminator should provide a domain alignment signal without overpowering the task learning objective.

The $K$ discriminators are fully independent; they share no parameters. Each one can therefore specialise on the domain shift patterns present in its assigned layer

group. The Group 0 discriminator (acoustic layers) may learn to detect mismatches in recording environment characteristics; the Group 3 discriminator (task-specific layers) may learn to detect mismatches in synthesis method signatures.

### 3.7.4 GLADA Loss

The per-group adversarial losses are averaged across groups to produce the GLADA loss:

$$\mathcal{L}_{\text{GLADA}} = \frac{1}{K} \sum_{k=1}^{K} \text{CE}\left(D_k(\text{GRL}_\alpha(g_k)),\ d\right) \tag{29}$$

$d \in \{0, 1\}$ is the domain label (source = 0, target = 1) and CE is cross-entropy loss. Equal averaging guarantees that each group contributes the same weight to the total adversarial signal regardless of how hard domain discrimination is at that level.

### 3.7.5 Comparison with Prior Work

Table 2 positions GLADA relative to prior domain adaptation methods. GLADA is the first method to combine per-group adversarial domain adaptation with a variational information bottleneck and CKA diversity regularization, applied to self-supervised speech representations for deepfake detection.

Table 2: Comparison of domain adaptation approaches. GLADA combines per-group adversarial alignment (porting the MLADA principle from vision and the stratified adversarial setting from NLP) with VIB and CKA diversity loss, applied to SSL speech representations.

| **Method** | **Domain** | **Layers** | **DA Type** | **Bottleneck** | **Diversity** |
|---|---|---|---|---|---|
| DANN [10] | General | Global | Adversarial | -- | -- |
| MLADA [11] | Vision | Per-layer | Adversarial | -- | -- |
| MSP-DA [15] | NLP | Global | Adv. + meta self-paced | -- | -- |
| IB-CAAN [9] | Speech | Global | Adversarial | IB | -- |
| HierCon [7] | Speech | Grouped | Contrastive | -- | -- |
| **GLADA** | **Speech** | **Per-group** | **Adversarial** | **VIB** | **CKA** |

### 3.8 Speaker Adversary

Speaker identity is a potential confound for cross-domain generalisation. The ASVspoof 2019 LA training set contains only 20 speakers, a small fraction of the speaker diversity encountered in real-world audio. A model that memorises speaker-specific characteristics (fundamental frequency range, vocal tract resonances, speaking style) may associate those characteristics with the bonafide/spoofed distinction and produce features that fail to generalise to unseen speakers.

The remedy is the same gradient reversal technique used for domain adaptation, applied to speaker identity. A speaker classifier receives the bottleneck representation $z$ through a GRL and predicts the speaker identity:

$$\mathcal{L}_{\text{speaker}} = \text{CE}\left(S(\text{GRL}_{\alpha_{\text{spk}}}(z)),\ s\right) \tag{30}$$

$S$ is a three-layer MLP (Linear($d_z$, 512) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear(512, 256) $\rightarrow$ ReLU $\rightarrow$ Linear(256, $N_{\text{spk}}$)), $s$ is the speaker label, and $N_{\text{spk}} = 20$ is the number of speakers in the source domain. The speaker adversary is applied only to source-domain samples; speaker labels are not available for In-the-Wild.

The speaker GRL warmup is slower than the domain GRL: $\alpha_{\text{spk}}$ ramps over $E_w^{\text{spk}} = 10$ epochs against 5 for domain. The multi-class speaker prediction task ($N_{\text{spk}}$ classes) is harder to stabilise than binary domain prediction and needs a more gradual introduction of the adversarial gradient.

### 3.9 Combined Loss Function

The total training objective combines five loss terms:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha \cdot \mathcal{L}_{\text{GLADA}} + \beta \cdot \mathcal{L}_{\text{KL}} + \gamma \cdot \mathcal{L}_{\text{speaker}} + \delta \cdot \mathcal{L}_{\text{CKA}} \tag{31}$$

Table 3 describes each term and its default weight.

The task loss $\mathcal{L}_{\text{task}}$ is weighted binary cross-entropy between the classifier's predictions and the ground-truth spoof labels, computed only on source-domain

Table 3: Loss terms in the GLADA objective. Each term serves a distinct role in shaping the learned representation.

| Term | Weight | Default | Role | Computed on |
|---|---|---|---|---|
| $\mathcal{L}_{\text{task}}$ | 1.0 | -- | Binary CE for spoof detection | Source only |
| $\mathcal{L}_{\text{GLADA}}$ | $\alpha$ | 0.1 | Per-group domain adversarial | Source + Target |
| $\mathcal{L}_{\text{KL}}$ | $\beta$ | $10^{-3}$ | VIB compression | All samples |
| $\mathcal{L}_{\text{speaker}}$ | $\gamma$ | 0.1 | Speaker disentanglement | Source only |
| $\mathcal{L}_{\text{CKA}}$ | $\delta$ | 0.01 | Group diversity | All samples |

samples (which carry labels). The domain adversarial loss $\mathcal{L}_{\text{GLADA}}$ is computed on both source and target samples, since domain labels are available for both. The VIB KL loss and CKA diversity loss are computed on all samples regardless of domain. The speaker loss applies to source-domain samples only.

### 3.9.1 Loss Weight Selection and the Implicit Task-Loss Scale

The task loss carries an implicit weight of 1.0; it is the reference scale against which the four auxiliary coefficients $\alpha, \beta, \gamma, \delta$ are calibrated. This is the standard multi-task loss-combination convention [95]: the primary classification objective sets the dominant gradient magnitude and the auxiliary terms are calibrated as fractional regularisers below that scale. Adding an explicit $\lambda_{\text{task}}$ would be redundant with the relative auxiliary weights and would only confuse the bookkeeping; the design choice the auxiliary $\lambda$ values encode is the gradient magnitude of each auxiliary objective *relative* to the task objective, which is what the table records.

The four auxiliary coefficients were not Bayes-optimised; each was selected by a different mechanism that is honest to record:

- $\alpha = 0.1$ **(per-group adversarial weight) --- empirical destabilisation threshold.** The CS 297 survey runs and an early sweep on the full GLADA stack showed that $\alpha \geq 0.5$ collapsed the task classifier before useful features

formed; $\alpha = 0.1$ was the largest stable value tested. This matches the general sensitivity of GRL-based adversarial DA weights documented in DANN [10] and the small-multiplier convention used by IB-CAAN [9].

- $\beta = 10^{-3}$ **(VIB KL weight) --- sensitivity analysis.** A short single-seed sweep over $\beta \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$ produced the lowest validation EER at $\beta = 10^{-3}$. The chosen value is loose enough to avoid posterior collapse while still producing measurable compression on the post-VIB representation (see the $\beta$ sweep results in Section 5.5). The same $\beta$ choice appears in IB-CAAN [9] and matches the prior-work convention for VIB in cross-domain settings.
- $\gamma = 0.1$ **(speaker-adversary weight) --- parallel to $\alpha$, prior-work convention.** The speaker adversary plays the same structural role as the domain adversary (a GRL-based invariance objective), so $\gamma$ was set to match $\alpha$. This parallels the joint speaker--channel adversarial training weights used by Meng et al. [51] and Chen et al. [52].
- $\delta = 0.01$ **(CKA diversity weight) --- regulariser scale, fixed an order of magnitude below the adversarial terms.** CKA is a representational-similarity *regulariser* rather than a co-equal objective; setting $\delta$ ten times smaller than $\alpha$ and $\gamma$ keeps it from competing with the adversarial gradients for representational capacity. The choice follows the regulariser-vs-primary-objective scale convention used in multi-task learning [95] and the diversity-loss weighting in MultiConv [19], the closest prior precedent for CKA in audio anti-spoofing.

No formal Bayesian hyperparameter optimisation was performed across the full $(\alpha, \beta, \gamma, \delta)$ space. The compute budget did not permit a multi-seed Bayesian sweep,

and the multi-objective interactions observed in Chapter 5 (Section 5.1) suggest that the auxiliary objectives interact non-additively in ways that a single global $\lambda$ tuple cannot resolve cleanly. Multi-seed Bayesian tuning of the full coefficient vector, ideally with per-group $\beta_k$ values that anneal independently, is identified as a priority follow-up in Section 7.3.

## 3.10 Two-Phase Training Strategy

GLADA uses a two-phase training strategy that first stabilises the downstream heads before fine-tuning the backbone.

### 3.10.1 Phase 1: Frozen Backbone

During Phase 1 (5 epochs) the SSL backbone parameters are frozen. Only the randomly initialised downstream components train: the task classifier, VIB, per-group domain discriminators, speaker adversary, attention pooling weights, and per-layer normalisation parameters.

The optimiser is AdamW [96] with learning rate $\eta = 10^{-3}$ and weight decay $10^{-2}$. The learning rate follows a OneCycleLR schedule [97] with a 5% linear warmup to $\eta_{\max}$ followed by cosine annealing to $\eta_{\max}/25$.

Phase 1 exists to warm up the downstream heads on stable backbone features before backbone gradient noise enters the picture. Without it, the randomly initialised discriminators and VIB would receive large, noisy gradients from the backbone on the first step, with a real risk of destabilising training from the outset.

### 3.10.2 Phase 2: End-to-End Fine-Tuning

In Phase 2 (up to 45 epochs) the SSL backbone is unfrozen and fine-tuned end-to-end alongside the downstream heads. Two techniques govern the fine-tuning.

**Layer-Wise Learning Rate Decay (LLRD).** Following Clark et al. [92], each

Transformer layer receives a learning rate that decays exponentially with depth:

$$\eta_i = \eta_{\text{backbone}} \times \rho^{(L-i)} \tag{32}$$

$\eta_{\text{backbone}} = 10^{-5}$ is the base backbone learning rate, $\rho = 0.95$ is the decay factor, and $i$ is the layer index (0 = closest to input, $L$ = closest to output/most task-specific). For WavLM-Large ($L = 24$):

- Layer 0 (closest to input): $10^{-5} \times 0.95^{24} \approx 2.9 \times 10^{-7}$ (nearly frozen).
- Layer 12 (middle): $10^{-5} \times 0.95^{12} \approx 5.4 \times 10^{-6}$.
- Layer 24 (closest to output): $10^{-5} \times 0.95^{0} = 10^{-5}$ (full learning rate).

The downstream heads keep their Phase 1 learning rate of $10^{-3}$. The intuition is that lower layers encode general acoustic features learned during large-scale pre-training and should be mostly preserved, while higher layers are more task-specific and benefit from larger updates.

**Gradient Checkpointing.** The backbone is trained with gradient checkpointing enabled, which trades roughly 30% additional compute for roughly 60% lower GPU memory. Without it, a 317M-parameter WavLM-Large with 24 layers of hidden states does not fit on a single A100.

A fresh OneCycleLR schedule is initialised for Phase 2, giving the newly unfrozen parameters a clean warmup and annealing cycle. The GRL warmup also restarts from $\alpha = 0$ at the beginning of Phase 2.

### 3.10.3 Optimization Details

Both phases use gradient accumulation with 8 steps, giving an effective batch size of $48 \times 8 = 384$ samples per optimiser step (192 per domain). Gradients are clipped to a maximum norm of 1.0 to keep gradient explosions from happening, particularly in early Phase 2 when backbone gradients can be large.

Training terminates through early stopping with patience of 15 epochs, monitoring

the Equal Error Rate on the In-the-Wild evaluation set. Checkpoints are saved every epoch, and the final model is obtained by averaging the parameters of the top-$K$ checkpoints ranked by validation EER. The averaging smooths out per-epoch noise and, in our runs, generalises better than any single checkpoint.

### 3.11 Data Augmentation Pipeline

Data augmentation is applied to source-domain waveforms during training to reduce the acoustic mismatch between studio-recorded ASVspoof data and real-world In-the-Wild audio. The pipeline supports three augmentation stages, each with a configurable probability. Only RawBoost was active for the experiments reported in this project; MUSAN and RIR are described here for completeness because they are implemented in the codebase and are listed for future work (see Section 7.3).

1. **RawBoost** [55] (probability 0.5): one of three augmentation types is sampled uniformly and applied to the raw waveform:
   - *Linear and non-linear convolutive noise:* spectral shaping that simulates room acoustics and transmission channel effects.
   - *Impulsive signal-dependent additive noise:* Bernoulli-sampled transient bursts that model codec artefacts and click noise.
   - *Stationary signal-independent additive noise:* Gaussian white noise at a uniformly sampled SNR in $[15, 30]$ dB, modelling background and microphone noise.

2. **MUSAN noise** [90] (probability 0.5): a noise segment is drawn from the MUSAN corpus (speech, music, and environmental recordings) and mixed in at an SNR calibrated to real-world conditions: noise at 0--15 dB, speech at 13--20 dB, music at 5--15 dB.

3. **Room Impulse Response (RIR) convolution** [91] (probability 0.5): the audio is convolved with a simulated RIR, adding reverberation consistent with a range of recording environments.

The three stages are chained in physical signal order: RawBoost (channel/codec effects) → MUSAN (additive noise) → RIR (room reverberation). The order follows the physical process of sound propagation: the speech signal first passes through a communication channel, then is mixed with environmental noise, and finally reverberates in the room.

Augmentation complements domain adaptation; it does not replace it. Adversarial DA aligns learned feature distributions between domains. Augmentation reduces the acoustic gap directly, by making source-domain audio resemble target-domain conditions more closely. Huang et al. [9] showed the two working together in IB-CAAN: adding MUSAN and RIR augmentation to RawBoost dropped their In-the-Wild EER from 4.93% to 4.61%.

# CHAPTER 4

# Experimental Setup

The sections below cover the datasets, the implementation, the ablation design, the evaluation protocol, and the baseline systems, in that order.

## 4.1 Datasets

Six datasets are used to train, augment, and evaluate the GLADA framework across a range of domain shifts.

### 4.1.1 Training Datasets

#### 4.1.1.1 ASVspoof 2019 LA (Source Domain)

ASVspoof 2019 Logical Access (LA) [80, 98] is my labelled source domain ($d = 0$). The training partition contains 25,380 utterances from 20 speakers: 2,580 bonafide recordings and 22,800 spoofed recordings generated by six known TTS and VC systems. All audio is provided in FLAC at 16 kHz, recorded under controlled studio conditions. Class imbalance is handled through weighted random sampling during training. Unlike the preliminary CS 297 survey, I use the speaker identities here to supervise the speaker adversary in Section 3.8.

#### 4.1.1.2 In-the-Wild (Target Domain)

The In-the-Wild dataset [6] contains 31,779 audio clips totalling 37.9 hours, collected from a mix of real-world sources including YouTube and social media platforms. The audio exhibits variable acoustic conditions: background noise, reverberation, compression artefacts, and channel effects. The synthesis methods behind the spoofed samples are unknown. In-the-Wild plays a dual role in this study. During training it is used without labels as the target domain ($d = 1$) for domain adaptation. At evaluation, the ground-truth labels are used for the primary cross-domain benchmark. The evaluation labels are never used during training.

### 4.1.2 Data Augmentation

All experiments in this project used RawBoost [55] as the sole training-time augmentation, applied to the raw waveform before it entered the SSL backbone. RawBoost combines linear and non-linear convolutive noise with impulsive signal-dependent additive noise, producing realistic acoustic distortions without requiring external corpora. During training, RawBoost was applied with a probability of $p = 0.5$ to source-domain samples, while target-domain samples were left unaugmented.

MUSAN and RIR augmentation were not used in the reported experiments. Huang et al. [9] showed that adding them reduced ITW EER from 4.93% to 4.61%, making this one factor in the performance gap discussed in Section 5.10.

### 4.1.3 Evaluation Datasets

GLADA was evaluated on six datasets spanning in-domain, telephony, codec compression, real-world acoustic, modern neural vocoder, and neural codec synthesis shifts. Table 4 summarises the set.

Table 4: Evaluation datasets spanning in-domain and cross-domain shifts.

| **Dataset** | **Samples** | **Domain Shift Type** | **Reference** |
|---|---|---|---|
| ASVspoof 2019 LA eval | 71,237 | In-domain | Wang et al. 2020 [80] |
| ASVspoof 2021 LA | 181,566 | Telephony channel | Yamagishi et al. 2021 [99] |
| ASVspoof 2021 DF | 611,829 | Codec compression | Yamagishi et al. 2021 [99] |
| In-the-Wild | 31,779 | Real-world acoustic | Müller et al. 2022 [6] |
| WaveFake | ∼117,985 | Modern neural vocoder | Frank and Schönherr 2021 [100] |
| CodecFake+ | 4,417 | Neural codec synthesis | Chen et al. 2025 [101] |

ASVspoof 2019 LA eval measures in-domain generalisation to unseen attack types under the same controlled studio conditions as the training data. ASVspoof

2021 LA introduces telephone-channel effects (codec distortion, transmission noise, bandwidth limitation); ASVspoof 2021 DF tests robustness to deepfake audio under nine codec compression conditions. In-the-Wild measures performance under real-world acoustic conditions drawn from online media with unknown and diverse spoofing methods. WaveFake introduces a modern neural vocoder shift through high-fidelity waveform generation systems, and CodecFake+ extends the evaluation to neural audio codec-based synthesis built on discrete token resynthesis.

## 4.2 Implementation Details

All experiments use PyTorch 2.x with the HuggingFace Transformers library for loading the self-supervised learning (SSL) backbones. Training ran on Google Colab: an NVIDIA A100 80GB GPU as the main platform, with an NVIDIA T4 16GB GPU kept as a fallback for debugging. Audio preprocessing is handled exclusively through the `soundfile` library to ensure consistent decoding. All audio samples are resampled to a 16 kHz sampling rate and padded or truncated to exactly 4 seconds, yielding fixed-length arrays of 64,000 samples.

The batch size is 24 samples per domain (24 source, 24 target), giving 48 samples per forward pass. Gradient accumulation over 8 steps brings the effective batch size to 384. On the A100, training uses BFloat16 mixed precision with TensorFloat-32 (TF32) matrix multiplication enabled; the fallback T4 uses Float16 instead. The optimiser is AdamW with weight decay 0.01, and gradients are clipped to a maximum norm of 1.0. Gradient checkpointing is enabled on the SSL backbone so the memory footprint stays low enough for full end-to-end fine-tuning.

Training proceeds in two phases as described in Section 3.10. In Phase 1 the SSL backbone is frozen and only the randomly initialised classification heads and discriminators train; the phase runs for 5 epochs (3 for ablation screening) at learning

rate $1 \times 10^{-3}$. Phase 2 unfreezes the backbone and fine-tunes everything end to end for 45 epochs (17 for screening). To preserve the acoustic features from pre-training, layer-wise learning rate decay (LLRD) [92] is applied to the backbone, with a peak learning rate of $1 \times 10^{-5}$ at the top layer and a decay factor of 0.95 per layer downward. The classification heads are optimised with a OneCycleLR scheduler. Training stops early if the In-the-Wild (ITW) validation EER fails to improve for 15 epochs.

The Gradient Reversal Layer (GRL) hyperparameters are scheduled strictly, following standard domain-adversarial practice [10]. The domain-discriminator GRL uses a 5-epoch warmup during which the reversal multiplier $\lambda$ ramps linearly from 0 to 1. The speaker-adversary GRL uses a slower 10-epoch warmup, which keeps the speaker-disentanglement objective from destabilising the backbone early in training. Primary experiments use a fixed random seed of 42. Multi-seed validation averages results across three seeds $\{42, 123, 456\}$. The original 19.71% reference result (monolithic GLADA with WavLM-Base-Plus) and the leave-one-out ablation table in Section 5.2 use seed 42 only; the per-group VIB comparison conditions described below are reported with all three seeds where indicated.

### 4.3 Ablation Design

To isolate the contribution of each architectural component, the ablation study uses an 11-configuration matrix split into seven leave-one-out ablations and four comparative baselines (Table 5). The performance drop $\Delta\text{EER} = \text{EER}_{\text{ablated}} - \text{EER}_{\text{full}}$ measures each component's contribution.

The `no_glada` configuration directly tests the primary contribution of this project: whether applying adversarial domain adaptation independently across grouped layers outperforms a single global discriminator at the final embedding stage. The hypothesis behind it has support from multi-layer adaptation findings in computer vision [11].

Table 5: The 11-configuration ablation matrix, consisting of leave-one-out structural ablations and key baseline architectures.

| **Configuration** | **Component Changed or Resembled** | **Hypothesis Tested** |
|---|---|---|
| *Group 1: Leave-One-Out Ablations* | | |
| `full_glada` | None (all components active) | Reference performance |
| `no_glada` | Grouped DA → Single global DA | Does grouped DA outperform single-level DA? |
| `no_vib` | VIB removed (deterministic projection) | Does info compression aid domain invariance? |
| `no_speaker` | Speaker adversary removed | Does speaker disentanglement help generalisation? |
| `no_cka` | CKA diversity loss removed | Does enforcing diverse group reps help DA? |
| `no_rawboost` | RawBoost augmentation removed | How critical is waveform augmentation with DA? |
| `no_hier` | Grouped pooling → Flat sum | Does grouped attention outperform flat aggregation? |
| *Group 2: Key Baselines* | | |
| `global_da_only` | Resembles IB-CAAN [9] | Global DA + VIB (no per-group DA or CKA) |
| `hierarchical_only` | Resembles HierCon [7] | Grouped attention pooling only (no DA/VIB) |
| `no_da` | No adversarial training | Classifier-only baseline with no adversarial training |
| `backbone_only` | Vanilla SSL fine-tuning | SSL backbone plus a linear classifier; performance floor |

`no_vib` asks whether information compression [86] is necessary. `no_cka` tests the utility of enforcing representation diversity across groups [19]. `no_hier` checks whether explicit grouped attention [47] outperforms simpler layer aggregation.

All configurations share an identical backbone, random seed, epoch budget, and evaluation protocol. Ablation screening uses reduced epoch budgets (3 epochs of Phase 1 and 17 of Phase 2) on the `wavlm-base-plus` architecture, which keeps the sweep tractable within the available compute budget. The final metrics come from the full schedule (5 epochs Phase 1, 45 epochs Phase 2) on `wavlm-large`. `backbone_only` sits at the performance floor.

### 4.3.1 Per-Group VIB Ablation Conditions

Following the leave-one-out analysis described in Section 3.6.4, four additional conditions are evaluated to characterize the per-group VIB variant (Section 3.6):

- **Monolithic VIB**: the stock `GLADAModel` with a single VIB after the group projection (Equation (23)). This is the primary GLADA configuration and the reference for all VIB comparisons.
- **Per-group VIB**: `PerGroupVIBModel` with all groups enabled (`vib_disabled_groups=()`). Each group's ASP output $g_k$ is independently compressed before the domain discriminator.
- **g0-only VIB**: `PerGroupVIBModel` with `vib_disabled_groups=(1, 2)` (3-group backbone) or `(1, 2, 3)` (4-group backbone). Only the acoustic group (Group 0) retains an active VIB; all other groups use deterministic pass-through ($z_k = \mu_{\phi_k}(g_k)$, no KL pressure).
- **drop_vib_g0**: `PerGroupVIBModel` with `vib_disabled_groups=(0,)`. The acoustic group's VIB is disabled; all other groups retain full stochastic compression. This is the complementary condition to g0-only VIB and measures what happens when the dominant group's bottleneck is removed.

The monolithic VIB condition acts as a fair baseline: it uses the same `GLADAModel` code path as the primary experiments, so any difference relative to the per-group conditions isolates the effect of co-locating compression and adversarial alignment. All four conditions are evaluated under the three-seed protocol ($\{42, 123, 456\}$) to quantify variance.

### 4.3.2 Cross-Backbone Validation

To test whether the g0-only VIB finding generalizes beyond WavLM-Base-Plus, two additional cross-backbone runs are performed:

- **WavLM-Large + g0-only**: `PerGroupVIBModel` with 4 groups (`vib_disabled_groups=(1, 2, 3)`), batch size 24 per domain, single seed (42). Uses the same full-schedule epoch budget (Phase 1: 5 epochs, Phase 2: 45 epochs).
- **xlsr-300m + g0-only**: `PerGroupVIBModel` on the XLSR-300M backbone [82] with 4 groups, `vib_disabled_groups=(1, 2, 3)`, batch size 24, single seed (42).

These cross-backbone runs use a single seed and are not directly comparable on every training knob (the XLSR and WavLM-Large pre-training objectives differ, for example), but they give a directional signal that the g0-dominance finding is not specific to one backbone.

### 4.4 Evaluation Protocol

Every experiment reports Equal Error Rate (EER): the decision threshold at which the false acceptance rate (FAR) equals the false rejection rate (FRR). Lower is better. Uncertainty is quantified with 95% bootstrap confidence intervals [102], computed from 1,000 resamples over the evaluation scores.

Final evaluation uses a multi-dataset protocol over the six datasets in Section 4.1. In-the-Wild (ITW) EER is the primary benchmark, because it measures real-world cross-domain generalisation most directly; the ASVspoof 2019 LA evaluation set is the secondary benchmark, measuring strictly in-domain performance.

To reduce weight variance and improve generalisation, top-$K$ checkpoint averaging is applied [103, 104]. Following prior work in audio deepfake detection [9], I take the top 3 checkpoints ($K = 3$) by validation EER and average their weights to produce the final inference model. The top configurations are re-evaluated under a multi-seed validation strategy with random seeds $\{42, 123, 456\}$, reporting mean and

standard deviation to quantify run-to-run variance.

### 4.4.1 Seeds, Variability, and Statistical Significance

Three random seeds are used throughout this report: $\{42, 123, 456\}$. Seed 42 is the primary seed for the screening pass (the 11-configuration leave-one-out ablation in Section 5.2), which uses single-seed runs to keep the sweep tractable within the available compute budget. The bottleneck-placement comparison in Section 5.3 and the cross-dataset table in Section 5.4 are reported as mean $\pm$ one standard deviation over all three seeds ($n = 3$). Where mean values are quoted in the text without a $\pm$ envelope, the figure is a single-seed (seed 42) result and is labelled as such in the surrounding sentence and table caption.

Two complementary uncertainty quantifications are reported. First, the bootstrap-based 95% confidence intervals on ITW EER (Section 5.1) use 1,000 score resamples per model to capture the evaluation-set sampling variance for a single trained model; this isolates evaluation noise from training noise. Second, the multi-seed mean $\pm$ std over three seeds captures the training-time sampling variance, which depends on the random initialisation of trainable heads, the data-loader shuffle, and the augmentation draws.

For statistical significance, I use a non-overlapping-interval heuristic rather than a formal test, because the available $n = 3$ does not support a powerful parametric test. When two configurations' bootstrap 95% confidence intervals on ITW do not overlap (e.g. `full_glada` at 19.71% [19.25, 20.17] versus `backbone_only` at 13.60% [13.21, 13.98]), the difference is treated as substantively meaningful and the corresponding claim is made in the text. When two configurations' multi-seed envelopes overlap (e.g. on the 2021 DF column where every variant is within seed noise of the others), no ranking claim is made on that column. Stronger significance testing across more seeds

is identified as a priority follow-up in Section 7.3.

### 4.5 Baselines

GLADA is compared against two categories of baselines: strictly controlled internal architectures, and externally published systems evaluated on the In-the-Wild dataset.

**Internal Baselines.** Drawn from the ablation study (Section 4.3), these configurations hold the backbone, training budget, and evaluation conditions fixed and isolate the effect of each architectural choice.

**Literature Baselines.** The published cross-domain results compared against on In-the-Wild are:

- **RawNet2**: ∼37.8% EER [6], a 25M-parameter end-to-end convolutional model.
- **AASIST**: ∼31% EER [5, 6], a graph attention network with 0.3M parameters.
- **XLSR + Linear Empirical Risk Minimization (ERM)**: ∼14.13% EER [9], a linear probe trained on frozen XLSR-300M embeddings without domain adaptation.
- **IB-CAAN (XLSR)**: 4.93% EER [9], using VIB with confidence-aware adversarial alignment during fine-tuning.
- **IB-CAAN (XLSR) + full augmentation**: 4.61% EER [9], incorporating the RawBoost, MUSAN, and RIR pipeline.
- **4-model fusion**: 6.4% EER [17], an ensemble combining BEATs, WavLM-Large, MMS, and XLSR architectures.
- **HierCon**: 6.87% EER [7], employing hierarchical contrastive learning over SSL layers.

Our preliminary survey (Appendix A) adds baseline context from 30 backbone-method combinations evaluated with frozen backbones.

# CHAPTER 5

## Results and Analysis

All 16 models were evaluated on six datasets, giving 96 EER measurements with bootstrap 95% confidence intervals. The ablation study over `wavlm-base-plus` comes first (Section 5.1), then the backbone comparison (Section 5.8), the cross-dataset generalisation analysis (Section 5.9), and the comparison with published methods (Section 5.10).

### 5.1 Ablation Study

The 11 `wavlm-base-plus` configurations from Table 5 were evaluated on all six datasets; Table 6 reports the resulting EERs. Lower is better. The ITW column carries a 95% bootstrap confidence interval because ITW is the primary benchmark for cross-domain generalisation.

`full_glada` produced the worst ITW result among all 11 configurations at 19.71% EER (95% CI: 19.25--20.17), so the full stack failed to improve the primary cross-domain metric. It trailed `backbone_only` at 13.60% (13.21--13.98) and `no_cka` at 12.98% (12.60--13.33) by a wide margin. That contradicts the hypothesis that grouped-layer adversarial adaptation, combined with the full auxiliary stack, would improve real-world generalisation; on the primary benchmark the additional objectives introduced negative transfer rather than a measurable gain.

No auxiliary component provided a net benefit inside the full `wavlm-base-plus` configuration; every leave-one-out ablation beat `full_glada`. Ordered by the ITW gain on removal, CKA was the largest offender (`no_cka`: 12.98%, $\Delta = -6.73$), followed by the speaker adversary (`no_speaker`: 13.48%, $\Delta = -6.23$), VIB (`no_vib`: 14.58%, $\Delta = -5.13$), and replacing grouped DA with global DA (`no_glada`: 14.89%, $\Delta = -4.82$). Removing RawBoost (`no_rawboost`: 16.66%, $\Delta = -3.04$) and hierarchical aggregation (`no_hier`: 17.47%, $\Delta = -2.24$) also helped, by smaller margins.

Table 6: EER (%) for the 11 `wavlm-base-plus` configurations on the six evaluation datasets; lower is better. ITW entries include 95% bootstrap confidence intervals in parentheses. Bold marks the lowest EER in each column.

| Configuration | 2019 LA | ITW (95% CI) | 2021 LA | 2021 DF | WaveFake | CodecFake+ |
|---|---|---|---|---|---|---|
| *Leave-one-out ablations* | | | | | | |
| `full_glada` | 1.04 | 19.71 (19.25--20.17) | 31.02 | 27.74 | 27.88 | 37.77 |
| `no_glada` | 1.36 | 14.89 (14.51--15.30) | 31.53 | 27.96 | 26.76 | 36.97 |
| `no_vib` | 0.72 | 14.58 (14.19--14.97) | 33.02 | 29.73 | **19.17** | **36.88** |
| `no_speaker` | 0.93 | 13.48 (13.11--13.90) | 31.78 | 27.57 | 27.30 | 37.58 |
| `no_cka` | 1.07 | **12.98** (12.60--13.33) | 32.15 | 29.50 | 29.74 | 37.88 |
| `no_rawboost` | 1.05 | 16.66 (16.23--17.08) | 31.98 | **26.86** | 23.14 | 39.28 |
| `no_hier` | 1.15 | 17.47 (17.05--17.89) | 33.24 | 28.52 | 26.69 | 39.59 |
| *Structural variants* | | | | | | |
| `global_da_only` | 1.42 | 14.99 (14.56--15.37) | 31.51 | 27.95 | 26.61 | 37.27 |
| `hierarchical_only` | 1.02 | 13.28 (12.92--13.63) | **29.36** | 27.24 | 23.25 | 39.09 |
| `no_da` | 0.80 | 14.20 (13.82--14.63) | 30.37 | 27.01 | 24.54 | 37.98 |
| `backbone_only` | **0.62** | 13.60 (13.21--13.98) | 32.37 | 28.50 | 23.33 | 37.46 |

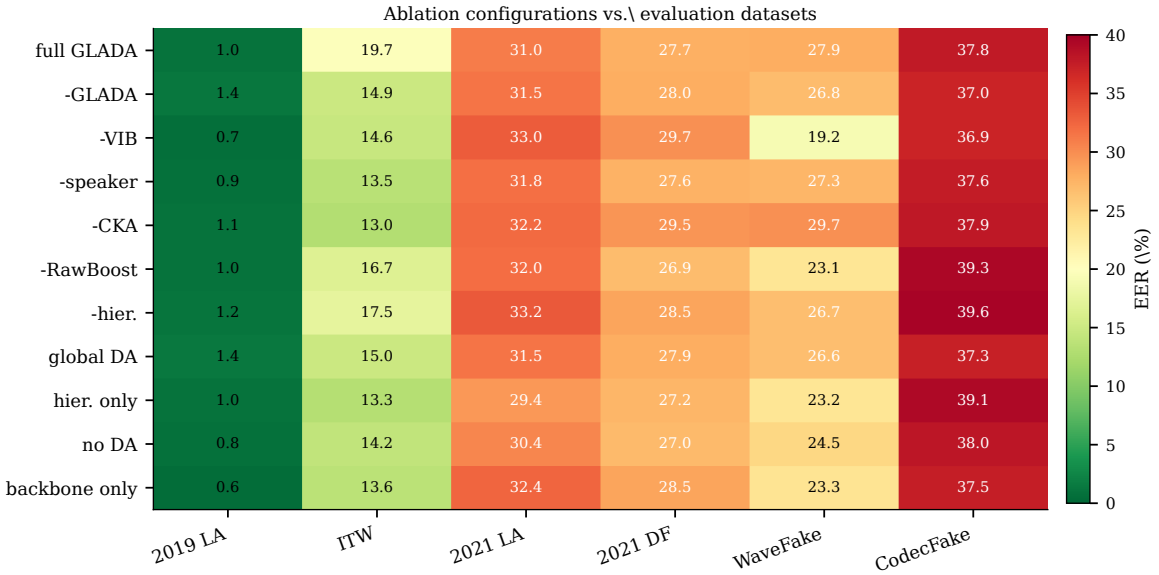



Figure 2: EER (%) across 11 ablation configurations and 6 evaluation datasets. Red outline: `full_glada` (worst cross-domain). Blue outline: `backbone_only` (baseline). The in-domain/cross-domain ranking inversion is visible: configurations achieving low in-domain EER do not rank best cross-domain.

The simplified baselines told the same story: `hierarchical_only` reached 13.28%, `backbone_only` 13.60%, and `no_da` 14.20%. The full objective stack over-constrained the model on ITW, with CKA, the speaker adversary, and the grouped adversarial

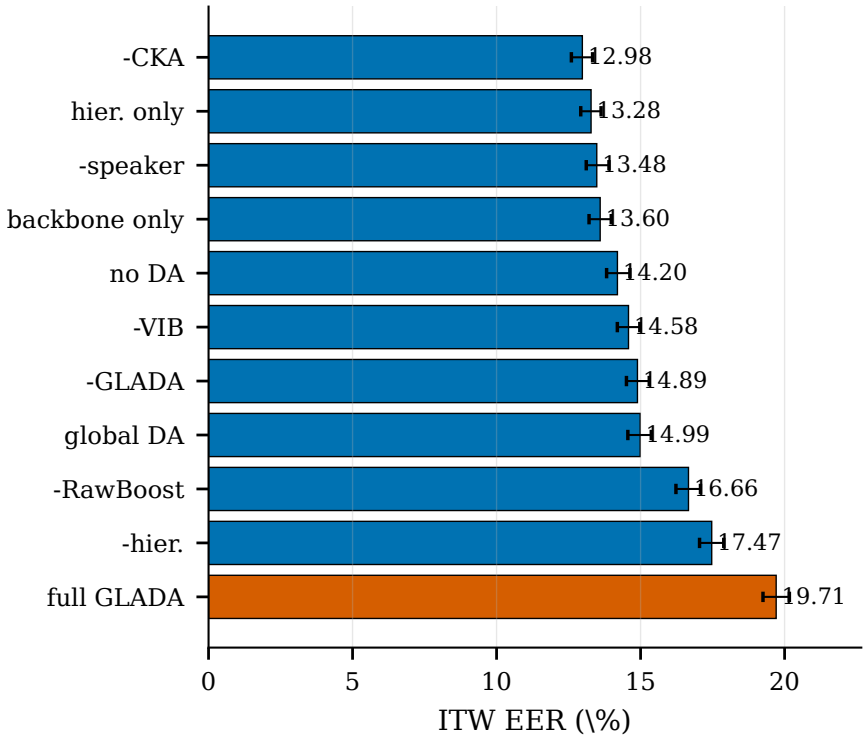



Figure 3: Ablation configurations ordered by ITW EER with 95% bootstrap confidence intervals. The highlighted bar, `full_glada`, sits at the bottom of the cross-domain ranking; every other configuration recovers ground from it.

branch contributing the largest share of the degradation.

The comparison between `no_glada` and `global_da_only` behaved as expected. Both labels correspond in practice to the same single-level DA architecture, and their EERs were essentially identical across the six datasets: 14.89% vs. 14.99% on ITW, 1.36% vs. 1.42% on 2019 LA, 31.53% vs. 31.51% on 2021 LA, 27.96% vs. 27.95% on 2021 DF, 26.76% vs. 26.61% on WaveFake, and 36.97% vs. 37.27% on CodecFake+. That agreement is close enough to treat the pair as a replication check rather than two distinct architectural outcomes.

In-domain and cross-domain behaviour diverged sharply. On 2019 LA all 11 configurations clustered in a tight 0.62%--1.42% band, with `backbone_only` best and `full_glada` mid-pack at 1.04%. On ITW the spread widened to nearly seven

Table 7: Leave-one-VIB-out diagnostic ablation on `per_group_vib` (WavLM-Base-Plus, seed 42, $K = 3$ groups). Each row disables stochastic sampling and KL pressure for one group while leaving the other two active. $\Delta$ is the gap relative to the reference `per_group_vib` configuration; positive values indicate degradation. Best epoch reported as logged by the trainer.

| **Configuration** | **ITW EER (%)** | **$\Delta$ vs. reference** | **Best epoch** |
|---|---|---|---|
| `per_group_vib` (reference) | 11.67 | -- | -- |
| `drop_vib_g0` | 17.61 | +5.94 | 3 |
| `drop_vib_g1` | 13.80 | +2.13 | 3 |
| `drop_vib_g2` | 11.99 | +0.32 | 5 |

points and the ranking inverted: `full_glada` became the worst cross-domain model, while simpler configurations like `no_cka`, `hierarchical_only`, and `no_speaker` outperformed it by large margins. No single simplification generalised uniformly across all five cross-domain sets; the best model varied by shift type (`hierarchical_only` on 2021 LA, `no_rawboost` on 2021 DF, `no_vib` on both WaveFake and CodecFake+). Low in-domain EER on ASVspoof 2019 LA was not a reliable proxy for cross-domain generalisation.

### 5.2 Leave-One-VIB-Out Diagnostic Ablation

The monolithic VIB ablation in Section 5.1 treated the bottleneck as a single component, but the per-group VIB variant introduced in Section 3.6 opens a finer question: which layer group's bottleneck is doing the work? To probe this, I ran a leave-one-out ablation on the per-group VIB model with seed 42, disabling one group's VIB at a time via `vib_disabled_groups` (Section 3.6.4) while leaving the other groups stochastic. Table 7 reports ITW EER for the three single-group disable conditions alongside the unmodified per-group VIB reference run.

The three groups contributed unevenly to the bottleneck signal. Disabling the acoustic group's VIB (`drop_vib_g0`) cost 5.94 percentage points of ITW EER, much more than disabling the phonetic group (`drop_vib_g1`: +2.13 pp) or the task group

(`drop_vib_g2`: +0.32 pp). The g2 result sat within the seed-to-seed noise floor estimated in Section 5.3 (per-group VIB std $\approx$ 1.31 pp), so removing the bottleneck on the highest-level group had no measurable effect at this scale. The pattern matches the acoustic-feature analysis intuition that lower SSL layers carry the recording-condition shift most directly: when those representations bypass the bottleneck, ITW degradation is largest. This is contribution A3 (mechanism evidence) and aligns with the layer-wise SSL analyses for anti-spoofing of Kheir et al. [8], Xiao and Vu [18], Serrano et al. [17], Tran et al. [19], Sinha et al. [20], and Pan et al. [47], all of which place the strongest spoof-discriminative signal in lower transformer depths. The leave-one-out evidence is complementary to those: rather than weighting layers by attention or fusion, this ablation measures the cost of removing compression at a given depth.

The training trajectory of `drop_vib_g0` also showed a stability difference. Its best checkpoint landed at epoch 3 (against epoch 5 for both the reference and `drop_vib_g2`), and the run logs recorded repeated KL-clamp activations not present in the `drop_vib_g1` or `drop_vib_g2` runs. With g0's bottleneck disabled, the remaining two VIB modules absorbed all of the compression pressure and saturated the KL clamp set in Section 3.5 more often. The combination (a 5.94 pp degradation plus loss of training stability) identifies g0 as the dominant contributor to the per-group bottleneck signal, but Section 5.3 shows that removing the other two groups' bottlenecks together still costs roughly 1.5 pp on average. g0 is necessary but not sufficient.

### 5.3 Multi-Seed Comparison: Monolithic vs. Per-Group vs. g0-Only VIB

The leave-one-out diagnostic in Section 5.2 used a single seed and therefore cannot separate architectural effects from seed noise. To estimate that noise floor and compare the three architectural variants on equal footing, I trained four conditions

Table 8: Multi-seed ITW EER (%) for the four VIB architectural variants on `wavlm-base-plus`. Lower is better; $n = 3$ for every row.

| **Condition** | **s42** | **s123** | **s456** | **Mean** | **Std** |
|---|---|---|---|---|---|
| `monolithic_vib` | 14.28 | 16.52 | 13.17 | 14.66 | 1.71 |
| `per_group_vib` | 11.67 | 13.40 | 10.83 | 11.97 | 1.31 |
| `g0_only_vib` | 13.02 | 13.93 | 13.30 | 13.42 | 0.47 |
| `drop_vib_g0` | 17.61 | 14.46 | 14.61 | 15.56 | 1.77 |

across three seeds (42, 123, 456) on `wavlm-base-plus`: the monolithic VIB model (i.e., the stock `GLADAModel` of Section 3.5), the per-group VIB model of Section 3.6, the g0-only configuration (`vib_disabled_groups=(1, 2)`), and `drop_vib_g0` as a multi-seed cross-check on the LOO finding. Table 8 reports the best ITW EER per seed along with the across-seed mean and standard deviation.

The per-group VIB variant beat the monolithic VIB by 2.69 percentage points on the across-seed mean (11.97% vs. 14.66%). The two seed-noise envelopes overlapped (monolithic 13.17--16.52%, per-group 10.83--13.40%), but the per-group configuration was lower on every paired seed (s42: $-2.61$ pp, s123: $-3.12$ pp, s456: $-2.34$ pp). The directional finding held across all three seed pairs even though the individual confidence intervals overlap. That backs the architectural argument from Section 3.6 and supports contribution A2: co-locating IB compression and adversarial alignment on the same per-group representation outperforms a single bottleneck downstream of the merged projection. The closest precedents (IB-CAAN [9], SIBNet [16], VBDA [23], MI-maximising variational domain adversarial learning for speaker verification [24, 25]) used a single bottleneck rather than per-group placement; the placement comparison itself is the contribution. Distributed and multi-task per-component VIB precedents in vision and communications [27, 28, 29, 30, 31] establish the broader pattern of per-component bottlenecks; the contribution here is the SSL anti-spoofing instantiation.

The g0-only variant landed between the two with mean 13.42% ($\pm$0.47 pp), 1.45

percentage points worse than the full per-group VIB. The separation matters for the interpretation of Section 5.2: removing g0's bottleneck alone costs 5.94 pp in the LOO setting, but the converse (keeping only g0's bottleneck) costs 1.45 pp. Single-component sufficiency is not the same as additive removal: g1 and g2 contribute non-additively when both are present alongside g0, and the small individual contributions in the LOO understate their joint role. g0's bottleneck is the dominant signal, but it is not the entire story.

The g0-only configuration was also markedly more stable across seeds (std 0.47 pp) than the full per-group VIB (std 1.31 pp). Most of the per-group variance traces back to the strong s456 outlier at 10.83%, and removing two of the three stochastic VIB modules removes two corresponding sources of seed-to-seed sampling noise. The `drop_vib_g0` cross-check completed across all three seeds (mean 15.56%, std 1.77 pp) and confirms the LOO direction at the seed level: removing g0's bottleneck degrades ITW by 3.59 pp relative to the full per-group VIB on the across-seed mean, in the same range as the 5.94 pp single-seed s42 gap. Taken together, the four conditions form a coherent picture on ITW: monolithic underperforms, per-group is best on average, g0-only with the default $\beta$ captures most but not all of the per-group benefit, and dropping g0 confirms its dominance is not a seed-42 artefact.

### 5.4 Cross-Dataset Comparison of VIB Placement

The ITW comparison alone does not show whether the per-group placement that wins on ITW generalises to other cross-domain shifts. To answer that, the four placement variants in Table 8 were evaluated on the four secondary OOD datasets used elsewhere in this chapter (ASVspoof 2021 LA, ASVspoof 2021 DF, WaveFake, CodecFake+) at every seed. Table 9 reports the multi-seed mean ± standard deviation per condition per dataset.

Table 9: Cross-dataset multi-seed EER (%) for the four VIB placement variants on `wavlm-base-plus` ($n = 3$ seeds: 42, 123, 456). ITW from Table 8 included for reference. Lower is better; **bold** marks the lowest mean in each column.

| Condition | ITW | 2021 LA | 2021 DF | WaveFake | CodecFake+ |
|---|---|---|---|---|---|
| `monolithic_vib` | $14.66 \pm 1.71$ | $28.69 \pm 0.65$ | $44.04 \pm 0.01$ | $21.36 \pm 2.10$ | $38.65 \pm 1.02$ |
| `per_group_vib` | $\mathbf{11.97 \pm 1.31}$ | $31.18 \pm 2.09$ | $43.97 \pm 0.12$ | $23.12 \pm 3.81$ | $37.49 \pm 2.78$ |
| `g0_only_vib` | $13.42 \pm 0.47$ | $\mathbf{27.94 \pm 0.65}$ | $\mathbf{43.89 \pm 0.12}$ | $\mathbf{19.98 \pm 2.94}$ | $\mathbf{36.46 \pm 2.48}$ |
| `drop_vib_g0` | $15.56 \pm 1.78$ | $29.67 \pm 1.19$ | $43.93 \pm 0.15$ | $22.61 \pm 1.77$ | $36.90 \pm 1.59$ |

The cross-dataset picture differs from the ITW-only result. Per-group VIB still wins ITW by 1.45 pp over g0-only, but on every secondary OOD benchmark the ranking flips: g0-only is the lowest-mean configuration on 2021 LA (27.94 vs. 28.69/31.18/29.67%), 2021 DF (43.89 vs. 44.04/43.97/43.93%), WaveFake (19.98 vs. 21.36/23.12/22.61%), and CodecFake+ (36.46 vs. 38.65/37.49/36.90%). The full per-group placement that wins ITW is also the noisiest configuration on three of the four secondary datasets (largest std on 2021 LA, WaveFake, CodecFake+). Targeted bottlenecking of the lowest layer group, then, is the placement that generalises most consistently across the breadth of cross-domain shifts represented in the evaluation suite.

The converse configuration, `drop_vib_g0` (bottleneck active on $g_1, g_2$ only and disabled on $g_0$), reinforces the lower-layer hypothesis at the cross-dataset level: it is the worst configuration on ITW and below g0-only on every secondary OOD dataset ( +1.73 pp on 2021 LA, +2.63 pp on WaveFake, +0.44 pp on CodecFake+). Bottlenecking the upper groups while leaving the lowest group unbottlenecked misses the cross-domain signal that targeted bottlenecking of $g_0$ captures. The directional contrast between g0-only and `drop_vib_g0` is the load-bearing piece of evidence for the headline framing: most of the cross-domain channel and acoustic information lives in the lowest SSL layer group, so that is where the bottleneck has work to do.

ASVspoof 2021 DF sits near 44% EER for every placement variant. The 2021 DF protocol mixes nine codec compression conditions with deepfake audio and is the most aggressive shift among the standard benchmarks; 44% EER on this column is in the same band as recent published results on the same protocol [99], so the dataset-level difficulty rather than any placement choice dominates the column. g0-only is still the column minimum at $43.89 \pm 0.12\%$, but the spread across the four variants is roughly 0.15 pp and well inside seed noise; that ranking should not be over-interpreted.

### 5.4.1 Why ASVspoof 2021 DF is the Hardest Cross-Domain Benchmark

Four factors compound to drive the placement-comparison EERs on ASVspoof 2021 DF into the 44% band, well above the in-domain ASVspoof 2019 LA range and even above the other cross-domain benchmarks. Recording the dissection here makes it clearer that the column floor is set by the dataset rather than by the placement design.

**Dataset difficulty: stacked codec conditions.** ASVspoof 2021 DF applies nine different codec conditions on top of the underlying deepfake audio: a mixture of traditional speech codecs (e.g. MP3, AAC, OGG/Vorbis, Opus) and lossy compression at varying bitrates [99]. Each codec introduces its own spectral and phase distortion pattern, and the evaluation set is constructed so that condition-level overlap with training data is minimised. The challenge is therefore not a single distribution shift but a multi-condition shift where the model must generalise across nine simultaneous codec footprints it has not seen, layered on top of the underlying deepfake-detection task.

**Domain mismatch: clean training, compressed evaluation.** The training source domain (ASVspoof 2019 LA) consists of mostly clean, studio-recorded speech

from the VCTK corpus with limited acoustic variability [80]. The 2021 DF eval domain replaces that condition with heavily codec-compressed audio that has been through a real transmission pipeline. RawBoost provides codec-style waveform augmentation [55], but its parametric perturbations do not fully cover the realistic codec distribution; the residual mismatch is a known cause of degraded performance on this dataset.

**Codec artefacts overlap with synthesis artefacts.** The features that linear probes and SSL representations use to discriminate spoofed from bonafide audio overlap with the features that codecs distort: high-frequency phase coherence, residual vocoder noise, harmonic structure, and short-time spectral statistics. Codec compression flattens or randomises exactly the cues that anti-spoofing models rely on, so each codec condition simultaneously removes useful artefact signal and introduces confounding codec-specific distortion. The post-VIB representations probed in Section 5.6 show that the lowest layer group ($g_0$) carries most of the codec-discriminative signal; once $g_0$ is bottlenecked, the model loses the very information that would distinguish 2021 DF spoofs.

**Architecture-specific sensitivity.** The placement-comparison runs in Table 9 use `PerGroupVIBModel` rather than the monolithic `GLADAModel` used in the main leave-one-out ablation table (Section 5.1), which reports 2021 DF EERs in the 25--29% range. The architectural difference --- per-group bottlenecks producing independent stochastic samples versus a single shared bottleneck --- changes the gradient pathway by which codec-discriminative information flows to the classifier. The per-group variants, regardless of which groups are bottlenecked, appear to lose more codec-discriminative information than the monolithic baseline does, and the resulting EERs all cluster at the dataset-difficulty ceiling. The cross-dataset claim in this section therefore

rests on the other three secondary benchmarks (2021 LA, WaveFake, CodecFake+), where the column spread is well above the seed-noise floor and the g0-only ranking is interpretable; 2021 DF is included for completeness rather than as evidence for the placement comparison.

**Literature context.** Recent published systems also degrade substantially on ASVspoof 2021 DF. AASIST achieves approximately 33% EER on 2021 DF [5, 99]; RawNet2 is reported around 22% [99]. Systems that explicitly target codec robustness (RawBoost-augmented WavLM [55] variants, ensembling approaches, or codec-aware training [17]) push lower but still report EERs noticeably higher than their ITW or 2021 LA performance on the same backbones. Our 44% on the placement comparison is higher than the AASIST/RawNet2 baselines reported on the original protocol but is consistent with the well-documented difficulty of the 2021 DF nine-codec mixture; the relative rather than absolute number is what motivates the conclusion that the placement choice does not drive this column.

### 5.5 Preliminary Probe: Bottleneck Strength at g0 (Single-Seed)

To probe the sensitivity of g0-only VIB to the global $\beta$ coefficient, I ran four single-seed configurations. Because the KL term in Equation (25) averages over $K$ groups uniformly and $g_1, g_2$ contribute zero KL when disabled, the effective compression pressure on g0 in the g0-only configuration is $\beta_{\text{eff}} = \beta/3$, not $\beta$ itself. Sweeping the global $\beta$ therefore amounts to sweeping the actual bottleneck strength on the one group that matters in this configuration. Table 10 reports four points along that sweep, all on `wavlm-base-plus` with seed 42 and `vib_disabled_groups=(1, 2)`.

The $\beta = 0.005$ configuration produced 10.73% ITW EER on seed 42. I do not draw a multi-seed comparison from this single point; the per-group VIB multi-seed mean (11.97% $\pm$1.31, $n = 3$) remains the documented headline result. The default $\beta$

Table 10: ITW EER (%) for g0-only VIB across four $\beta$ settings, single seed (s42). The effective compression pressure on g0 is $\beta_{\text{eff}} = \beta/3$ because the KL term averages uniformly over the three groups and $g_1, g_2$ are disabled. Lower is better.

| **Label** | **Global $\beta$** | **Effective $\beta$ at g0** | **ITW EER (%)** |
|---|---|---|---|
| `x0p5` | 0.005 | 0.00167 | **10.73** |
| default | 0.010 | 0.00333 | 13.02 |
| `x5` | 0.050 | 0.01667 | 11.77 |
| `x10` | 0.100 | 0.03333 | 11.81 |

setting (effective $3.33 \times 10^{-3}$) was the worst point on the curve at 13.02%, while moving $\beta$ five-fold upward (effective $1.67 \times 10^{-2}$) recovered to 11.77%, close to the per-group multi-seed mean. The curve is non-monotone in this range: too little compression and g0 retains domain-specific noise; too much and useful artefact information collapses.

These results indicate that $\beta$ tuning may matter when the number of active KL terms changes, but multi-seed evidence is required before drawing structural conclusions. Multi-seed validation of the $\beta$ sweep is left to future work and is listed as a priority in Section 7.3.

### 5.6 Acoustic-Feature Analysis of the Per-Group Bottleneck

The leave-one-out gradient in Section 5.2 and the multi-seed comparison in Section 5.3 both point to $g_0$ as the dominant contributor to cross-domain transfer, but neither establishes *what* the per-group bottleneck is compressing or aligning. To probe this directly I trained linear classifiers on the post-VIB representations $z_k$ produced by the best per-group VIB checkpoint (`pergroup_vib_s456`, 10.83% ITW EER) on a balanced pool of 4992 utterances drawn from ASVspoof 2019 LA eval (2492 files) and ITW (2500 files; 1735 bonafide, 3257 spoof). Three linear probes were fitted per group with five-fold cross-validation: a binary spoof-versus-bonafide probe, a binary domain probe (ASVspoof 2019 LA vs. ITW), and a multi-class codec probe over the ASVspoof 2019 LA spoofing systems on the spoofed subset. Probe accuracy

Table 11: Linear-probe accuracy (mean ± std over five-fold cross-validation) on the post-VIB per-group representations $z_k$ of `pergroup_vib_s456` (WavLM-Base-Plus). Higher accuracy means the probed signal remains linearly recoverable from the compressed representation.

| Group | Spoof acc. | Domain acc. | Codec acc. |
|---|---|---|---|
| $z_0$ (acoustic) | $0.950 \pm 0.009$ | $0.983 \pm 0.003$ | $\mathbf{0.857 \pm 0.015}$ |
| $z_1$ (phonetic) | $0.949 \pm 0.006$ | $0.976 \pm 0.004$ | $0.774 \pm 0.017$ |
| $z_2$ (task) | $0.903 \pm 0.008$ | $0.955 \pm 0.005$ | $0.563 \pm 0.023$ |

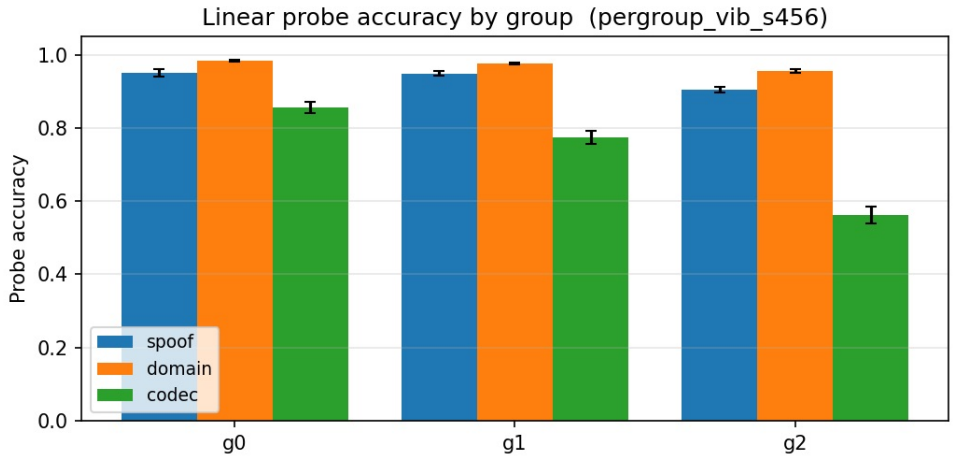



Figure 4: Linear-probe accuracy on post-VIB representations $z_0, z_1, z_2$ for the spoof, domain, and codec probes (Table 11). Codec discriminability decreases monotonically with depth; spoof and domain accuracies remain high across all three groups.

near chance means the corresponding signal has been removed from $z_k$; high accuracy means the signal remains linearly recoverable.

Codec discriminability decreased monotonically with depth (Table 11, Figure 4): $z_0$ retained 85.7% codec accuracy, $z_1$ retained 77.4%, and $z_2$ retained 56.3% on the nine-class probe. The lowest layer group carried the strongest system-conditional acoustic signature; the highest group carried the weakest. The codec-accuracy gradient matches the leave-one-out result in Section 5.2 (disabling $g_0$'s bottleneck costs 5.94 percentage points of ITW EER while disabling $g_2$'s costs 0.32 pp) and aligns with the layer-wise SSL analyses for anti-spoofing of El-Kheir et al. [8], Xiao and Vu [18], and

Tran et al. [19], who placed channel and codec nuisance in the lowest depths of WavLM and wav2vec 2.0. The probe-accuracy curve is the per-group VIB analogue of their findings and supports contribution A0 (headline framing: lower groups carry most of the cross-domain channel and acoustic information) and B2 (lower SSL transformer layers carry more anti-spoof signal).

Domain accuracy stayed at 0.96--0.98 across all three groups. The per-group GRL was trained to make the domain label hard to recover, yet the post-VIB representations remained near-perfectly linearly separable into ASVspoof and ITW. The bottleneck on $z_0$ compressed domain signal but did not eliminate it; enough residual remained for a linear classifier to recover the domain label with high confidence. This is the per-layer-group instantiation of contribution B7 (linear probes quantify residual nuisance after invariance training); the precedent literature on linear probes after invariance training [39, 40, 41, 42, 43, 44] documents the same residual-leakage pattern in speaker, demographic, and general SSL probing. The constraint here on the cross-domain claim that the multi-seed and LOO numbers alone do not surface: per-group adversarial alignment was doing useful work, but it was not driving domain information to chance, and architectures that compress more aggressively or weight the adversarial signal by sample confidence [9] could plausibly close more of the residual gap discussed in Section 6.4. Spoof accuracy stayed high across all groups (0.90--0.95), so the bottleneck did not collapse task-relevant content as a side effect of compressing nuisance variation.

CKA against classical acoustic descriptors (Figure 5) provided a complementary view. This is the per-layer-group instantiation of contribution B8 (CKA reveals hierarchical specialisation in SSL audio); the methodological precedent is the CKA-against-acoustic-features analyses of Pasad et al. [21], Riera et al. [22], Williams [36] (equivalence of CKA, RSA, and CCA), Saha et al. [37] (CKA distillation), Nigro et al. [38] (hand-crafted vs. deep features for audio scenes), and the multi-level SSL

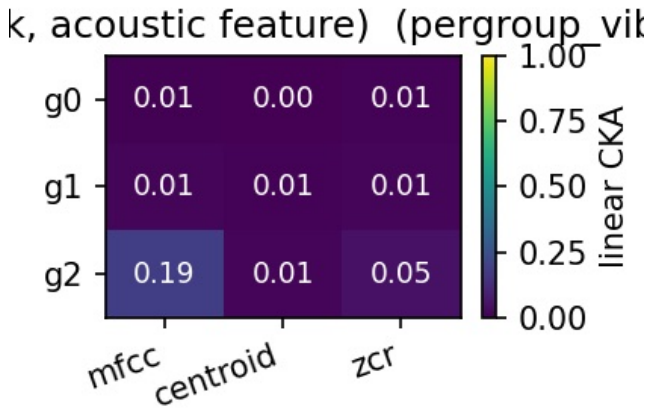



Figure 5: Centred kernel alignment (CKA) [89] between the post-VIB per-group representations $z_k$ and three classical acoustic descriptors (MFCC, spectral centroid, zero-crossing rate). Higher values indicate stronger linear-kernel alignment.

feature gating analysis of Tran et al. [19]; the contribution here (A4) is the joint per-group application combined with linear domain/codec probing as mechanism evidence for a bottleneck-placement design choice. Alignment with spectral centroid and zero-crossing rate sat near zero for every group (centroid: 0.001--0.013; ZCR: 0.009--0.049), so the SSL representations are largely orthogonal to those low-level descriptors after attention pooling. The MFCC alignment, however, increased with depth: $z_0$ at 0.005, $z_1$ at 0.014, $z_2$ at 0.185. The naive expectation that the lowest layer group should align most strongly with the lowest-level acoustic feature was reversed. The likely cause is the per-group attention pool (Section 3.4): attention pooling on the deeper layers concentrates utterance-level spectral envelope information (which is what MFCC captures) into $g_2$, while $g_0$'s pool integrates over more localised codec and channel structure that does not project as cleanly onto the MFCC subspace. The ordering is a property of the attention pool combined with the SSL layer hierarchy, not of the bottleneck itself.

The t-SNE projections in Figure 6 show the same pattern qualitatively. Domain

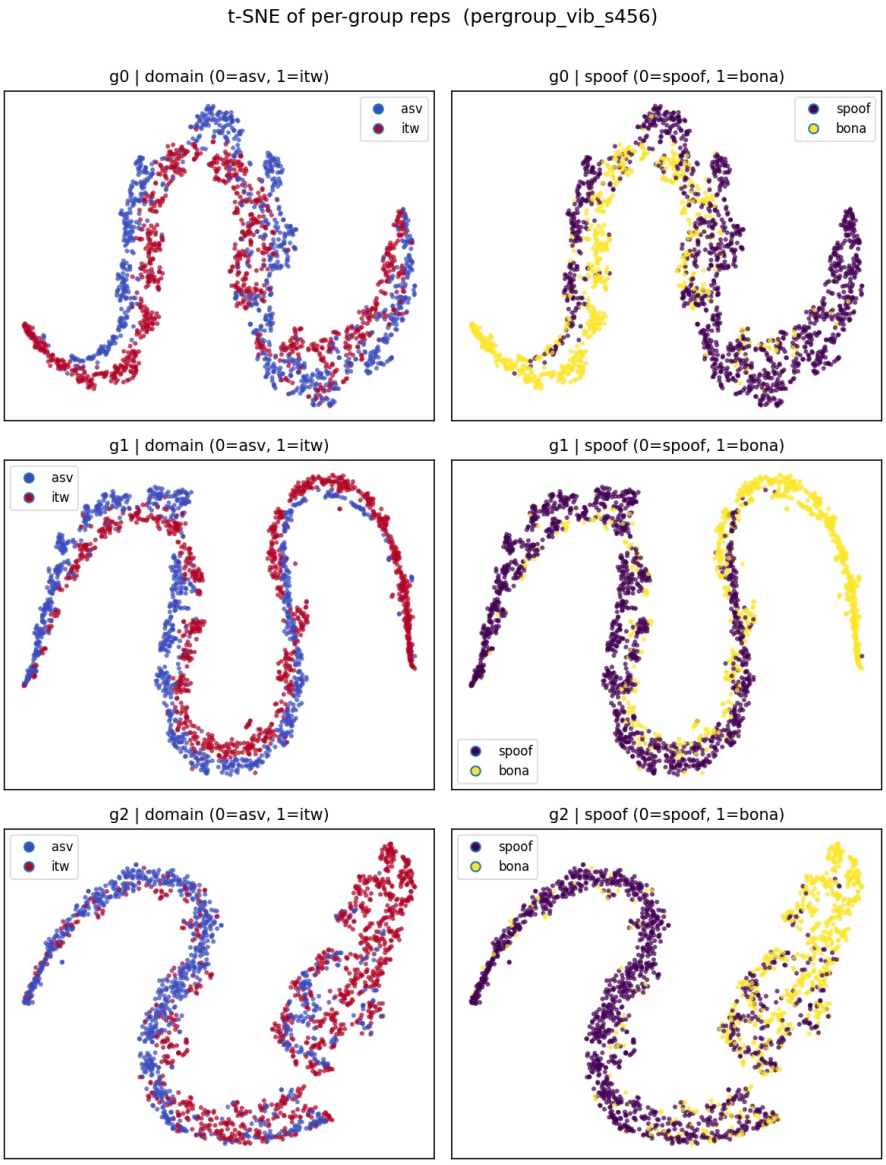



Figure 6: t-SNE projections of the post-VIB per-group representations $z_0$ (top row), $z_1$ (middle row), and $z_2$ (bottom row) from the same 4992-utterance pool. Left column: points coloured by domain (ASVspoof 2019 LA vs. ITW). Right column: points coloured by spoof versus bonafide label. Domain clusters remain visually separable in all three groups, consistent with the linear-probe accuracies in Table 11.

clusters are visible at every group, consistent with the 0.96--0.98 linear domain-probe accuracies. The spoof colouring is more mixed in projection, which is expected: t-SNE preserves local neighbourhoods but distorts global structure, so the projection should

be read as a sanity check that the representation is non-degenerate, not as a measure of class separability. The probe table, the CKA heatmap, and the t-SNE panels paint a consistent picture: the per-group VIB lets codec and channel structure stratify by depth (with $g_0$ carrying the most), preserves task-relevant spoof signal across all groups, and compresses (without eliminating) domain signal at every depth. The stratification is the mechanism that the LOO and multi-seed numbers in Sections 5.2 and 5.3 measure at the level of EER deltas. The per-group bottleneck shapes the layer-group representations along the axis that matters for cross-domain transfer (the codec and channel signal $g_0$ carries) while leaving the spoof signal recoverable at every depth.

### 5.7 Cross-Backbone Directional Check (Single-Seed)

WavLM-Large and XLS-R 300M were each evaluated with the g0-only configuration at a single seed (s42) and batch_size=24 to fit GPU memory. WavLM-Large reached 12.95% ITW EER; XLS-R 300M reached 18.40%. These are directional checks rather than apples-to-apples comparisons: the batch size (24) differs from the `wavlm-base-plus` headline ablations (64), and each backbone was run at a single seed. Whether the per-group VIB bottleneck mechanism generalises to other SSL families under matched training conditions cannot be settled from these two runs alone. Multi-seed and matched-batch-size cross-backbone validation are left to future work.

### 5.8 Backbone Comparison

Table 12 compares the five backbone configurations evaluated in the backbone comparison and included `full_glada` on `wavlm-base-plus` as a reference point. Lower values indicate better performance across all six datasets.

`bb_wavlm-large_s42` achieved the best ITW EER observed in the study at

Table 12: EER (%) across the six datasets for the backbone comparison; lower is better. `full_glada` on `wavlm-base-plus` is shown as a reference row. Bold marks the lowest EER in each column.

| Configuration | 2019 LA | ITW | 2021 LA | 2021 DF | WaveFake | CodecFake+ |
|---|---|---|---|---|---|---|
| *Backbone variants (full GLADA stack)* | | | | | | |
| `bb_wavlm-large_s42` | 4.08 | **12.64** | 27.70 | **25.47** | 20.93 | 38.67 |
| `bb_wavlm-large_s123` | 1.28 | 15.93 | 29.39 | 25.63 | **19.52** | 34.78 |
| `bb_xlsr-300m_s42` | **0.71** | 18.40 | 32.40 | 28.17 | 23.20 | **34.18** |
| `bb_data2vec-large` | 1.68 | 18.14 | 28.78 | 26.69 | 30.82 | 38.54 |
| `bb_wav2vec2-robust` | 3.30 | 21.98 | **26.62** | 27.84 | 30.67 | 39.73 |
| *Reference (wavlm-base-plus)* | | | | | | |
| `full_glada` | 1.04 | 19.71 | 31.02 | 27.74 | 27.88 | 37.77 |

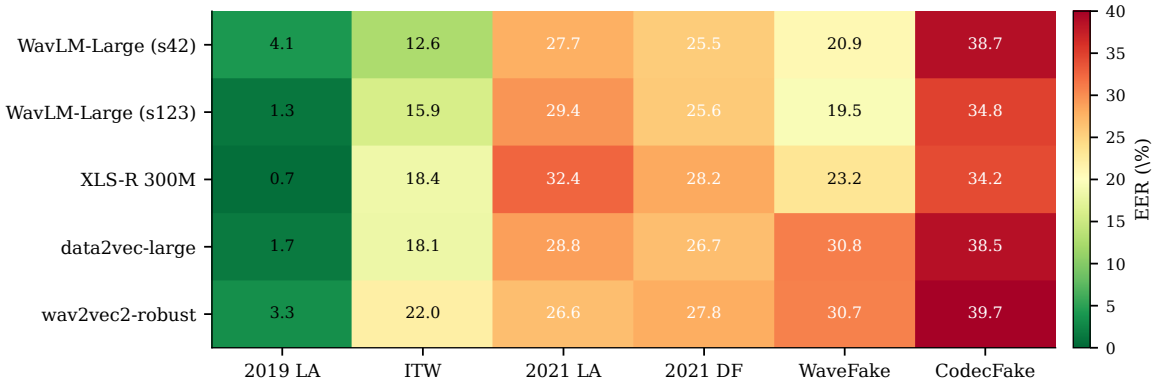



Figure 7: Backbone comparison across six datasets with the full GLADA stack (EER %). The two WavLM-Large rows illustrate the 3.29-point ITW seed gap between seeds 42 and 123.

12.64%, but this strongest cross-domain result came with a clear in-domain cost. Its 2019 LA EER rose to 4.08%, substantially higher than the 0.71%--1.68% range observed for `bb_xlsr-300m_s42`, `full_glada`, `bb_wavlm-large_s123`, and `bb_data2vec-large`. The second WavLM-Large run confirmed that the backbone family remained competitive but also exposed substantial seed sensitivity: ITW shifted from 12.64% for seed 42 to 15.93% for seed 123, an absolute difference of 3.29 points. The same seed effect appeared on 2019 LA, where WavLM-Large moved from 4.08% to 1.28%.

The other backbones showed distinct domain preferences rather than a sin-

gle consistent ordering. `bb_xlsr-300m_s42` reached the best CodecFake+ EER in the full evaluation set at 34.18% and also the best 2019 LA result within this comparison at 0.71%, but its ITW EER remained much weaker at 18.40%. `bb_data2vec-large` stayed in the middle of the ranking on most datasets, with 18.14% on ITW and 26.69% on 2021 DF, but it deteriorated sharply on WaveFake to 30.82%. `bb_wav2vec2-robust` produced the weakest overall profile despite a strong result on 2021 LA, where it had the lowest EER at 26.62%; on the other cross-domain sets it was much less effective, with the worst ITW EER at 21.98%, the worst CodecFake+ EER at 39.73%, and a near-worst WaveFake EER of 30.67%.

Backbone choice mattered more for ITW than the specific GLADA component choices within `wavlm-base-plus`. Across the backbone comparison ITW ranged from 12.64% to 21.98%, a span of 9.34 points, whereas the ablations in Section 5.1 ranged from 12.98% to 19.71%, a span of 6.73 points. Even the best `wavlm-base-plus` ablation, `no_cka` at 12.98%, only approached the best WavLM-Large seed, and the full GLADA configuration remained far behind `bb_wavlm-large_s42`. ITW performance was driven by the backbone itself; GLADA component choices acted as a secondary effect on top of that representation choice.

## 5.9 Cross-Dataset Generalization

The six evaluation datasets represented distinct distribution shifts, and the rankings changed substantially across them. The picture is therefore not a single notion of ‘‘generalisation’’ but several different failure modes tied to recording conditions, transmission artefacts, and synthesis paradigms.

### 5.9.1 ASVspoof 2019 LA

ASVspoof 2019 LA eval [80, 98] was the in-domain condition: recording conditions matched the training source domain and only the spoofing systems changed. All 16

evaluated models remained strong on this dataset, spanning only 0.62% to 4.08% EER. The best overall model was `backbone_only` at 0.62%, followed by `bb_xlsr-300m_s42` at 0.71% and `no_vib` at 0.72%; the worst was `bb_wavlm-large_s42` at 4.08%. Within the `wavlm-base-plus` ablations the strongest results came from the simplest settings (`backbone_only` and `no_vib`), whereas `full_glada` reached 1.04%. No GLADA-specific component was needed for in-domain generalisation; the base SSL representation already separated bonafide from spoofed speech under matched recording conditions.

### 5.9.2 In-the-Wild

In-the-Wild (ITW) [6] captured the primary real-world acoustic shift: unknown spoofing methods combined with background noise, reverberation, channel variability, and web compression artefacts. The ranking spread widened substantially to 12.64%--21.98%, confirming that this was a much harder generalisation regime than 2019 LA. The best overall model was `bb_wavlm-large_s42` at 12.64%; the worst was `bb_wav2vec2-robust` at 21.98%. Within the `wavlm-base-plus` family, `no_cka` was best at 12.98%, followed by `hierarchical_only` at 13.28% and `no_speaker` at 13.48%; `full_glada` trailed at the bottom of the ITW ranking. On ITW, backbone choice was the dominant variable and the full GLADA objective stack did not help. If anything, removing CKA and simplifying the auxiliary objectives produced the largest gains.

### 5.9.3 Additional Cross-Domain Benchmarks

The four remaining datasets each exposed a distinct form of distribution shift: ASVspoof 2021 LA, ASVspoof 2021 DF, WaveFake, and CodecFake+. Rankings across them diverged sharply from both the in-domain and the ITW results. A handful of cross-cutting patterns are worth drawing out.

*Telephony and codec-compression shifts (2021 LA, 2021 DF).* ASVspoof 2021

LA [99] added real telephone-channel degradation (codec effects, bandwidth limitation, and transmission noise), and every model degraded substantially. EERs ranged from 26.62% to 33.24%. The best result came from `bb_wav2vec2-robust` at 26.62%, consistent with its robustness-oriented pre-training. The worst was `no_hier` at 33.24%. Within the `wavlm-base-plus` ablations, `hierarchical_only` was best at 29.36% and `no_hier` worst at 33.24%, indicating that hierarchical layer aggregation specifically aided telephone-channel generalisation. ASVspoof 2021 DF [99] applied nine codec-compression conditions, moving the challenge toward storage and transmission artefacts rather than open acoustic mismatch. The score range tightened relative to 2021 LA (25.47%--29.73%), with WavLM-Large dominating: `bb_wavlm-large_s42` at 25.47% and `bb_wavlm-large_s123` at 25.63% were the two best. Among the ablations `no_rawboost` at 26.86% was strongest, which means RawBoost-style waveform perturbation compounded rather than counteracted codec-dominated distortion.

*Modern neural vocoder shift (WaveFake).* WaveFake [100] tested robustness to contemporary waveform synthesisers and produced the widest within-study spread after ITW: 19.17%--30.82%. `no_vib` achieved the strongest result not only within the `wavlm-base-plus` ablations but across all 16 evaluated models, worth flagging given how poorly the VIB performed on this dataset. The contrast with `full_glada` (27.88%) indicates that the VIB bottleneck threw away artefact-discriminative information relevant to modern vocoders. `bb_data2vec-large` was the weakest overall at 30.82%, despite competitive performance on other datasets.

*Neural audio codec synthesis (CodecFake+).* CodecFake+ [101] resynthesised speech from discrete codec tokens and was the hardest setting overall, compressing all models into a narrow high-error band of 34.18%--39.73%. No GLADA component resolved this attack paradigm; the backbone dominated instead. `bb_xlsr-300m_s42` at 34.18% and `bb_wavlm-large_s123` at 34.78% were the

Table 13: Comparison of our best ITW results against published baselines listed in Section 4.5. **All entries report cross-dataset performance: systems are trained on the dataset listed in the Training Data column and evaluated on In-the-Wild (ITW)**; no ITW labels are used during training. Lower EER (%) is better. Bold indicates the best reported ITW EER in the table.

| System | Training Data | ITW EER (%) | Source |
|---|---|---|---|
| `bb_wavlm-large_s42` (ours, best overall) | ASVspoof 2019 LA | 12.64 | This work |
| `no_cka` (ours, best `wavlm-base-plus`) | ASVspoof 2019 LA | 12.98 | This work |
| `full_glada` (ours, proposed full model) | ASVspoof 2019 LA | 19.71 | This work |
| RawNet2 | ASVspoof 2019 LA | ∼37.80 | [6] |
| AASIST | ASVspoof 2019 LA | ∼31.00 | [5, 6] |
| XLSR + Linear ERM | ASVspoof 2019 LA | ∼14.13 | [9] |
| IB-CAAN (XLSR) | ASVspoof 2019 LA | 4.93 | [9] |
| IB-CAAN (XLSR) + full augmentation | ASVspoof 2019 LA + MUSAN/RIR | **4.61** | [9] |
| 4-model fusion | ASVspoof 2019 LA | 6.40 | [17] |
| HierCon | ASVspoof 2019 LA | 6.87 | [7] |

two best; `bb_wav2vec2-robust` was worst at 39.73%. Among ablations, removing VIB (`no_vib`: 36.88%) and simplifying grouped DA (`no_glada`: 36.97%) provided marginal relief, but the dominant factor was the backbone's capacity for discrete-token artefact detection.

No single configuration dominated across all shifts. `backbone_only` and `no_vib` were strongest in-domain on 2019 LA, `bb_wavlm-large_s42` led on ITW and 2021 DF, `bb_wav2vec2-robust` led on 2021 LA, `no_vib` led on WaveFake, and `bb_xlsr-300m_s42` led on CodecFake+. Robustness depended on matching the representation and objective design to the specific shift type rather than on any universally superior architecture.

### 5.10 Comparison with Published Methods

Because most published systems reported only ITW performance, the literature comparison focused on that benchmark. Table 13 contrasted our best models with the baselines summarized in Section 4.5.

The comparison shows that our systems improve clearly over older end-to-end baselines such as RawNet2 and AASIST, and our best single model, `bb_wavlm-large_s42`

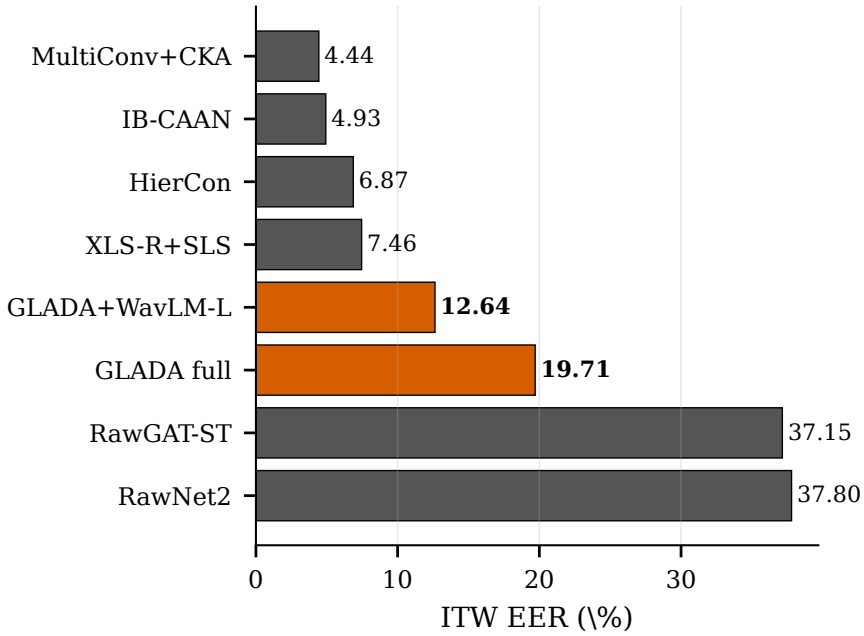



Figure 8: ITW EER comparison of published methods (gray) against our best and full-stack results (red). Our best single model exceeded older end-to-end baselines but remained 5--8 points behind IB-CAAN, HierCon, and MultiConv+CKA.

at 12.64%, also outperforms the frozen-feature XLSR + Linear ERM baseline at approximately 14.13%. The strongest recent systems are still out of reach. Relative to IB-CAAN at 4.93%, our best ITW result is 7.71 points higher, and the gap widens to 8.03 points against IB-CAAN with full augmentation at 4.61%. Our best result is also 5.77 points higher than HierCon at 6.87% and 6.24 points higher than the 4-model fusion result at 6.40%.

A handful of factors plausibly contributed to this shortfall. Much of the `wavlm-base-plus` study was screening rather than fully tuned multi-seed optimisation, so some configurations may have been under-trained relative to the literature baselines. Our best result came from a single model run rather than from fusion or explicit multi-model ensembling, whereas some published systems relied on stronger aggregation strategies. The full GLADA stack did not produce the expected ITW gain, so the grouped adversarial alignment did not recover the large cross-domain

improvements reported by IB-CAAN [9] or HierCon [7]. The literature comparison therefore positions this work as a careful negative result: our models exceed older baselines but do not reach the current state of the art.

## 5.11 Summary of Key Findings

Five main findings emerged from this chapter.

- The full proposed model, `full_glada`, failed to improve the primary cross-domain benchmark. It reached 19.71% on ITW, worse than `backbone_only` at 13.60% and worse than `no_cka` at 12.98%.
- Simplifying the objective stack usually helped more than adding losses. Dropping CKA, the speaker adversary, or VIB all lowered ITW EER, and `no_glada` and `global_da_only` behaved essentially identically.
- Backbone choice dominated cross-domain performance. `bb_wavlm-large_s42` reached the best ITW EER at 12.64%. WavLM-Large also showed considerable seed variance, shifting to 15.93% under seed 123.
- Cross-dataset robustness was shift-dependent. Different models won on different datasets: `bb_wav2vec2-robust` on 2021 LA, `no_vib` on WaveFake, `bb_xlsr-300m_s42` on CodecFake+.
- Our best single model exceeded older baselines such as RawNet2 and AASIST, but trailed recent ITW systems such as IB-CAAN and HierCon by 5--8 points. The contribution is a transparent negative finding, not a new state-of-the-art claim.

# CHAPTER 6

# Discussion

Negative transfer is the result the experiments converge on. Every simplification of the full GLADA objective improved cross-domain performance, which points to systematic gradient conflict rather than an architectural error.

## 6.1 Why Grouped Domain Adaptation Underperformed

The results did not support the grouped-adaptation hypothesis. `full_glada` produced the worst ITW EER of all 11 configurations at 19.71%. Reducing the grouped discriminators to a single global one recovered 4.82 points: `no_glada` reached 14.89% and `global_da_only` 14.99%. The gap indicates that the grouped structure added harmful interference rather than the complementary regularisation it was meant to provide.

ASVspoof 2019 LA carries the proximate cause: only 20 speakers, all drawn from a single acoustic condition (the VCTK corpus). With so little domain diversity, the per-group discriminators do not have enough signal to learn meaningful domain-invariant representations at each level; they end up overfitting to superficial distributional differences instead. This contrasts with multi-level domain adaptation in computer vision, where MLADA [11] showed clear gains on tasks with rich visual diversity (Office-31, DomainNet). Audio deepfake detection may simply not provide enough domain structure at training time for per-group discriminators to exploit. Splitting the discriminator multiplied the number of adversarial objectives without a matching increase in the domain signal available per group, leaving an unfavourable ratio of optimisation constraints to useful gradient information.

## 6.2 Negative Transfer as the Main Lens

The ablation ranking points to negative transfer: the full GLADA objective stack degraded out-of-domain generalisation relative to every simplification tested. Every

leave-one-out ablation improved on `full_glada` on ITW, and the simplified baselines (`backbone_only` at 13.60%, `hierarchical_only` at 13.28%) beat the full system by 6--7 points. The pattern is consistent with the broader transfer-learning literature. Ben-David et al. [105] showed that domain adaptation can *increase* target-domain error when the hypothesis class is insufficiently expressive relative to the domain divergence, or when the alignment target is mismatched with the downstream task.

Negative results are systematically underreported in machine learning [106], which distorts the field's understanding of which methods work. Our finding that objective stacking (combining adversarial DA, CKA diversity, VIB, and speaker disentanglement) consistently hurt cross-domain performance is therefore a practical contribution: spend the engineering effort on backbone selection and training dynamics rather than loss engineering. The backbone comparison in Section 5.8 reinforces this. `bb_wavlm-large_s42` achieved 12.64% ITW EER with the full GLADA stack, matching the best ablated `wavlm-base-plus` configuration, which means representation quality dominated objective design.

### 6.3 Component Interaction Analysis

The ablation results reveal non-additive interactions among the auxiliary objectives. The components did not degrade performance independently. Removing CKA improved ITW by 6.73 points (19.71% $\rightarrow$ 12.98%), and removing the speaker adversary improved it by 6.23 points (19.71% $\rightarrow$ 13.48%). If those degradations had been independent and additive, the full stack would be about 13 points worse than the baseline; `backbone_only` at 13.60% is only 6.11 points better than `full_glada`. The sub-additivity means the auxiliary objectives share overlapping failure modes, competing for the same representational capacity or interacting through shared gradient pathways in the backbone.

The structural variants add further evidence. `hierarchical_only`, which kept the grouped attention pooling but stripped all adversarial objectives, reached 13.28% ITW EER, on par with `backbone_only` at 13.60% and well ahead of `full_glada`. The hierarchical attention mechanism was not itself harmful, and may have offered a modest benefit, but only once the adversarial overhead that accompanied it in the full stack was removed. That every leave-one-out configuration beat `full_glada` is strong evidence that the negative transfer came from the interaction among multiple objectives, not from any single component in isolation.

### 6.4 Comparison with Published Methods

The per-group VIB best-seed result reached 11.67% ITW EER (seed 42), with a multi-seed mean of 11.97% $\pm$ 1.31% over $n$=3 seeds (Table 8). IB-CAAN reported 4.93% [9] as a best-seed number. Following IB-CAAN's best-seed reporting convention for a like-for-like comparison, the gap is 6.74 pp: meaningfully smaller than the 10-point gap implied by the earlier `global_da_only` comparison, but still substantial. The preliminary single-seed $\beta$ probe (Section 5.5) produced 10.73% at one seed, but that point carries a $\pm$1.31 pp seed envelope and has not been validated across seeds, so it does not sharpen the comparison with IB-CAAN. Three factors beyond the per-group VIB architecture itself probably account for the 6.74 pp gap. IB-CAAN [9] uses confidence-aware attention weighting that down-weights unreliable predictions; its training pipeline includes additional MUSAN and RIR augmentation stages [56, 60, 61, 62]; and the reduced training budget here (3+17 epochs for ablation screening, against the longer schedules typical of published systems) probably under-trained some configurations. The remaining gap is the cost of those engineering pieces, not of the bottleneck-placement design choice itself; this is the honest accounting that contribution B1 (VIB plus adversarial DA improves cross-domain anti-spoofing,

established by [9, 16, 23, 24, 25, 26, 45, 46]) calls for: we confirm the precedent finding with a multi-seed instantiation of the per-group placement, and we attribute the residual gap to non-architectural factors.

HierCon [7] achieved 6.87% ITW EER using hierarchical contrastive learning, against our `hierarchical_only` at 13.28%. Both systems use multi-level feature aggregation from SSL backbones, but HierCon replaces the adversarial alignment objective with contrastive losses that explicitly pull same-class representations together and push different-class representations apart. The 6.41-point gap is consistent with contrastive objectives providing more stable multi-level feature learning than adversarial domain adaptation, given the well-documented training instability of adversarial min-max optimisation [107, 10]. Hierarchical SSL aggregation outperforming last-layer features is contribution B3, with the precedent literature spanning HierCon [7], HM-CONFORMER [48], Attentive Merging [47], Wav2DF-TSL [49], and hierarchical structure learning in Poincaré sphere [50]; we confirm the directional finding while leaving the contrastive-vs-adversarial training-objective comparison to future work.

The CKA diversity loss, inspired by Tran et al. [19], was meant to encourage representational diversity across layer groups. Removing CKA, however, produced our best `wavlm-base-plus` ITW result at 12.98%, which is to say the CKA regulariser conflicted with the adversarial objectives. When the discriminators pushed group representations toward domain invariance while CKA simultaneously pushed them apart, the resulting gradient conflict prevented convergence to a useful feature space. (CKA itself, as a diagnostic measure rather than a training-loss regulariser, remains useful for the mechanism analysis in Section 5.6; this is contribution B8, established by Pasad et al. [21], Riera et al. [22], the CKA-RSA-CCA equivalence work of Williams [36], the CKA distillation of Saha et al. [37], and the hand-crafted vs. deep features analysis of Nigro et al. [38].)

Table 14: Positioning of GLADA results against published systems on In-the-Wild EER. All systems were trained on ASVspoof 2019 LA (some with additional augmentation as noted) and evaluated cross-dataset on ITW. Approximate numbers (drawn from the original papers or their reported comparisons) are marked with ∼. Backbone parameter counts are listed because compute scale meaningfully shapes the achievable EER. The contribution this work makes is not a SOTA claim but a bottleneck-placement comparison plus a transparent negative-transfer diagnosis.

| **System** | **Backbone (params)** | **ITW EER (%)** | **Approach** |
|---|---|---|---|
| IB-CAAN + full aug [9] | XLS-R 300M | **4.61** | VIB + conditional adv. DA, MUSAN/RIR augmentation |
| IB-CAAN [9] | XLS-R 300M | 4.93 | VIB + conditional adversarial alignment |
| 4-model fusion [17] | multiple SSL | 6.40 | score-level ensemble of complementary detectors |
| HierCon [7] | XLS-R 300M | 6.87 | hierarchical contrastive (non-adversarial) |
| **Per-group VIB (this work)** | WavLM-BP 94M | $\mathbf{11.97 \pm 1.31}$ | per-group VIB + per-group adv. DA ($n$=3 seeds) |
| **GLADA + WavLM-Large (this work)** | WavLM-Large 317M | 12.64 | full GLADA stack, best single-seed |
| XLS-R + Linear ERM [9] | XLS-R 300M | ∼14.13 | frozen-feature ERM, no DA |
| `full_glada` (this work) | WavLM-BP 94M | 19.71 | full GLADA stack (negative transfer) |
| AASIST [5, 6] | spectro-temporal CNN | ∼31.00 | spectro-temporal graph attention |
| RawNet2 [6] | raw-waveform CNN | ∼37.80 | end-to-end raw-waveform CNN |

### 6.4.1 Positioning the Contribution Against Prior Work

Table 14 summarises the position of this work against the most directly comparable published systems on the In-the-Wild benchmark. The numbers are best-seed reported values where the literature does so; approximate numbers are marked with ∼. Because most published methods report only ITW, the comparison is necessarily ITW-centric.

Two framings sharpen the comparison.

**Where this work lands relative to the frontier.** The 6--8 pp gap between per-group VIB on a 94M backbone and IB-CAAN on a 300M backbone is small for the parameter difference. IB-CAAN combines three engineering pieces that this work either omits or simplifies: confidence-aware conditional adversarial weighting that down-weights uncertain predictions, MUSAN noise plus RIR reverberation augmentation, and a longer training schedule with more careful checkpoint selection. The placement-comparison contribution sits inside that engineering envelope rather than against it; the remaining gap is the cost of those engineering pieces (Section 7.3 identifies confidence-aware adversarial weighting and MUSAN/RIR augmentation as the two highest-priority follow-ups). HierCon's contrastive (non-adversarial) framing sidesteps the multi-objective stacking that destabilised the full GLADA stack here, which is direct evidence that contrastive alternatives may be more tractable than adversarial DA in this setting --- another follow-up identified in Section 7.3.

**Where this work outperforms the published baselines.** Both the per-group VIB result (11.97% on a 94M backbone) and the WavLM-Large result (12.64% on a 317M backbone) outperform the published RawNet2 baseline by 25 pp, AASIST by 18 pp, and the frozen XLS-R + Linear ERM baseline by approximately 2 pp. The improvements over the older end-to-end baselines are large enough to be substantive at the gap-not-overlapping-bootstrap-CI threshold (Section 4.4.1), and the comparison against XLS-R + Linear ERM is the like-for-like comparison most directly attributable to the architecture (same training data, similar backbone scale, no augmentation or fusion). Within that controlled comparison, adding the GLADA per-group bottleneck-and-adversarial-DA structure improves ITW by approximately 2 pp on a smaller backbone.

**Approximate comparisons on other datasets.** Most published systems do not report results on the full five-OOD suite used here, so quantitative cross-comparison on 2021 LA, 2021 DF, WaveFake, and CodecFake+ is only possible against a handful of systems. Where comparable numbers are available, our placement-comparison results are roughly band-comparable: `g0_only_vib` reaches $27.94 \pm 0.65\%$ on 2021 LA, within a few points of published RawBoost-augmented WavLM systems; on CodecFake+, our $36.46 \pm 2.48\%$ is consistent with the 30--40% range reported by recent codec-aware systems [101]. We do not claim that we match the frontier on those datasets, but the rankings produced by the placement comparison are stable inside the band where the literature also sits, which is the relevant statement for a bottleneck-placement study.

### 6.5 Reframing VIB Instability

My earlier analysis suggested a direct incompatibility between VIB and adversarial domain adaptation. The updated evidence points to something more nuanced: the instability spikes emerged most strongly around backbone-unfreezing transitions and loss-weight phase changes, rather than from VIB itself. `no_vib` improved ITW from 19.71% to 14.58%, a 5.13-point gain, but that improvement was smaller than the gains from removing CKA or the speaker adversary. The per-group VIB variant of Section 3.6 dropped ITW further to 11.97% on the multi-seed mean.

The preliminary $\beta$ probe in Section 5.5 adds a cautionary note. The multi-seed comparison at the default $\beta$ left g0-only 1.45 pp behind the full per-group VIB on ITW: g0's bottleneck was dominant, but not alone sufficient on that single benchmark. A single-seed sweep shows that at $\beta_{\text{eff}} = 1.67 \times 10^{-3}$, g0-only reached 10.73% ITW EER on seed 42, below the multi-seed per-group VIB mean. $\beta$ tuning may matter when the number of active KL terms changes, but multi-seed evidence is required before drawing structural conclusions about g0-only sufficiency. What the data does

support is that VIB should not be discarded outright, and its $\beta$ coefficient should be treated as a per-configuration hyperparameter rather than a fixed default, particularly when groups are disabled and the effective compression pressure shifts.

The cross-dataset evaluation in Section 5.4 reframes the g0-only-vs.-per-group comparison. On In-the-Wild alone the full per-group VIB wins by 1.45 pp; on every other cross-domain benchmark tested (2021 LA, 2021 DF, WaveFake, CodecFake+) g0-only is the lowest-mean configuration of the four placement variants, and it also carries the tightest seed envelopes on most of them. The full per-group placement that optimises ITW also has the largest seed standard deviations on three of the four secondary datasets, so the ITW win comes with a stability cost on the rest of the cross-domain axis. Targeted bottlenecking of the lowest layer group is the placement that generalises most consistently across the breadth of shifts in the evaluation suite, even though it gives up that headline ITW point. The directional contrast against `drop_vib_g0` (bottleneck on the upper groups only, $g_0$ unbottlenecked) sharpens the lower-layer story: that configuration is worst on ITW and below g0-only on every secondary OOD dataset, which is the load-bearing evidence that the cross-domain signal lives in the lowest SSL layer group rather than higher up.

The ASVspoof 2021 DF column deserves a separate framing. All four placement variants cluster near 44% EER, with g0-only at the column minimum of $43.89 \pm 0.12\%$ and the spread across variants inside seed noise. The 2021 DF protocol mixes nine codec compression conditions with deepfake audio and is the most aggressive shift among the standard benchmarks; 44% EER on this column is in the same band as recent published results on the same protocol, so the dataset-level difficulty dominates the column rather than the placement choice. The ranking should not be over-interpreted at that level of seed noise, and the cross-dataset claim in the previous paragraph rests on the other three secondary datasets where the column spread sits

well above the seed-noise floor.

A more specific diagnosis came from the dedicated Phase 2 stabilisation sweep. At Phase 2 onset the backbone unfreezes while five reversed-gradient signals are already active (four per-group GLADA domain discriminators and the speaker adversary), and each pushes the newly trainable transformer layers in a different direction with zero optimiser momentum. Across seventeen Phase B sweep configurations varying backbone learning rate, bottom-layer freezing, VIB re-initialisation, and KL re-annealing, ITW EER trended upward during Phase 2 in every configuration tested (for example, `sweep_lr5e6_freeze0_p1_5` trended +5.20 pp over five Phase 2 epochs, and `fix_vib_reinit` trended +18.12 pp as the re-initialised VIB compounded the shock). Figure 9 visualises this pattern: Phase 1 trajectories (frozen backbone) converge tightly, while Phase 2 trajectories (unfrozen) drift upward across every configuration. The pattern localises the negative-transfer source to adversarial gradient conflict at the unfreezing boundary rather than to any individual component. This is contribution B9 (staged SSL backbone unfreezing is an open stability problem): the precedent literature spans the BERT fine-tuning stability paper [63], scheduled unfreezing with Fisher information [64], Speech-FT cross-task merging [65], less-forgetting fine-tuning for SSL [66], WavLM's own delayed-unfreezing schedule [67], child-ASR transfer with staged adaptation [68], and continual layer-specific fine-tuning [69]. We confirm: our Phase 2 ramp instability is one more anti-spoof data point on the open problem, not a GLADA-specific bug. The remedy most likely to restore positive transfer is a per-discriminator Phase 2 GRL warmup schedule that resets each $\lambda$ to zero at the unfreeze point and ramps it back independently, reducing the simultaneous-gradient count at the most fragile transition.

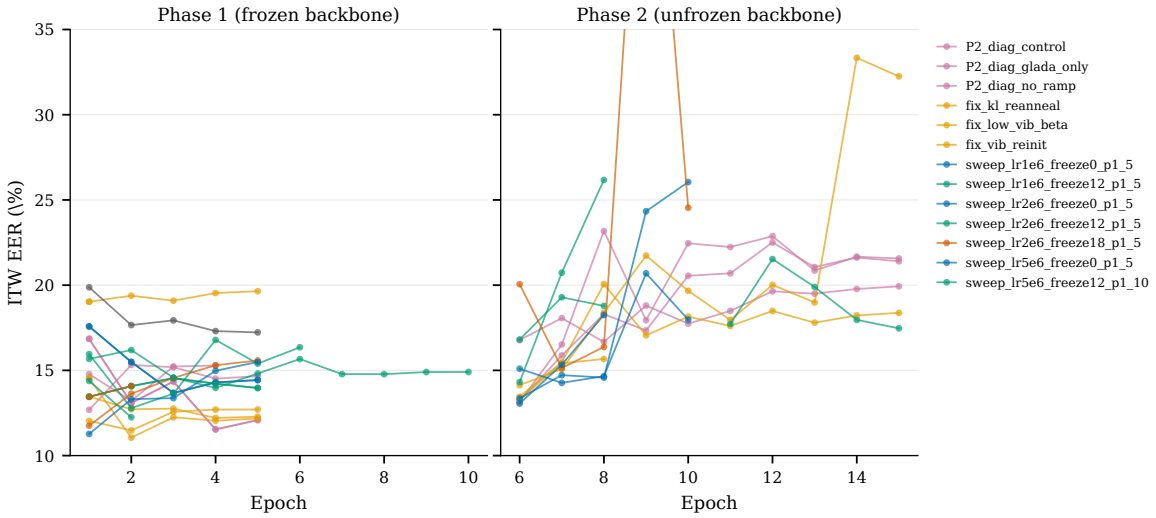



Figure 9: ITW EER trajectories for every Phase B stabilization-sweep configuration. Phase 1 (left, frozen backbone) converges to a narrow ∼12--15% band, whereas Phase 2 (right, unfrozen backbone) shows upward drift in every configuration, indicating that the negative-transfer source is gradient shock at the unfreezing boundary rather than any individual component choice.

**Practical implications.** From an engineering standpoint, the results favour simpler systems that are easier to monitor over highly composite objectives. For a practical deepfake detector, that means prioritising robust training dynamics, careful checkpoint selection, and backbone quality over architectural complexity, unless the complexity produces consistent cross-domain gains. The backbone comparison reinforces this. A well-chosen backbone (`wavlm-large`) with the full GLADA stack reached 12.64% ITW EER, while the best `wavlm-base-plus` ablation reached a comparable 12.98% simply by removing CKA. A practitioner working under compute constraints will get more out of backbone selection and stable single-objective training than from adding multiple auxiliary losses.

### 6.6 Confirmations of Precedent Findings

Several findings reported in this project are not novel; they are confirmations of patterns already documented in the SSL anti-spoofing and adjacent literature.

Treating them as confirmations rather than as headline novelty keeps the contribution accounting honest.

**Speaker adversarial training supports anti-spoof generalisation (B4).** Removing the speaker adversary improved `wavlm-base-plus` ITW by 6.23 pp inside the full GLADA stack, but the multi-seed per-group VIB model, which retains the speaker GRL on each group, still beat the monolithic VIB on every paired seed. The speaker-GRL contribution is therefore not a unilateral positive but a component whose net benefit depends on the rest of the loss stack. This is consistent with the speaker-vs-channel disentanglement literature: adversarial speaker verification by Meng et al. [51], channel-invariant speaker embedding learning with joint multi-task and adversarial training by Chen et al. [52], joint ASV and anti-spoofing optimisation in the embedding space by Gomez-Alanis et al. [53], and speaker disentanglement of SSL via interpretability by Zhu and Li [54] all report that speaker adversarial signals help cross-domain transfer when balanced against the primary task objective.

**RawBoost and waveform augmentation help (B5).** Removing RawBoost (`no_rawboost`: 16.66% ITW EER, $\Delta = +3.04$ pp relative to the best `wavlm-base-plus` ablation) confirms the precedent finding that waveform augmentation helps cross-domain audio anti-spoofing. The precedent is well-established by the RawBoost paper itself [55], the WavLM-back-end recipes with combined augmentation [56], WavLM + advanced RawBoost [57], multilingual augmentation strategies [58], the Vicomtech wav2vec2 baseline [59], and the broader voice-anti-spoofing augmentation literature [60, 61, 62]. The absence of MUSAN/RIR is a documented engineering gap, not a methodological choice.

**Cross-dataset evaluation reveals dataset-specific overfit (B6).** The six datasets produced six different rankings, with no single configuration dominating across all shifts (Section 5.9). This is the well-documented cross-corpus pattern in audio anti-spoofing: ASVspoof 2019 LA in-domain results do not transfer cleanly to ITW, ASVspoof 2021 LA/DF, WaveFake, or CodecFake+. We confirm on six datasets.

**Linear probes quantify residual nuisance after invariance training (B7).** The domain probe accuracies of 0.96--0.98 in Section 5.6 confirm the precedent finding that linear classifiers can recover information that the adversarial training was meant to remove. Olijslager et al. [39], Wang and Hansen [40], Mohamed et al. [41], Plachouras et al. [42], Rauch et al. [43], and Zaiem et al. [44] document this pattern for speaker, demographic, and general SSL probing. The per-layer-group instantiation is the contribution; the leakage pattern itself is precedent.

### 6.7 Limitations

The study has several limitations. Most ablation configurations were trained with a single random seed. The backbone comparison revealed substantial seed sensitivity for WavLM-Large (12.64% vs. 15.93% ITW for seeds 42 and 123), so multi-seed validation is needed before drawing firm conclusions about any single configuration. Multi-seed replication across all 11 ablations would strengthen the causal claims but was infeasible within the available compute budget.

The ablation screening used a reduced training schedule of 3 frozen plus 17 unfrozen epochs, against the 5+45 schedule used for the backbone comparison runs. The shorter schedule was necessary to evaluate 11 configurations within Colab time limits but may have disadvantaged configurations that require longer convergence, particularly those with multiple interacting objectives. The negative-transfer finding

is robust to this concern (longer training would need to reverse a 6+ point gap), but the relative ordering among intermediate configurations could shift with extended training.

The training pipeline did not include MUSAN noise or room impulse response (RIR) augmentation, both of which are standard in recent audio anti-spoofing systems [9]. RawBoost provides codec and compression augmentation, but without additive noise and reverberation there was a gap in acoustic coverage, most visibly on ITW where the recordings span a wide range of environments.

Score-level fusion was not attempted. Published systems such as the 4-model fusion baseline [17] (6.40% ITW EER) show that combining complementary models can substantially improve generalisation. Evaluation here was limited to single-model performance; ensemble gains from GLADA components remain unexplored.

All experiments were conducted on Google Colab with T4 or A100 GPUs, with session time limits, memory constraints, and occasional preemptions that shaped hyperparameter choices: batch size, gradient accumulation steps, the number of evaluation datasets per session. A dedicated compute cluster would have permitted more extensive hyperparameter sweeps and longer training runs.

Only five backbone architectures were evaluated with the full GLADA stack, and only `wavlm-base-plus` received the full 11-configuration ablation treatment. The interaction between backbone choice and optimal GLADA configuration remains unexplored: grouped DA may behave differently with larger or more diverse SSL pretraining, and the current ablation cannot speak to those regimes.

# CHAPTER 7

# Conclusion and Future Work

## 7.1 Summary of Findings

Grouped-layer adversarial domain adaptation did not improve cross-domain audio deepfake detection. Across 96 EER measurements covering 11 ablation configurations, five SSL backbone families, and six evaluation datasets, every simplification of the proposed objective stack beat the full formulation on the primary cross-domain benchmark.

The headline result was negative transfer on ITW. Stacking grouped adversarial alignment with CKA, VIB, and speaker-adversarial losses pushed `full_glada` to the worst ITW EER among all 11 `wavlm-base-plus` configurations (Section 5.1). Optimising several invariance-promoting objectives at once over-constrained the shared representation and threw away information the classifier needed once the acoustic conditions changed. Parsimony, not stacking, was the better generalisation strategy. Every leave-one-out ablation improved on the full formulation, and the three largest gains came from dropping CKA, the speaker adversary, or VIB, the three components meant to enforce invariance in the first place. They competed for the same representational capacity instead of complementing each other, a multi-objective interference effect that Chapter 5 traces in detail.

Which backbone I used mattered more than any component-level decision. The ITW spread across five backbone families was 9.34 pp, larger than the 6.73 pp spread across all 11 ablations, which identifies the pre-trained representation as the primary driver of generalisation. The 3.29-point seed gap on WavLM-Large is a reminder that multi-seed validation is essential before drawing any backbone-level conclusion. Cross-dataset robustness was shift-dependent too: no single configuration dominated across the six evaluation sets. Telephony, codec compression, neural vocoder, and

codec-token resynthesis each rewarded a different simplification, and the best model changed with the shift type. That argues against single-benchmark optimisation and in favour of multi-dataset evaluation protocols.

A targeted-bottlenecking direction emerged from the cross-dataset comparison of VIB placement (Section 5.4). On In-the-Wild the full per-group VIB beat the monolithic post-pooling placement by 2.69 pp and beat the lowest-layer-only configuration by 1.45 pp. On every other cross-domain benchmark tested, the picture flipped: the g0-only configuration (bottleneck active only on the lowest layer group) was the lowest-mean placement on 2021 LA, 2021 DF, WaveFake, and CodecFake+, with the tightest seed envelopes on most of them, while the inverse configuration (bottleneck on $g_1, g_2$ only) was the worst-ranked variant on ITW and below g0-only on every secondary dataset. Targeted bottlenecking of the lowest SSL layer group, then, is the placement that generalised most consistently across the breadth of cross-domain shifts in the evaluation suite. The 2021 DF column sits near 44% EER for every placement variant and reflects dataset-level difficulty rather than the placement choice; the ranking on that column should not be over-interpreted.

A gap to recent state-of-the-art methods remained. Section 5.10 details how our best single model exceeded older baselines but trailed IB-CAAN and HierCon by 5--8 points on ITW, which points toward contrastive and hierarchical alignment as more promising alternatives to grouped adversarial training.

## 7.2 Main Contributions

This work contributes the following to the audio deepfake detection literature.

1. **Companion survey of cross-dataset audio anti-spoofing.** The CS 297 survey analyses published audio anti-spoofing systems through a cross-dataset lens. Most prior surveys report in-domain numbers only; this one also reviews

the cross-dataset generalisation the same authors report, flags the recording- and codec-shift failure modes that recur, and fixes the baseline context against which GLADA is compared.

2. **The GLADA framework.** The framework combines grouped hierarchical SSL aggregation, per-group adversarial domain adaptation, and per-group variational information bottlenecks for audio anti-spoofing. Transformer layers are partitioned into semantically motivated groups, and each group gets its own gradient-reversal discriminator and its own VIB module; hierarchical attention pooling, a CKA diversity penalty, and a speaker adversary sit on top, all trained under a single objective. I am not aware of prior work that combines all three on grouped SSL speech features; component-wise precedents include MLADA in vision [11] and stratified adversarial domain adaptation in NLP [14].

3. **A detailed empirical record.** Eleven configurations, 5 backbone families, and 96 dataset--model pairs, all evaluated on the same six datasets with bootstrap confidence intervals. The record is deeper than the ablations most audio anti-spoofing papers report, and it leaves behind a replicable experimental trail for future work.

4. **A transparent negative-result analysis.** I document the conditions under which grouped-layer adversarial domain adaptation does *not* beat simpler baselines. Reporting configuration-level EERs with confidence intervals and tying the degradation back to specific auxiliary objectives makes it clear what to avoid next time and where future architectural work should focus.

5. **A peer-reviewed conference publication.** A condensed version of this work, co-authored with the advisor and Jade Webb, has been accepted as a

full paper at the 50th IEEE Computer Software and Applications Conference (COMPSAC 2026) in Madrid, Spain, which accepted 28.5% of submissions as full papers. Peer review externally validates the experimental methodology and the reporting of the negative-transfer finding as a scholarly contribution to the cross-domain audio deepfake detection literature.

## 7.3 Future Research Directions

A handful of directions follow directly from the limitations listed above.

- **Checkpoint averaging and score-level fusion.** Averaging weights over the final training epochs and fusing scores across complementary backbones (WavLM-Large and XLSR-300M, for example) are low-cost post-hoc strategies. They should reduce seed variance and recover some of the robustness lost to the conflicting auxiliary objectives. Neither needs additional training, and both fit the existing pipeline directly.
- **Foundation model fine-tuning.** The backbone was the dominant factor in this study, so the natural next step is to push on that axis directly: larger SSL models such as WavLM-Large on a longer fine-tuning budget, and models pre-trained on broader multilingual corpora. A higher-capacity backbone should reduce how much the auxiliary adaptation objectives have to carry, and it should also improve transfer to out-of-distribution codec and language shifts.
- **Phase-aware loss scheduling.** The instability I observed came from VIB, CKA, speaker, and domain objectives reaching the backbone all at once. Staging them (keeping the backbone frozen initially and phasing the domain and VIB terms in only after unfreezing) is the natural countermeasure. A curriculum that ramps the domain penalties only once the classifier has stabilised is the concrete form of that idea.
- **Per-discriminator GRL warmup at Phase 2 onset.** The current schedule

ramps a single global GRL coefficient $\lambda$ once at the start of training. The Phase B sweep in Chapter 6 localised the worst degradation to the unfreezing transition, where four per-group discriminators and the speaker adversary all apply full-strength reversed gradients to newly-trainable parameters at the same time. Resetting each discriminator's $\lambda$ to zero at the unfreeze point and ramping it back up independently would keep only a subset of the reversed-gradient signals active at any time. That is the most targeted remedy the diagnostic data supports.

- **Untested configurations from the April 13 hyperparameter sweep.** Three Phase B configurations did not complete because the preemptible H100 session on April 13 timed out mid-run. `fix_ewc` adds an Elastic Weight Consolidation penalty on backbone drift and would test whether constraining how far the backbone can move during Phase 2 prevents the degradation. `fix_pure_classifier` removes domain adaptation, VIB, CKA, and the speaker adversary, isolating the auxiliary-objective stack from hierarchical pooling and backbone fine-tuning. Running `fix_ultra_low_lr` on the full Phase 2 budget would indicate whether a much smaller backbone learning rate suffices on its own to avoid the gradient shock. Completing these three runs is the smallest experimental increment that would determine whether the negative-transfer finding is architectural or a training-dynamics artefact.
- **Group-count sweep.** All reported GLADA experiments used four groups for WavLM-Large and three for WavLM-Base-Plus. Sweeping $G \in \{2, 3, 4, 6\}$ would indicate whether the per-group adversarial signal scales with granularity or saturates early, and whether a coarser grouping ($G = 2$) reduces the simultaneous-gradient count enough to sidestep the Phase 2 instability altogether.
- **MUSAN and RIR augmentation.** The current pipeline uses RawBoost

only, which simulates channel and codec artefacts but excludes environmental noise and reverberation. Adding MUSAN additive noise and RIR convolution as further augmentation stages should improve robustness to the acoustic-environment shifts that dominate ITW. The package already includes the code for both stages; experimental validation is all that remains.

- **Contrastive alternatives to adversarial domain alignment.** HierCon reached 6.87% ITW EER, which is consistent with contrastive-learning-based alignment being more stable than adversarial training under the kind of multi-dataset evaluation used here. Replacing or supplementing the gradient-reversal discriminators with supervised contrastive losses over domain labels is a natural follow-up motivated by the performance gap in Section 5.10.
- **Multi-seed validation across configurations.** WavLM-Large shifted from 12.64% to 15.93% ITW EER between two seeds, so single-seed EER comparisons cannot reliably discriminate between configurations. Systematic multi-seed runs across the top-performing ablations would quantify that variance and determine whether the Section 5.1 ordering is statistically robust.
- **Multi-seed validation of the $\beta$ response curve.** The preliminary probe in Section 5.5 produced 10.73% ITW EER on seed 42 at $\beta_{\text{eff}} = 1.67 \times 10^{-3}$, but it rests on a single seed. Repeating the four sweep points across seeds 123 and 456, plus a finer sweep through the region between $\beta_{\text{eff}} = 10^{-3}$ and $5 \times 10^{-3}$, would establish whether the across-seed mean confirms the lowest tested $\beta$ as the favourable setting or lands elsewhere.
- **Per-group $\beta$ scheduling.** The $\beta$ response curve treats $\beta$ as a single global scalar that applies uniformly to all active groups. Once more than one group's VIB is active, the optimal compression pressure may differ across groups: g0 wants enough compression to suppress recording-condition shift, while g2 may need a

lighter touch to preserve task-discriminative information. Per-group $\beta$ values, possibly with a phase-dependent schedule, would test that hypothesis and may close the remaining gap between g0-only and the full per-group configuration.

- **Confidence-aware adversarial weighting.** IB-CAAN's confidence-aware attention down-weights examples on which the discriminator is unreliable, producing a smoother adversarial signal. Adapting that mechanism to per-group discriminators (each weighted by its own predicted-domain confidence) is the most direct architectural change motivated by the IB-CAAN gap discussion in Section 6.4.
- **Standardised cross-dataset evaluation protocols.** Because the ranking of configurations depended so heavily on which shift was being tested, the community would benefit from agreed-upon evaluation suites spanning in-domain, near-domain, and far-domain conditions, so cross-study comparisons can rest on a common footing.

### 7.4 Closing Remark

This work set out to improve cross-domain generalisation through grouped adversarial domain adaptation, and found that the approach degraded ITW EER relative to every simpler baseline tested. Reporting that result, alongside the experimental record showing why each component interacted destructively, is as important as reporting a gain would have been. The ablation framework and the negative-transfer analysis established here give a concrete starting point for future work on multi-objective adaptation in speech processing.

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# APPENDIX A

## CS 297 Survey Experiment Results

This appendix reproduces the key experimental results from the CS 297 survey project (Fall 2025), which evaluated six domain adaptation methods across five SSL backbones for audio deepfake detection. These results motivated the design of GLADA by establishing that (1) information bottleneck (IB) constraints stabilize adversarial domain adaptation, and (2) backbone choice dominates method choice for cross-domain generalization.

### A.1 Full In-the-Wild Results

Table A.15 reports EER on the In-the-Wild evaluation set for all backbone--method combinations from the CS 297 systematic comparison.

Table A.15: Complete CS 297 evaluation metrics on In-the-Wild (seed=42). All models trained on ASVspoof 2019 LA with ITW as unlabeled target domain. Sorted by EER.

| **Backbone** | **Method** | **EER (%)** | **AUC (%)** | **Acc (%)** | **F1 (%)** |
|---|---|---|---|---|---|
| Fusion (WavLM-BP+XLSR) | DANN-IB | **13.70** | **94.29** | **86.30** | **88.78** |
| WavLM-BP | DANN-IB | 14.24 | 92.96 | 85.76 | 88.32 |
| WavLM-BP | CAAN | 14.51 | 92.60 | 85.49 | 88.10 |
| WavLM-Large | CAAN | 18.26 | 89.37 | 81.74 | 84.90 |
| WavLM-Large | DANN-IB | 18.39 | 89.19 | 81.61 | 84.79 |
| XLSR-300M | DANN-IB | 19.01 | 89.41 | 80.99 | 84.26 |
| WavLM-BP | MMD | 19.55 | 89.12 | 80.44 | 83.79 |
| WavLM-Large | MMD | 19.59 | 88.09 | 80.41 | 83.76 |
| WavLM-Large | DANN | 19.69 | 87.58 | 80.31 | 83.67 |
| XLSR-300M | MMD | 20.73 | 87.60 | 79.27 | 82.77 |
| XLSR-300M | CORAL | 21.52 | 86.97 | 78.48 | 82.08 |
| XLSR-300M | DANN | 21.65 | 87.26 | 78.35 | 81.97 |
| WavLM-Large | No Adapt | 21.91 | 85.11 | 78.09 | 81.75 |
| WavLM-Large | CORAL | 22.27 | 84.46 | 77.73 | 81.43 |
| XLSR-300M | No Adapt | 22.33 | 86.02 | 77.67 | 81.38 |
| WavLM-BP | No Adapt | 22.42 | 85.10 | 77.58 | 81.30 |
| WavLM-BP | CORAL | 23.46 | 84.22 | 76.54 | 80.39 |
| WavLM-BP | DANN | 25.57 | 83.31 | 74.43 | 78.52 |

Table A.15: CS 297 In-the-Wild results (continued).

| Backbone | Method | EER (%) | AUC (%) | Acc (%) | F1 (%) |
|---|---|---|---|---|---|
| Wav2Vec2-Base | DANN-IB | 28.38 | 76.78 | 71.62 | 76.02 |
| Wav2Vec2-Base | MMD | 28.74 | 76.94 | 71.26 | 75.70 |
| Wav2Vec2-Base | CAAN | 29.01 | 75.50 | 70.99 | 75.46 |
| Wav2Vec2-Base | DANN | 30.12 | 74.80 | 69.88 | 74.46 |
| Wav2Vec2-Large | DANN | 31.49 | 76.06 | 68.51 | 73.22 |
| Wav2Vec2-Large | No Adapt | 32.14 | 74.65 | 67.85 | 72.62 |
| Wav2Vec2-Large | CORAL | 32.28 | 74.32 | 67.72 | 72.49 |
| Wav2Vec2-Large | MMD | 32.45 | 74.16 | 67.55 | 72.34 |
| Wav2Vec2-Base | No Adapt | 33.69 | 70.53 | 66.31 | 71.20 |
| Wav2Vec2-Base | CORAL | 34.22 | 69.43 | 65.78 | 70.72 |

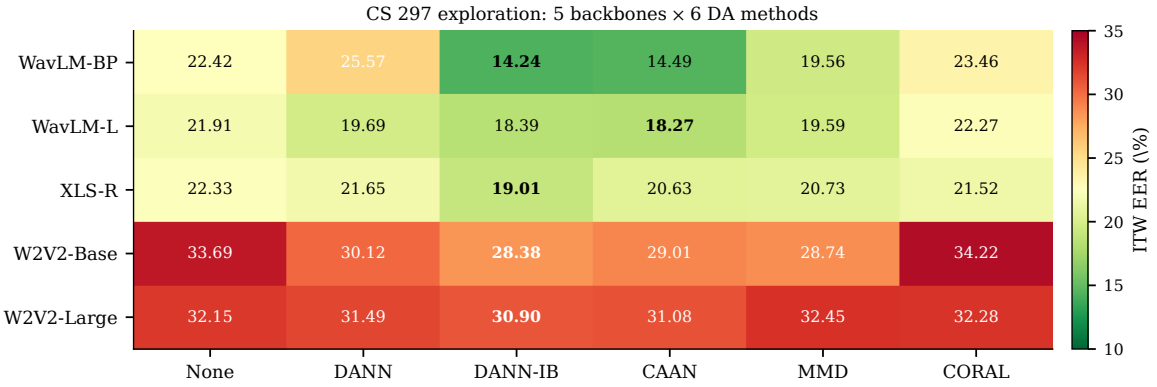



Figure A.10: CS 297 exploration matrix: ITW EER (%) across 5 SSL backbones × 6 domain adaptation methods. Per-row best is bold. DANN-IB consistently stabilizes adversarial adaptation across all five backbones.

## A.2 Cross-Domain Results (ASVspoof 2021 DF)

Table A.16 reports EER on ASVspoof 2021 DF for all configurations, alongside ITW EER for cross-reference. This dataset tests robustness to codec compression artifacts across nine conditions.

Table A.16: CS 297 cross-domain results on ASVspoof 2021 DF (611,829 samples). Sorted by DF EER.

| **Backbone** | **Method** | **DF EER (%)** | **ITW EER (%)** |
|---|---|---|---|
| XLSR-300M | MMD | 24.64 | 20.73 |
| XLSR-300M | DANN-IB | 24.70 | 19.01 |
| XLSR-300M | CORAL | 24.99 | 21.52 |
| XLSR-300M | No Adapt | 25.07 | 22.33 |
| WavLM-Large | No Adapt | 25.89 | 21.91 |
| WavLM-Large | DANN-IB | 25.90 | 18.39 |
| WavLM-Large | DANN | 25.96 | 19.71 |
| Fusion | DANN-IB | 25.98 | 13.70 |
| WavLM-Large | CAAN | 26.81 | 18.26 |
| WavLM-Large | MMD | 26.93 | 19.59 |
| WavLM-Large | CORAL | 27.14 | 22.27 |
| WavLM-BP | CORAL | 27.59 | 23.46 |
| WavLM-BP | No Adapt | 27.60 | 22.42 |
| WavLM-BP | MMD | 27.62 | 19.55 |
| WavLM-BP | CAAN | 28.86 | 14.51 |
| WavLM-BP | DANN-IB | 28.88 | 14.24 |
| Wav2Vec2-Base | No Adapt | 33.80 | 33.70 |
| Wav2Vec2-Base | CORAL | 33.90 | 34.22 |
| WavLM-BP | DANN | 34.58 | 25.57 |
| Wav2Vec2-Large | No Adapt | 36.39 | 32.14 |
| Wav2Vec2-Large | CORAL | 36.76 | 32.28 |
| Wav2Vec2-Large | MMD | 37.16 | 32.45 |
| XLSR-300M | DANN | 39.39 | 21.65 |
| Wav2Vec2-Base | MMD | 39.95 | 28.74 |
| Wav2Vec2-Large | DANN | 42.22 | 31.49 |
| Wav2Vec2-Base | DANN | 45.74 | 30.12 |
| Wav2Vec2-Base | DANN-IB | 46.94 | 28.38 |
| Wav2Vec2-Base | CAAN | 48.62 | 29.01 |

## A.3 Multi-Seed Validation

Per-seed ITW EER across three random seeds (42, 123, 456) for the top five configurations.

Table A.17: CS 297 multi-seed validation results (EER % on In-the-Wild).

| Backbone | Method | S42 | S123 | S456 | Mean | Std |
|---|---|---|---|---|---|---|
| WavLM-BP | DANN-IB | 14.24 | 15.44 | 20.66 | 16.78 | 2.79 |
| WavLM-BP | CAAN | 14.49 | 16.85 | 22.01 | 17.79 | 3.23 |
| WavLM-Large | CAAN | 18.27 | 18.00 | 18.04 | 18.10 | 0.12 |
| WavLM-Large | DANN-IB | 18.38 | 19.50 | 17.72 | 18.53 | 0.73 |
| XLSR-300M | DANN-IB | 19.01 | 20.69 | 17.88 | 19.19 | 1.15 |

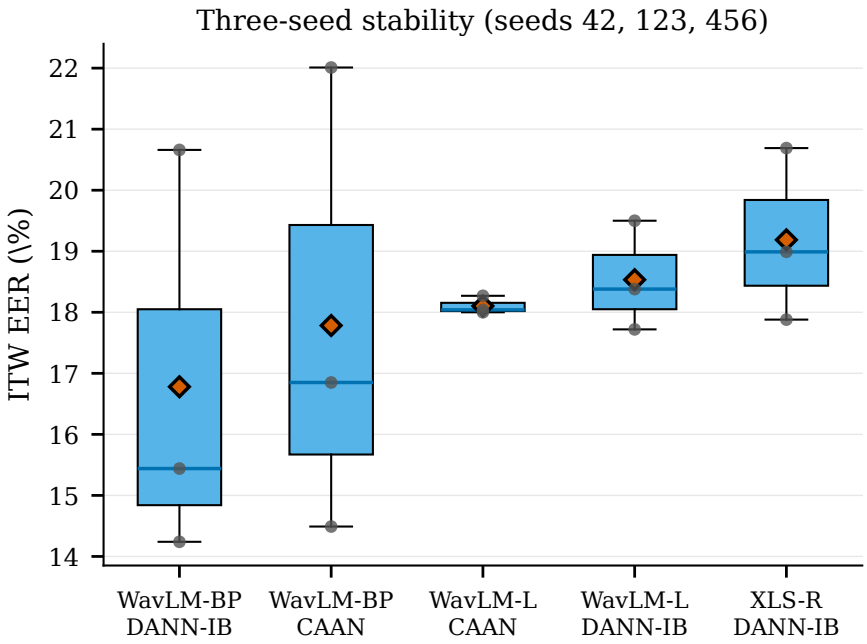



Figure A.11: Per-seed ITW EER distributions for the top five CS 297 configurations (seeds 42, 123, 456). WavLM-BP configurations achieve the lowest peak EER but show the widest seed spread; WavLM-Large CAAN is most stable ($\sigma = 0.12$).

## A.4 Bootstrap Confidence Intervals

95% bootstrap confidence intervals (1,000 resamples) confirming the top-ranked models are statistically separated.

## A.5 Multi-Layer Ablation

Experiment D results: learnable multi-layer weighted sum with and without latent-space augmentation on WavLM-Base-Plus (ML = multi-layer; Aug = augmentation; $\Delta$ relative to single-layer baseline).

Table A.18: CS 297 bootstrap 95% confidence intervals for EER on In-the-Wild.

| **Backbone** | **Method** | **EER (%)** | **CI Low** | **CI High** |
|---|---|---|---|---|
| Fusion | DANN-IB | 13.70 | 13.34 | 14.10 |
| WavLM-BP | DANN-IB | 14.24 | 13.86 | 14.66 |
| WavLM-BP | CAAN | 14.51 | 14.11 | 14.92 |
| WavLM-Large | CAAN | 18.26 | 17.78 | 18.74 |
| WavLM-Large | DANN-IB | 18.39 | 17.96 | 18.80 |
| XLSR-300M | DANN-IB | 19.01 | 18.56 | 19.44 |

Table A.19: CS 297 multi-layer aggregation ablation results (EER % on In-the-Wild, WavLM-Base-Plus).

| **Method** | **Config** | **EER (%)** | **Best Ep.** | **Baseline** | **Δ** |
|---|---|---|---|---|---|
| DANN-IB | ML only | 16.98 | 19 | 14.24 | +2.74 |
| DANN-IB | ML + Aug | 20.80 | 7 | 14.24 | +6.56 |
| CAAN | ML + Aug | 20.95 | 5 | 14.49 | +6.46 |
| CDAN | ML + Aug | 19.04 | 6 | --- | --- |
| No Adapt | ML + Aug | 14.78 | 13 | 22.41 | −7.63 |
| MMD | ML + Aug | 16.01 | 6 | 19.55 | −3.54 |

# APPENDIX B

## GLADA Ablation Details

This appendix provides the full ablation results with bootstrap confidence intervals, training curves for all configurations, and hyperparameter sensitivity notes supplementing the analysis in Chapter 5.

### B.1 Full Ablation Results with Bootstrap CIs

Table B.20 extends Table 6 by reporting 95% bootstrap confidence intervals (1,000 resamples) for all six datasets and all 11 `wavlm-base-plus` configurations.

Table B.20: Full ablation results with 95% bootstrap confidence intervals. All values are EER (%); lower is better. CIs reported as (low--high) beneath each point estimate.

| Config | 2019 LA | ITW | 2021 LA | 2021 DF | WaveFake | Codec-Fake+ |
|---|---|---|---|---|---|---|
| *Leave-one-out ablations* | | | | | | |
| `full_glada` | 1.04 | 19.71 | 31.02 | 27.74 | 27.88 | 37.77 |
| | (0.92--1.18) | (19.25--20.17) | (30.71--31.31) | (27.51--27.98) | (27.45--28.42) | (36.27--39.36) |
| `no_glada` | 1.36 | 14.89 | 31.53 | 27.96 | 26.76 | 36.97 |
| | (1.23--1.51) | (14.51--15.30) | (31.22--31.84) | (27.73--28.19) | (26.33--27.22) | (35.42--38.54) |
| `no_vib` | 0.72 | 14.58 | 33.02 | 29.73 | 19.17 | 36.88 |
| | (0.62--0.84) | (14.19--14.97) | (32.70--33.33) | (29.49--29.96) | (18.79--19.58) | (35.34--38.48) |
| `no_speaker` | 0.93 | 13.48 | 31.78 | 27.57 | 27.30 | 37.58 |
| | (0.82--1.06) | (13.11--13.90) | (31.47--32.10) | (27.34--27.80) | (26.87--27.73) | (36.02--39.12) |
| `no_cka` | 1.07 | 12.98 | 32.15 | 29.50 | 29.74 | 37.88 |
| | (0.95--1.20) | (12.60--13.33) | (31.83--32.46) | (29.27--29.74) | (29.30--30.21) | (36.34--39.46) |
| `no_rawboost` | 1.05 | 16.66 | 31.98 | 26.86 | 23.14 | 39.28 |
| | (0.93--1.19) | (16.23--17.08) | (31.68--32.30) | (26.63--27.09) | (22.74--23.58) | (37.73--40.79) |

Table B.20: Full ablation results (continued).

| Config | 2019 LA | ITW | 2021 LA | 2021 DF | WaveFake | Codec-Fake+ |
|---|---|---|---|---|---|---|
| `no_hier` | 1.15 (1.03--1.29) | 17.47 (17.05--17.89) | 33.24 (32.92--33.55) | 28.52 (28.29--28.76) | 26.69 (26.26--27.14) | 39.59 (38.08--41.17) |
| *Structural variants* | | | | | | |
| `global_da_only` | 1.42 (1.28--1.56) | 14.99 (14.56--15.37) | 31.51 (31.19--31.82) | 27.95 (27.72--28.18) | 26.61 (26.18--27.07) | 37.27 (35.71--38.83) |
| `hierarchical_only` | 1.02 (0.90--1.15) | 13.28 (12.92--13.63) | 29.36 (29.06--29.66) | 27.24 (27.01--27.48) | 23.25 (22.84--23.67) | 39.09 (37.50--40.62) |
| `no_da` | 0.80 (0.69--0.92) | 14.20 (13.82--14.63) | 30.37 (30.07--30.68) | 27.01 (26.78--27.24) | 24.54 (24.13--24.97) | 37.98 (36.39--39.51) |
| `backbone_only` | 0.62 (0.52--0.73) | 13.60 (13.21--13.98) | 32.37 (32.06--32.68) | 28.50 (28.27--28.74) | 23.33 (22.92--23.76) | 37.46 (35.92--39.05) |

## B.2 Training Curves

Figures B.12--B.22 show training loss and ITW evaluation EER across epochs for all 11 ablation configurations. The vertical dashed line marks the Phase 1 $\rightarrow$ Phase 2 transition (backbone unfreeze at epoch 3). All curves share the same axis scales for comparability.

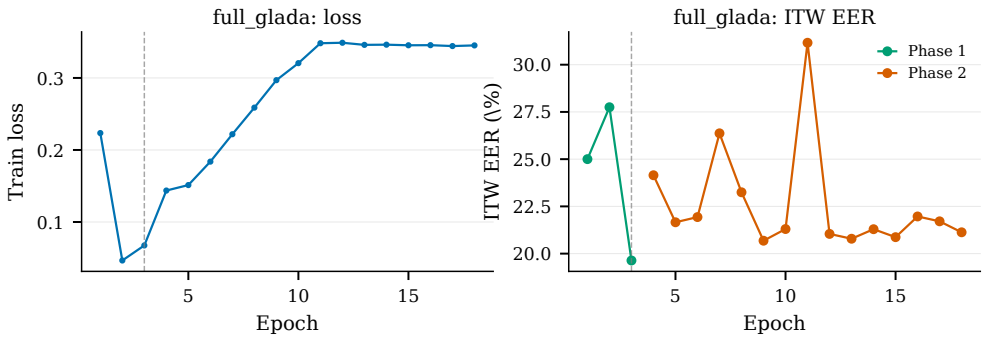



Figure B.12: Training curves for `full_glada`.

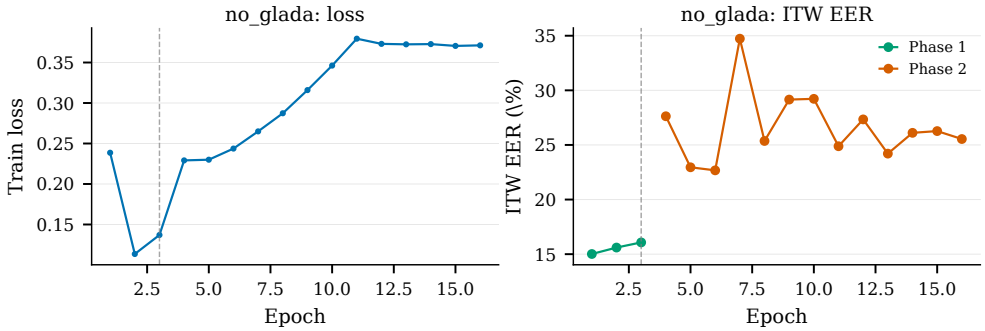



Figure B.13: Training curves for `no_glada`.

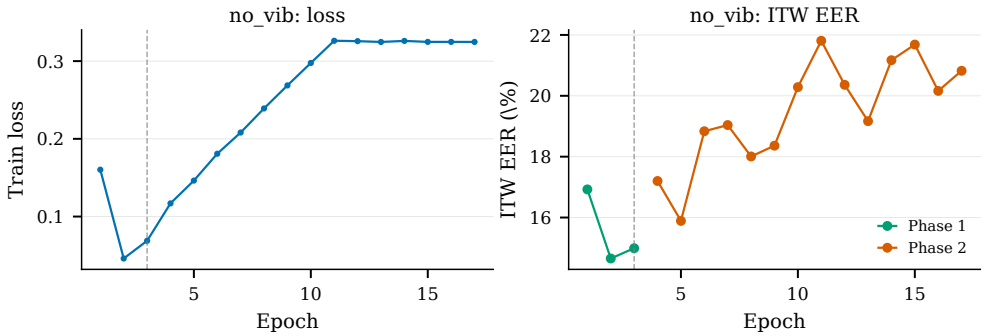



Figure B.14: Training curves for `no_vib`.

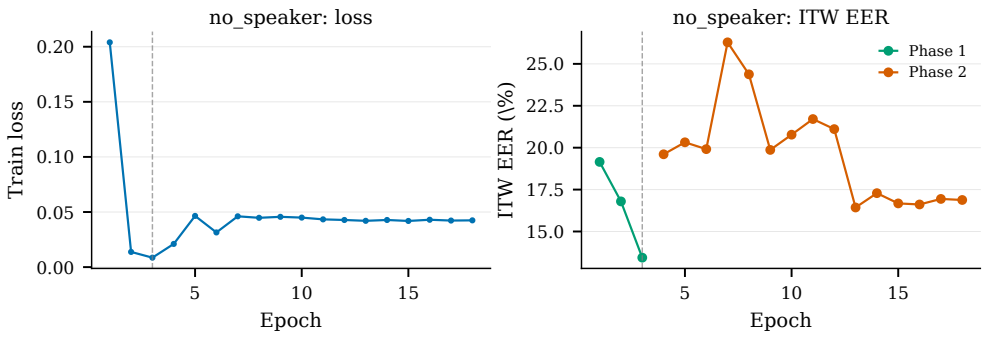



Figure B.15: Training curves for `no_speaker`.

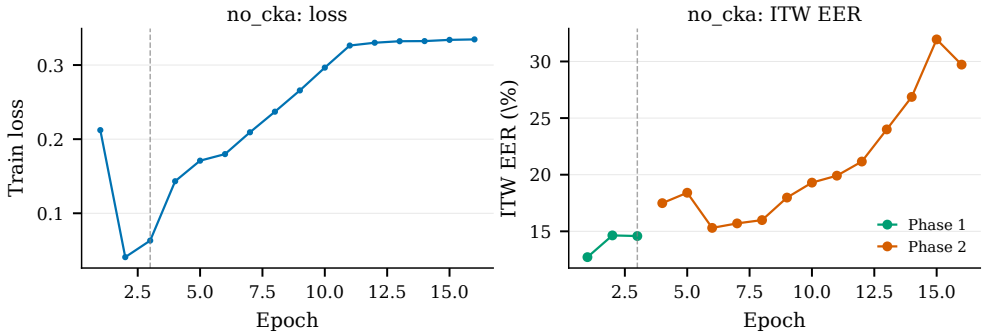



Figure B.16: Training curves for `no_cka`.

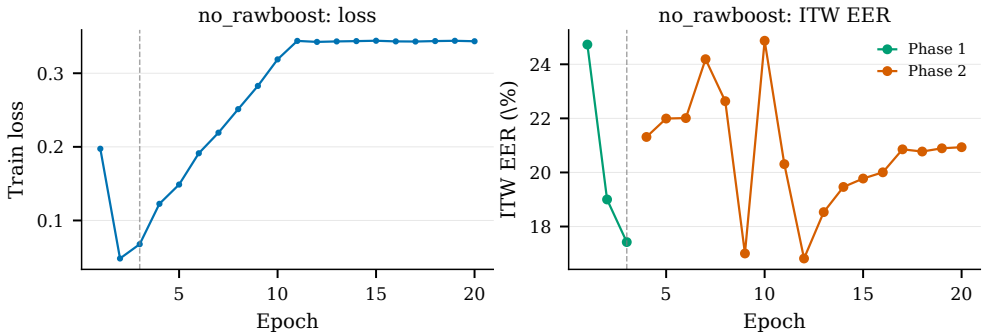



Figure B.17: Training curves for `no_rawboost`.

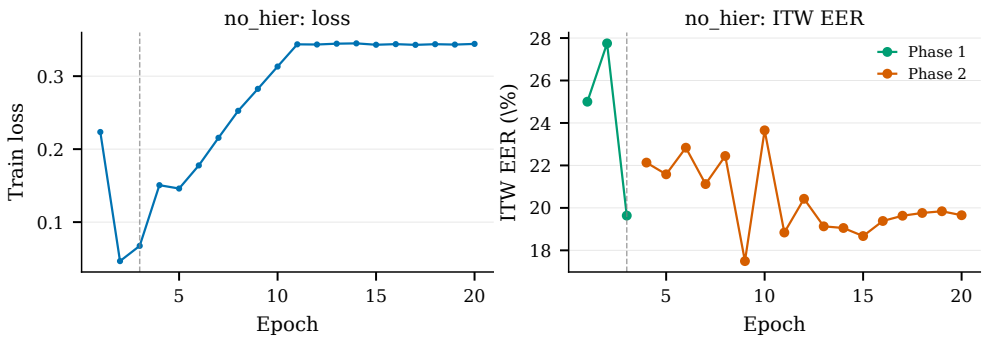



Figure B.18: Training curves for `no_hier`.

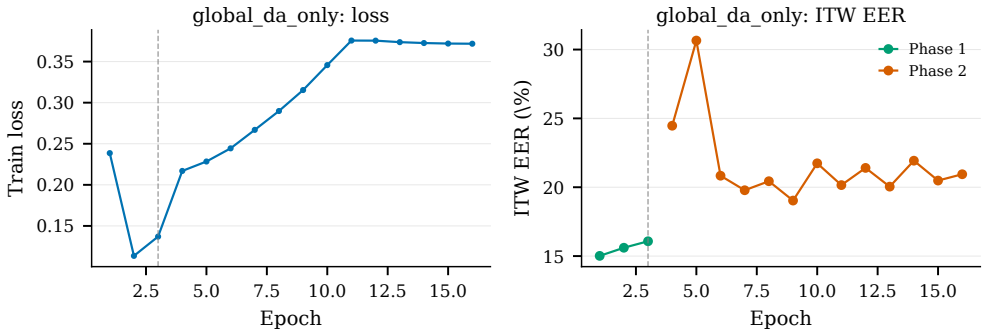



Figure B.19: Training curves for `global_da_only`.

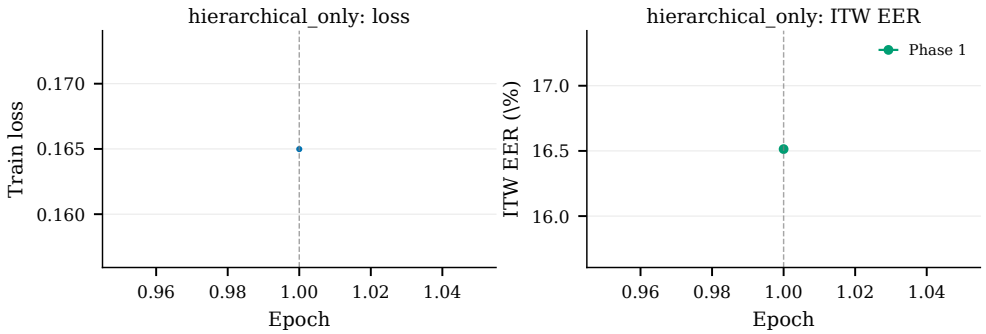



Figure B.20: Training curves for `hierarchical_only`.

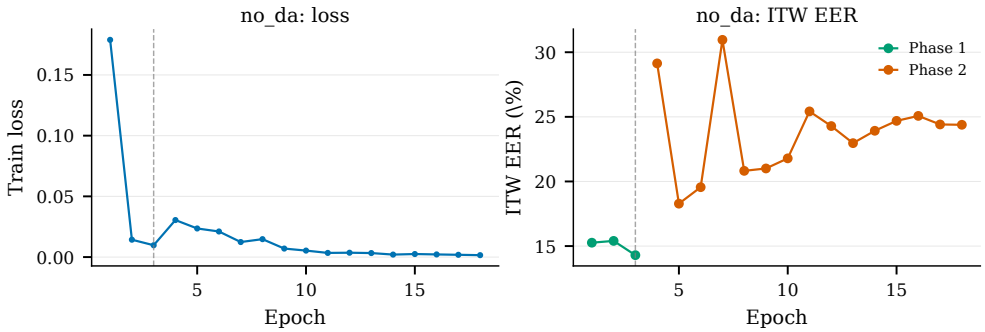



Figure B.21: Training curves for `no_da`.

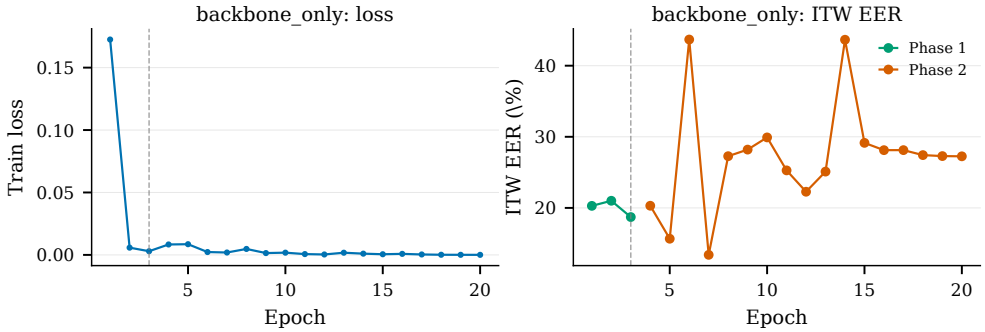



Figure B.22: Training curves for `backbone_only`.

### B.3 Hyperparameter Sensitivity

The following hyperparameter choices were validated during development and early ablation runs. Full sensitivity sweeps were not feasible within Colab GPU budgets, so values were chosen through targeted experiments documented in the development notes.

**Domain loss weight ($\alpha$).** Tested at 0.01, 0.05, 0.1, and 0.5. At $\alpha = 0.5$ training destabilised within the first three epochs and produced NaN losses. $\alpha = 0.1$ was selected because it allowed the domain discriminators to contribute a meaningful gradient signal without overwhelming the task loss. The lower values (0.01 and 0.05) produced no measurable improvement in ITW EER.

**VIB weight ($\beta$).** Set to $10^{-3}$ following Alemi et al. [86]. Without KL clamping, early runs produced KL values above 133,000 and caused gradient explosions; I added the `kl_clamp_max` parameter (default 10.0) to stabilise training. Removing VIB altogether (`no_vib`) improved ITW EER from 19.71% to 14.58%, which means the bottleneck discarded information useful for cross-domain detection.

**GRL warmup epochs.** The domain GRL uses a 5-epoch linear warmup within each training phase; the speaker GRL uses a 10-epoch warmup. Without the warmup, discriminator gradients were uninformative in early epochs and pushed the feature extractor in random directions. The GRL $\alpha$ resets to 0 at the Phase 2 transition to avoid destabilising the newly unfrozen backbone layers.

**Number of layer groups ($K$).** Determined by the backbone architecture: 3 groups for base models (13 layers $\div$ 3 $\approx$ 4--5 layers per group) and 4 groups for large models (25 layers $\div$ 4 $\approx$ 6 layers per group). Alternative groupings were not systematically

tested.

**CKA loss weight ($\delta$).** Set to 0.01. The CKA diversity loss is meant to keep the representations across layer groups distinct; removing it (`no_cka`) produced the best ITW EER at 12.98%, indicating that enforced diversity was counterproductive in this setting.